

The Complete Daily Integral

Field Guide

Structure before algebra

96 core generator archetypes, twelve active problem slots, and the full screenshot-learned technique index

Scope and evidence

This edition is keyed to the current local function `daily_integral_archetype_warmups()`, not only to the older archived generator skill. The live generator contains twelve active slots: four integrals, three derivatives, three differentials, one limit, and one logic problem. Each slot rotates through eight non-repeating families, giving **96 core archetypes**. The screenshot catalogue supplies a second layer of learned variants; those are indexed in Part III.

How to study a family

Read the recognition cue before the execution recipe. Then close the book and answer three questions: What feature triggers the method? What is the first written line? What independent check would expose a sign, branch, or endpoint error?

Universal diagnostic pass

1. **Read the request:** exact value, proof, count, asymptotic, derivative order, initial value, or model prediction.
2. **Inventory the structure:** bounds, symmetry, singularities, composite layers, polynomial roots, recurrence, boundary data, or group action.
3. **Name one family:** substitution, reflection, coefficient extraction, integrating factor, fixed-point rate, invariant, and so on.
4. **Write the first irreversible line:** transformed bounds, characteristic equation, logarithm of a power, or recurrence with boundary conditions.
5. **Check independently:** differentiate, substitute into the equation, inspect units and sign, test a boundary case, or compare a numerical approximation.

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PART I

Foundations and decision rules

Integral toolkit

Order of attack

Simplify and inspect symmetry before choosing substitution. On a definite integral, transform bounds immediately. On an improper integral, replace every singular endpoint by a limit before manipulating it. A successful antiderivative is checked by differentiation; a successful definite integral is also checked for sign, scale, and convergence.

Substitution, parts, and rational structure

Use $u = g(x)$ when $g'(x) dx$ is present. Use integration by parts when differentiating one factor makes it structurally simpler. Divide improper rational functions first, factor denominators, and let the discriminant decide between logarithms and arctangents. For rational trigonometric expressions, $t = \tan(x/2)$ turns the entire problem rational.

Symmetry and reflection

On $[-a, a]$, odd parts vanish and even parts double. More generally,

$$\int_{-a}^a f(x) dx = \int_0^a (f(x) + f(-x)) dx.$$

On $[0, a]$, compare $f(x)$ with $f(a - x)$. If x multiplies a reflection-invariant kernel K , then

$$\int_0^a xK(x) dx = \frac{a}{2} \int_0^a K(x) dx.$$

Special-function ladder

Gamma and beta encode moments; digamma and polygamma appear after parameter differentiation; polylogarithms are shifted by division by x and differentiation. Mellin, Fourier, and Laplace transforms should be recognised before a direct antiderivative is attempted.

Derivative toolkit

Simplify, then differentiate

Inspect the evaluation point first. Combine conjugate logarithms, use half-angle or product-to-sum identities, and replace a high derivative at zero by coefficient extraction: $f^{(n)}(0) = n![x^n]f(x)$. For products and quotients with nonzero factors, logarithmic derivatives organise the work.

Implicit and inverse structure

For $y = f^{-1}(x)$, locate the preimage before using $y' = 1/f'(y)$. For an implicit relation, differentiate once, collect the requested derivative, and verify the denominator is nonzero at the point. For $u(x)^{v(x)}$, take logs before differentiating.

Higher structure

Leibniz handles high product derivatives; Bell polynomials organise composite derivatives; Lagrange inversion extracts inverse-series coefficients. Jacobi's formula handles determinants, while Schwarzian invariance can remove an entire Möbius wrapper.

Differential-equation toolkit

Classification

Record order, linearity, autonomy, separability, and data type before solving. First-order equations may be separable, linear, Bernoulli, exact, homogeneous, or Riccati. Second-order constant-coefficient equations are controlled by characteristic roots; Cauchy-Euler equations use power trials.

Models and checks

Cooling, circuits, mixing, and queues become equilibrium plus transient. Logistic growth becomes linear in the odds. Always substitute the final function back into the equation and data; for models, also check units, monotonicity, and long-time behaviour.

Advanced operators

Characteristics solve first-order transport PDEs. Fourier eigenmodes diagonalise heat, wave, Laplace, and Sturm-Liouville operators. Variational problems produce Euler-Lagrange equations. Nonlinear oscillators often yield a first energy integral and an elliptic period.

Limit and puzzle toolkit

Asymptotic hierarchy

Try substitution, algebra, a standard limit, logarithms for variable powers, and Taylor expansion to the first surviving term. Discrete problems may require Stirling, Wallis, Stolz-Cesaro, Euler-Maclaurin, or a transformed recurrence. Never exchange a sum, limit, or integral without a convergence argument.

Puzzle recognition

Products over roots suggest strategic polynomial evaluation. Restricted walks suggest reflection. Repeated operations suggest an invariant. Finite impartial games suggest nim-sums. Symmetry classes suggest Burnside. Huge modular powers suggest orders and CRT. Conditional uncertainty suggests Bayes rather than plug-in estimates.

PART II

The 96 core generator archetypes

Integral 01 - Beginner

One structural move should finish the problem. The eight numbered entries correspond to the generator's eight-day rotation.

1. Logarithmic u -substitution

11

RECOGNITION CUE

Look for a denominator whose derivative is present in the numerator.

MATHEMATICAL SIGNATURE

$$\int \frac{g'(x)}{g(x)} dx = \log|g(x)| + C, \quad u = g(x)$$

METHOD

Set the entire denominator equal to u , change definite bounds immediately, and check by differentiation.

2. Power inside an exponential

11

RECOGNITION CUE

A factor x accompanies a base raised to x^2 .

MATHEMATICAL SIGNATURE

$$\int x a^{x^2} dx, \quad u = x^2, \quad du = 2x dx$$

METHOD

Use $u = x^2$; remember that the antiderivative of a^u contains $1/\log a$.

3. Trigonometric-square substitution

11

RECOGNITION CUE

$\sin(2x)$ sits beside a function of $\sin^2 x$.

MATHEMATICAL SIGNATURE

$$\int F(\sin^2 x) \sin(2x) dx, \quad u = \sin^2 x$$

METHOD

Use $u = \sin^2 x$ and translate both endpoints before integrating.

4. Finite geometric quotient

11

RECOGNITION CUE

The apparent singularity $(1 - x^n)/(1 - x)$ is removable.

MATHEMATICAL SIGNATURE

$$\frac{1 - x^n}{1 - x} = \sum_{k=0}^{n-1} x^k$$

METHOD

Expand it as $1 + x + \dots + x^{n-1}$, integrate termwise, and recognise a harmonic partial sum.

5. Reciprocal-power substitution**11****RECOGNITION CUE**

The derivative of $1 + ax^2$ appears over its square.

MATHEMATICAL SIGNATURE

$$u = 1 + ax^2, \quad du = 2ax \, dx$$

METHOD

Set $u = 1 + ax^2$ and integrate u to a negative power.

6. Nested-log logarithmic derivative**11****RECOGNITION CUE**

$\log(-\log x)$ appears with $1/(x \log x)$.

MATHEMATICAL SIGNATURE

$$\frac{d}{dx} \log(-\log x) = \frac{1}{x \log x}$$

METHOD

Use $u = \log(-\log x)$; control signs and evaluate the unusual lower bound carefully.

7. Trigonometric quotient collapse**11****RECOGNITION CUE**

$\tan x$ and $\cot x$ appear in the same rational expression.

MATHEMATICAL SIGNATURE

$$\frac{1}{\tan x + \cot x} = \sin x \cos x$$

METHOD

Rewrite in sine and cosine before integrating; the quotient collapses to a basic squared trig function.

8. Power-denominator substitution**11****RECOGNITION CUE**

x^{n-1} appears over $1 + x^n$.

MATHEMATICAL SIGNATURE

$$u = 1 + x^n, \quad du = nx^{n-1} \, dx$$

METHOD

Use $u = 1 + x^n$; the missing constant factor is the only adjustment.

Integral 02 - Easy

The radical chooses the trigonometric or hyperbolic substitution. The eight numbered entries correspond to the generator's eight-day rotation.

1. Sine substitution with x^2 moment

12

RECOGNITION CUE

A radical $\sqrt{a^2 - x^2}$ appears on a bounded interval.

MATHEMATICAL SIGNATURE

$$x = a \sin \theta, \quad \sqrt{a^2 - x^2} = a \cos \theta$$

METHOD

Set $x = a \sin(\theta)$, cancel the cosine, and reduce the remaining power with half-angle identities.

2. Secant substitution with inverse powers

12

RECOGNITION CUE

$\sqrt{x^2 - a^2}$ occurs for x at least a .

MATHEMATICAL SIGNATURE

$$x = a \sec \theta, \quad \sqrt{x^2 - a^2} = a \tan \theta$$

METHOD

Set $x = a \sec(\theta)$, track every secant power, then use a Wallis integral if needed.

3. Quarter-circle recognition

12

RECOGNITION CUE

The integrand is $\sqrt{a^2 - x^2}$ from 0 to a .

MATHEMATICAL SIGNATURE

$$\int_0^a \sqrt{a^2 - x^2} dx = \frac{\pi a^2}{4}$$

METHOD

Use geometry for speed or $x = a \sin(\theta)$ for a derivation; both must give one quarter of πa^2 .

4. Hyperbolic arcosh substitution

12

RECOGNITION CUE

The integrand is $1/\sqrt{x^2 - a^2}$ with noncompact algebra.

MATHEMATICAL SIGNATURE

$$\int \frac{dx}{\sqrt{x^2 - a^2}} = \operatorname{arcosh}\left(\frac{x}{a}\right) + C$$

METHOD

Use $x = a \cosh(t)$, or quote $\operatorname{arcosh}(x/a) = \log((x + \sqrt{x^2 - a^2})/a)$ with domain care.

5. Tangent substitution for quadratic sum

12

RECOGNITION CUE

A power of $x^2 + a^2$ appears.

MATHEMATICAL SIGNATURE

$$x = a \tan \theta, \quad a^2 + x^2 = a^2 \sec^2 \theta$$

METHOD

Set $x = a \tan(\theta)$; secant powers cancel and leave an elementary sine or cosine integral.

6. Wallis moment after sine substitution

12

RECOGNITION CUE

A high even power of x is divided by $\sqrt{a^2 - x^2}$.

MATHEMATICAL SIGNATURE

$$\int_0^{\pi/2} \sin^{2m} \theta \, d\theta = \frac{(2m-1)!!}{(2m)!!} \frac{\pi}{2}$$

METHOD

Use $x = a \sin(\theta)$, factor out a powers, and apply the even-power Wallis recurrence.

7. Tangent substitution with tan-squared rewrite

12

RECOGNITION CUE

$x^2/(a^2 + x^2)^{3/2}$ survives the first cancellation.

MATHEMATICAL SIGNATURE

$$\tan^2 \theta = \sec^2 \theta - 1$$

METHOD

After $x = a \tan \theta$, rewrite $\tan^2 \theta$ as $\sec^2 \theta - 1$ before integrating.

8. Secant radical-to-variable ratio

12

RECOGNITION CUE

$\sqrt{x^2 - a^2}/x$ suggests a clean secant ratio.

MATHEMATICAL SIGNATURE

$$\frac{\sqrt{x^2 - a^2}}{x} \xrightarrow{x=a \sec \theta} \sin \theta$$

METHOD

Set $x = a \sec(\theta)$; the ratio becomes $\sin(\theta)$, while dx supplies the remaining secant-tangent factors.

Integral 03 - Medium

Two coordinated moves are expected. The eight numbered entries correspond to the generator's eight-day rotation.

1. Algebraic numerator alignment

13

RECOGNITION CUE

The numerator nearly matches a repeated quadratic denominator.

MATHEMATICAL SIGNATURE

$$\frac{x^2}{(x^2 + a^2)^2} = \frac{1}{x^2 + a^2} - \frac{a^2}{(x^2 + a^2)^2}$$

METHOD

Add and subtract the denominator inside the numerator, split the integral, then use arctangent and a standard squared-denominator form.

2. Integration by parts with rational tidy

13

RECOGNITION CUE

A polynomial multiplies arctangent.

MATHEMATICAL SIGNATURE

$$\int u \, dv = uv - \int v \, du, \quad u = \arctan x$$

METHOD

Use arctangent as u ; after parts, rewrite x^2 as a denominator-aligned term minus a constant.

3. Weierstrass tangent-half-angle

13

RECOGNITION CUE

A rational expression in sine and cosine resists ordinary identities.

MATHEMATICAL SIGNATURE

$$t = \tan \frac{x}{2}, \quad \sin x = \frac{2t}{1+t^2}, \quad \cos x = \frac{1-t^2}{1+t^2}$$

METHOD

Set $t = \tan(x/2)$, replace $\sin$, $\cos$, and dx systematically, transform bounds, and integrate the rational function.

4. Derivative-of-denominator camouflage

13

RECOGNITION CUE

A complicated polynomial numerator is exactly the derivative of the denominator.

MATHEMATICAL SIGNATURE

$$\int \frac{Q'(x)}{Q(x)} dx = \log|Q(x)| + C$$

METHOD

Verify the derivative explicitly, substitute the denominator, and retain absolute values for indefinite logarithms.

5. Parity plus semicircle geometry**13****RECOGNITION CUE**

An odd term and a constant share an even semicircle weight on symmetric bounds.

MATHEMATICAL SIGNATURE

$$\int_{-a}^a h_{\text{odd}}(x)w_{\text{even}}(x) dx = 0$$

METHOD

Discard the odd contribution, then interpret the remaining integral as a scaled semicircle area.

6. Inverse-trig self-substitution**13****RECOGNITION CUE**

$\arctan(g(x))$ is multiplied by its own derivative.

MATHEMATICAL SIGNATURE

$$u = \arctan g(x), \quad du = \frac{g'(x)}{1 + g(x)^2} dx$$

METHOD

Set $u = \arctan(g(x))$; the problem becomes an integral of u and usually ends as one half u^2 .

7. Quadratic rational discriminant**13****RECOGNITION CUE**

The denominator is a general quadratic.

MATHEMATICAL SIGNATURE

$$\Delta = b^2 - 4ac : \quad \Delta > 0 \Rightarrow \log, \quad \Delta < 0 \Rightarrow \arctan$$

METHOD

Check the discriminant: factor and use logarithmic partial fractions when positive; complete the square and use arctangent when negative.

8. Half-angle radical exposure**13****RECOGNITION CUE**

$\sqrt{\cos x}$ is multiplied by $\sqrt{1 + \cos x}$.

MATHEMATICAL SIGNATURE

$$1 + \cos x = 2 \cos^2 \frac{x}{2}$$

METHOD

Use $1 + \cos x = 2 \cos^2(x/2)$, then choose a half-angle substitution whose derivative is exposed.

Integral 04 - Hard

The first insight matters more than algebraic stamina. The eight numbered entries correspond to the generator's eight-day rotation.

1. King reflection with linear weight

14

RECOGNITION CUE

x multiplies a kernel invariant under $x \rightarrow a - x$.

MATHEMATICAL SIGNATURE

$$\int_0^a xK(x) dx = \frac{a}{2} \int_0^a K(x) dx, \quad K(a-x) = K(x)$$

METHOD

Reflect, add the two equations, replace $x + (a - x)$ by a , and evaluate the simpler unweighted integral.

2. Feynman parameter for the Gaussian transform

14

RECOGNITION CUE

A Fourier-cosine Gaussian integral contains a frequency parameter.

MATHEMATICAL SIGNATURE

$$F(t) = \int_0^\infty e^{-x^2} \cos(tx) dx, \quad F'(t) = -\frac{t}{2} F(t)$$

METHOD

Define $F(t)$, differentiate under the integral sign, integrate by parts in x to obtain an ODE for F , and use $F(0)$.

3. Euler beta plus gamma reflection

14

RECOGNITION CUE

$x^{s-1}/(1+x)$ appears on the positive half-line.

MATHEMATICAL SIGNATURE

$$B(s, 1-s) = \Gamma(s)\Gamma(1-s) = \frac{\pi}{\sin(\pi s)}$$

METHOD

Identify $B(s, 1-s)$, then use $\Gamma(s)\Gamma(1-s) = \pi/\sin(\pi s)$; verify $0 < s < 1$.

4. Double-integral geometric-series separation

14

RECOGNITION CUE

The denominator $1 - (xy)^a$ invites a geometric series.

MATHEMATICAL SIGNATURE

$$\frac{1}{1 - (xy)^a} = \sum_{n=0}^{\infty} (xy)^{an}$$

METHOD

Expand, split $-\log(xy)$ into $-\log x - \log y$, separate the square domain, and sum the resulting reciprocal powers.

5. Reciprocal Mellin log moment

14

RECOGNITION CUE

A palindromic quadratic denominator and log-squared x appear on $(0, \infty)$.

MATHEMATICAL SIGNATURE

$$F(s) = \int_0^\infty \frac{x^{s-1}}{Q(x)} dx, \quad F''(1) \leftrightarrow (\log x)^2$$

METHOD

Introduce $F(s) = \int_0^\infty x^{s-1}/Q(x)$, exploit $x \rightarrow 1/x$ symmetry, and recover the log moment from $F''(1)$.

6. Polylogarithm ladder

14

RECOGNITION CUE

$\text{Li}_n(x)/x$ appears, often paired with $\text{Li}_n(-x)$.

MATHEMATICAL SIGNATURE

$$\frac{d}{dx} \text{Li}_{n+1}(\pm x) = \frac{\text{Li}_n(\pm x)}{x}$$

METHOD

Use $d \text{Li}_n(n+1)(\text{plus-or-minus } x)/dx = \text{Li}_n(\text{plus-or-minus } x)/x$ and evaluate endpoint zeta or eta constants.

7. Complex gamma parameter differentiation

14

RECOGNITION CUE

$x^m e^{-x} \sin x \log x$ combines a gamma moment, oscillation, and a logarithm.

MATHEMATICAL SIGNATURE

$$\int_0^\infty x^{s-1} e^{-(1-i)x} dx = \frac{\Gamma(s)}{(1-i)^s}$$

METHOD

Embed $\sin x$ as an imaginary part, evaluate $\Gamma(s)/(1-i)^s$, differentiate in s , and take the correct component.

8. Nested-radical peeling

14

RECOGNITION CUE

arcsine contains square roots nested inside another root.

MATHEMATICAL SIGNATURE

$$\sqrt{A + \sqrt{B(x)}} \xrightarrow{\text{peel one layer}} u = \sqrt{B(x)}$$

METHOD

Peel the radical layers with successive substitutions, then finish the polynomial-times-arcsine integral by parts.

Derivative 01 - Easy

Organise elementary rules and exploit the evaluation point. The eight numbered entries correspond to the generator's eight-day rotation.

1. Clean-evaluation product/quotient cascade

D1

RECOGNITION CUE

A factor vanishes at the requested point.

MATHEMATICAL SIGNATURE

$$(uv)'(a) = u'(a)v(a) + u(a)v'(a)$$

METHOD

Inspect the point before expanding the derivative; retain only product-rule terms that can survive.

2. Log-derivative for a messy product quotient

D1

RECOGNITION CUE

Several nonzero factors are multiplied and divided.

MATHEMATICAL SIGNATURE

$$\frac{f'}{f} = \sum_j \frac{u'_j}{u_j} - \sum_k \frac{v'_k}{v_k}$$

METHOD

Differentiate $\log f$ as a sum and difference of log derivatives, then multiply by f and evaluate.

3. Hidden logarithmic identity

D1

RECOGNITION CUE

Two logs contain conjugate radical expressions.

MATHEMATICAL SIGNATURE

$$\log A(x) + \log B(x) = \log(A(x)B(x))$$

METHOD

Combine the logs first; their arguments multiply to 1, leaving only any explicit elementary remainder.

4. Inverse-function derivative

D1

RECOGNITION CUE

The requested derivative belongs to g inverse at a known output.

MATHEMATICAL SIGNATURE

$$(f^{-1})'(y) = \frac{1}{f'(x)}, \quad y = f(x)$$

METHOD

Find the preimage and use $(g \text{ inverse})'(y) = 1/g'(x)$; check that $g'(x)$ is nonzero.

5. Leibniz rule for moving bounds**D1****RECOGNITION CUE**

An integral has x -dependent upper and lower limits.

MATHEMATICAL SIGNATURE

$$\frac{d}{dx} \int_{a(x)}^{b(x)} F(t) dt = F(b)b' - F(a)a'$$

METHOD

Apply the fundamental theorem to both boundaries, multiply by their derivatives, and include parameter differentiation if the integrand also contains x .

6. Tangent-half-angle identity before differentiation**D1****RECOGNITION CUE**

$\arctan(x/(1+\sqrt{1+x^2}))$ is displayed.

MATHEMATICAL SIGNATURE

$$\arctan\left(\frac{x}{1+\sqrt{1+x^2}}\right) = \frac{1}{2} \arctan x$$

METHOD

Recognise it as one half $\arctan x$ on the appropriate branch, then differentiate the simpler expression.

7. Logarithmic derivative of rational products**D1****RECOGNITION CUE**

Many linear factors form a large quotient.

MATHEMATICAL SIGNATURE

$$\frac{d}{dx} \log\left(\frac{\prod_j u_j}{\prod_k v_k}\right) = \sum_j \frac{u'_j}{u_j} - \sum_k \frac{v'_k}{v_k}$$

METHOD

Use f'/f as a sum of reciprocals, evaluate at the point, and multiply by f there.

8. Evaluation-point annihilation**D1****RECOGNITION CUE**

$\cos x$ or another simple factor vanishes at a special angle.

MATHEMATICAL SIGNATURE

$$u(a) = 0 \implies (uv)'(a) = u'(a)v(a)$$

METHOD

Write the product rule but delete zero terms only after identifying which differentiated factor survives.

Derivative 02 - Medium

Use structure to avoid repeated mechanical differentiation. The eight numbered entries correspond to the generator's eight-day rotation.

1. High derivative via complex exponentials

D2

RECOGNITION CUE

$e^{ax} \sin(bx)$ is differentiated many times at zero.

MATHEMATICAL SIGNATURE

$$\left. \frac{d^n}{dx^n} (e^{ax} \sin bx) \right|_{x=0} = \Im(a + ib)^n$$

METHOD

Write it as the imaginary part of $e^{(a+ib)x}$; the answer is the imaginary part of $(a + ib)^n$.

2. Implicit curve differentiation

D2

RECOGNITION CUE

x and y are entangled algebraically.

MATHEMATICAL SIGNATURE

$$F(x, y) = 0 \implies y' = -\frac{F_x}{F_y}$$

METHOD

Differentiate every term, collect y' terms, solve symbolically, then substitute the point and check regularity.

3. Related rates with geometric constraint

D2

RECOGNITION CUE

Several geometric dimensions change but similarity relates them.

MATHEMATICAL SIGNATURE

$$G(r(t), h(t)) = 0 \implies \frac{dG}{dt} = 0$$

METHOD

Eliminate dependent dimensions before differentiating with respect to time; substitute numerical values last and keep units.

4. Self-referential sampled derivatives

D2

RECOGNITION CUE

f is defined using f' or f'' evaluated at fixed points.

MATHEMATICAL SIGNATURE

$$A = f'(a), B = f''(b), \dots \implies \text{solve a finite system in } A, B, \dots$$

METHOD

Name the sampled values, differentiate the definition until the dependence closes, solve the finite algebraic system, then evaluate f .

5. Polynomial reconstruction from stationary data

D2

RECOGNITION CUE

Extrema locations and function values are supplied.

MATHEMATICAL SIGNATURE

$$p'(x) = k \prod_j (x - r_j)$$

METHOD

Build $p'(x) = k \text{ product}(x - r_i)$, integrate, use value differences to find k , then determine any remaining constant.

6. Composition functional inversion

D2

RECOGNITION CUE

q and p composed with q are known, but p' is requested at an output.

MATHEMATICAL SIGNATURE

$$\frac{d}{dx} p(q(x)) = p'(q(x))q'(x)$$

METHOD

Find the input mapped to the requested argument, differentiate $p(q(x))$, and divide by $q'(x)$.

7. Derivative-sign change counting

D2

RECOGNITION CUE

$g = e^f$ and factored f' are given.

MATHEMATICAL SIGNATURE

$$g = e^f \implies \text{sgn } g' = \text{sgn } f'$$

METHOD

Because e^f is positive, g' has the sign of f' ; only odd-multiplicity roots change sign, and $+$ to $-$ marks a maximum.

8. Generalised Leibniz / coefficient extraction

D2

RECOGNITION CUE

A polynomial times an exponential is differentiated to high order at zero.

MATHEMATICAL SIGNATURE

$$(uv)^{(n)} = \sum_{k=0}^n \binom{n}{k} u^{(k)} v^{(n-k)}$$

METHOD

Use the n th-product rule, or read $n!$ times the x^n coefficient; only a small number of terms can survive.

Derivative 03 - Hard

Expect inverse series, invariance, or special-function structure. The eight numbered entries correspond to the generator's eight-day rotation.

1. Gamma-Lambert logarithmic differentiation

D3

RECOGNITION CUE

$\Gamma(x)$ is raised to a multiple of $W(x)$.

MATHEMATICAL SIGNATURE

$$\frac{d}{dx} \log(\Gamma(x)^{cW(x)}) = cW' \log \Gamma + cW\psi$$

METHOD

Take logs; at $x = 1$ use $\log \Gamma(1)=0$, $\psi(1)=-\gamma$, and $W(1)=\Omega$ so the W' term vanishes.

2. Second derivative of a parametric curve

D3

RECOGNITION CUE

x and y are both functions of t .

MATHEMATICAL SIGNATURE

$$\frac{d^2y}{dx^2} = \frac{d}{dt} \left(\frac{\dot{y}}{\dot{x}} \right) / \dot{x}$$

METHOD

Compute $dy/dx = (dy/dt)/(dx/dt)$, differentiate that with respect to t , then divide once more by dx/dt .

3. Jacobi determinant derivative

D3

RECOGNITION CUE

A parameterised determinant is differentiated.

MATHEMATICAL SIGNATURE

$$\frac{d}{dt} \det A = \text{tr}(\text{adj}(A)A')$$

METHOD

Use $\frac{d}{dt} \det A = \text{tr}(\text{adj}(A)A')$ or $\det(A) \text{tr}(A^{-1}A')$ when invertible; direct 2×2 expansion is a check.

4. Lagrange inverse series

D3

RECOGNITION CUE

The inverse is defined implicitly by $x = y + ay^3$.

MATHEMATICAL SIGNATURE

$$x = y + ay^3 \implies y = x + c_3x^3 + c_5x^5 + \dots$$

METHOD

Assume an odd series $y = x + cx^3 + dx^5 + \dots$, substitute, match coefficients, then multiply by $5!$ for $y^5(0)$.

5. Schwarzian invariance

D3

RECOGNITION CUE

A Möbius transformation wraps $\tan x$.

MATHEMATICAL SIGNATURE

$$\mathcal{S}(M \circ g) = \mathcal{S}(g) \quad (M \text{ Möbius})$$

METHOD

Use $\mathcal{S}(M \text{ composed with } g) = \mathcal{S}(g)$, reduce to $\mathcal{S}(\tan x) = 2$, and avoid quotient-rule expansion.

6. Functional double-angle identity

D3

RECOGNITION CUE

$$f(2x) = 2f(x)/(1 + f(x)^2).$$

MATHEMATICAL SIGNATURE

$$\tanh(2u) = \frac{2 \tanh u}{1 + \tanh^2 u}$$

METHOD

Recognise $\tanh(2u)$, or insert a Taylor series and compare coefficients through cubic order.

7. Implicit variable exponent

D3

RECOGNITION CUE

$y = x^y$ near a specified branch point.

MATHEMATICAL SIGNATURE

$$y = x^y \implies \log y = y \log x$$

METHOD

Take logs, differentiate $\log y = y \log x$, collect y' terms, and ensure the denominator does not vanish.

8. High local derivative by series

D3

RECOGNITION CUE

A layered elementary function is queried at zero.

MATHEMATICAL SIGNATURE

$$f^{(n)}(0) = n! [x^n] f(x)$$

METHOD

Expand only through the required order; the answer is $n!$ times the coefficient, not the coefficient itself.

Differential 01 - Easy

First-order modelling and one clean substitution. The eight numbered entries correspond to the generator's eight-day rotation.

1. Equilibrium shift in a linear ODE

DE1

RECOGNITION CUE

Q' = input-rate minus decay times Q .

MATHEMATICAL SIGNATURE

$$Q' = r - kQ, \quad Q(t) = \frac{r}{k} + Ce^{-kt}$$

METHOD

Subtract Q = input/decay, solve the homogeneous excess, and restore the equilibrium.

2. Separable growth with time-varying rate

DE1

RECOGNITION CUE

$P' = g(t)P$.

MATHEMATICAL SIGNATURE

$$P' = g(t)P, \quad P(t) = P_0 \exp\left(\int_0^t g(s) ds\right)$$

METHOD

Separate $dP/P = g(t)dt$, integrate, exponentiate, and use the initial value to fix the multiplicative constant.

3. Mixing equation

DE1

RECOGNITION CUE

A well-stirred tank has equal inflow and outflow.

MATHEMATICAL SIGNATURE

$$A' = c_{\text{in}}r_{\text{in}} - \frac{r_{\text{out}}}{V}A$$

METHOD

Write amount' = solute-in minus concentration times outflow, then use an integrating factor or equilibrium plus transient.

4. Exact product derivative

DE1

RECOGNITION CUE

The equation has $tP' + P$

MATHEMATICAL SIGNATURE

$$tP' + P = (tP)'$$

METHOD

Recognise $(tP)'$ immediately, integrate over the stated interval, and impose the data away from $t = 0$.

5. Square-root separable emptying**DE1****RECOGNITION CUE** h' is proportional to $-\sqrt{h}$.**MATHEMATICAL SIGNATURE**

$$h' = -k\sqrt{h} \implies 2\sqrt{h} = -kt + C$$

METHODSeparate $dh/\sqrt{h}$; $\sqrt{h}$ becomes affine in time, and the emptying time is where it reaches zero.**6. RL circuit step response****DE1****RECOGNITION CUE** $I' + aI = b$ with zero initial current.**MATHEMATICAL SIGNATURE**

$$I' + \frac{R}{L}I = \frac{V}{L}$$

METHODUse equilibrium b/a plus an exponentially decaying transient; verify initial and long-time limits.**7. Logistic odds substitution****DE1****RECOGNITION CUE** $A' = kA(K - A)$.**MATHEMATICAL SIGNATURE**

$$A' = kA(K - A), \quad \frac{d}{dt} \log \frac{A}{K - A} = kK$$

METHODSeparate by partial fractions or differentiate $\log(A/(K - A))$; solve for the odds before returning to A .**8. Newton cooling half-life****DE1****RECOGNITION CUE** $T' = -k(T - T_{\text{ambient}})$.**MATHEMATICAL SIGNATURE**

$$T' = -k(T - T_a), \quad T - T_a = Ce^{-kt}$$

METHODWork with excess temperature $U = T - T_{\text{ambient}}$; equal time intervals multiply U by the same factor.

Differential 02 - Medium

Choose the normal form before solving. The eight numbered entries correspond to the generator's eight-day rotation.

1. Bernoulli reciprocal substitution

DE2

RECOGNITION CUE

$$y' + y = a y^2.$$

MATHEMATICAL SIGNATURE

$$y' + y = a y^2, \quad v = y^{-1} \implies v' - v = -a$$

METHOD

Set $v = 1/y$; the nonlinear equation becomes linear in v . Restore y and check denominator zeros.

2. Homogeneous ratio y/x

DE2

RECOGNITION CUE

The right side depends only on y/x and x/y .

MATHEMATICAL SIGNATURE

$$y = vx, \quad y' = v + xv'$$

METHOD

Set $y = vx$, use $y' = v + xv'$, separate the resulting equation, and apply the initial ratio.

3. Harmonic oscillator

DE2

RECOGNITION CUE

$$y'' + \omega^2 y = 0.$$

MATHEMATICAL SIGNATURE

$$y'' + \omega^2 y = 0, \quad y = A \cos \omega x + B \sin \omega x$$

METHOD

Use cosine and sine, fit both initial conditions, and exploit special evaluation angles only at the end.

4. Cauchy-Euler power trial

DE2

RECOGNITION CUE

Coefficients scale as x^2 , x , and 1.

MATHEMATICAL SIGNATURE

$$x^2 y'' + a x y' + b y = 0, \quad y = x^m$$

METHOD

Try $y = x^m$, solve the indicial polynomial, use $x^m \log x$ for a repeated root, then fit conditions.

5. Riccati reduction from a particular solution

DE2

RECOGNITION CUE

A nonlinear first-order equation supplies y_1 .

CONTINUED

MATHEMATICAL SIGNATURE

$$y' = q_0 + q_1 y + q_2 y^2, \quad y = y_1 + \frac{1}{v}$$

METHOD

Set $y = y_1 + 1/v$; cancellation turns the Riccati equation into a linear equation for v .

6. Swap dependent and independent variables

DE2

RECOGNITION CUE

dx/dy is linear although dy/dx is not.

MATHEMATICAL SIGNATURE

$$\frac{dx}{dy} + P(y)x = Q(y)$$

METHOD

Treat x as the unknown function of y , find the integrating factor, then invert only if the problem requires it.

7. Exact differential equation

DE2

RECOGNITION CUE

$M dx + N dy = 0$ may be a total differential.

MATHEMATICAL SIGNATURE

$$M dx + N dy = 0, \quad M_y = N_x$$

METHOD

Check $M_y = N_x$, integrate one component, recover the missing single-variable function, and set the potential equal to C .

8. Repeated-sum substitution

DE2

RECOGNITION CUE

The derivative depends on $x + y$.

MATHEMATICAL SIGNATURE

$$u = x + y, \quad u' = 1 + y'$$

METHOD

Set $u = x + y$, so $u' = 1 + y'$; solve the separable u -equation and back-substitute.

Differential 03 - Hard

Advanced equations are solved by recognising the operator and boundary structure. The eight numbered entries correspond to the generator's eight-day rotation.

1. First-order PDE characteristics

DE3

RECOGNITION CUE

$u_t + a x u_x = -b u$ transports and decays data.

MATHEMATICAL SIGNATURE

$$u_t + axu_x = -bu : \quad \dot{x} = ax, \quad \dot{u} = -bu$$

METHOD

Solve $dx/dt = ax$ and $du/dt = -bu$ along characteristics; trace the target point back to $t = 0$.

2. Heat equation eigenmode

DE3

RECOGNITION CUE

Initial data are a single sine satisfying zero endpoint conditions.

MATHEMATICAL SIGNATURE

$$u_t = \kappa u_{xx}, \quad u(x, 0) = \sin(nx) \implies u = e^{-\kappa n^2 t} \sin(nx)$$

METHOD

Preserve the spatial eigenfunction and attach $e^{-n^2 t}$; direct differentiation verifies the PDE.

3. Wave equation d'Alembert

DE3

RECOGNITION CUE

The domain is the full line and the initial velocity is zero.

MATHEMATICAL SIGNATURE

$$u(x, t) = \frac{1}{2} [f(x - ct) + f(x + ct)]$$

METHOD

Average two translated copies of the initial displacement: one travels left and one right.

4. Forced boundary-value eigenfunction

DE3

RECOGNITION CUE

The forcing is exactly a Dirichlet sine mode.

MATHEMATICAL SIGNATURE

$$Ly = \lambda_n y_n \implies Ly = cy_n \text{ reduces to a scalar equation}$$

METHOD

Try the same mode with an unknown amplitude; the differential operator acts by multiplication.

5. Euler-Lagrange variational equation

DE3

RECOGNITION CUE

A functional depends on u and u' with fixed endpoints.

MATHEMATICAL SIGNATURE

$$\frac{d}{dx} \left(\frac{\partial L}{\partial u'} \right) - \frac{\partial L}{\partial u} = 0$$

METHOD

Compute $d/dx(L_{u'}) = L_u$, solve the resulting boundary-value ODE, and use convexity or second variation to certify a minimum.

6. Sturm-Liouville Robin eigenvalue

DE3

RECOGNITION CUE

One endpoint is Dirichlet and the other mixes y and y' .

MATHEMATICAL SIGNATURE

$$y(0) = 0, \quad \alpha y(\ell) + \beta y'(\ell) = 0 \implies \text{an eigenvalue equation}$$

METHOD

Treat positive, zero, and negative λ separately; for $\lambda = z^2$ derive the transcendental Robin equation.

7. Nonlinear pendulum elliptic period

DE3

RECOGNITION CUE

$\theta'' + \sin \theta = 0$ has finite amplitude.

MATHEMATICAL SIGNATURE

$$\frac{1}{2} \dot{\theta}^2 + 1 - \cos \theta = E, \quad T = 4K(k)$$

METHOD

Use energy for a first integral, isolate a quarter period, and set $\sin(\theta/2) = k \sin(\phi)$ to obtain $K(k)$.

8. Laplace equation on a disk

DE3

RECOGNITION CUE

Boundary data are Fourier modes.

MATHEMATICAL SIGNATURE

$$u(r, \theta) = r^n \cos(n\theta) \quad \text{for boundary mode } \cos(n\theta)$$

METHOD

Extend $\cos(n\theta)$ inward as $r^n \cos(n\theta)$; boundedness excludes negative powers.

Limit - Daily

Identify the asymptotic scale before calculating. The eight numbered entries correspond to the generator's eight-day rotation.

1. Riemann-sum conversion

L1

RECOGNITION CUE

A $1/n$ sum samples $f(k/n)$.

MATHEMATICAL SIGNATURE

$$\frac{1}{n} \sum_{k=1}^n f\left(\frac{k}{n}\right) \rightarrow \int_0^1 f(x) dx$$

METHOD

Convert to an integral on $[0,1]$, prove continuity/integrability, then evaluate with the appropriate integration tool.

2. Stirling logarithmic scale

L1

RECOGNITION CUE

Factorials appear under roots, logs, or powers.

MATHEMATICAL SIGNATURE

$$\log(n!) = n \log n - n + \frac{1}{2} \log(2\pi n) + o(1)$$

METHOD

Take logarithms, insert $\log(n!) = n \log n - n + O(\log n)$, retain only terms surviving the final normalisation, and exponentiate.

3. Wallis/Laplace concentration

L1

RECOGNITION CUE

$\sin^n x$ or $\cos^n x$ concentrates near its maximum.

MATHEMATICAL SIGNATURE

$$\int_0^{\pi/2} \sin^n x dx \sim \sqrt{\frac{\pi}{2n}}$$

METHOD

Use the gamma ratio or a quadratic expansion at the peak; multiply by the stated square-root scale.

4. Telescoping product by cube factorisation

L1

RECOGNITION CUE

$(k^3 - 1)/(k^3 + 1)$ appears.

MATHEMATICAL SIGNATURE

$$\frac{k^3 - 1}{k^3 + 1} = \frac{(k - 1)(k^2 + k + 1)}{(k + 1)(k^2 - k + 1)}$$

METHOD

Factor both cubes, shift indices to expose cancellation, retain boundary factors, then pass to the limit.

5. Taylor matching through second order**L1****RECOGNITION CUE**

Two radicals approach 1 and their constants cancel.

MATHEMATICAL SIGNATURE

$$f(x) = f(0) + f'(0)x + \frac{f''(0)}{2}x^2 + O(x^3)$$

METHOD

Expand inner and outer functions only to the first surviving order; divide by the matching power and keep the coefficient.

6. Recursive fixed-point rate**L1****RECOGNITION CUE**

$x_{n+1} = \sin x_n$ approaches zero too slowly for linearisation alone.

MATHEMATICAL SIGNATURE

$$x_{n+1} = x_n - cx_n^p + o(x_n^p) \implies x_n^{1-p} \asymp n$$

METHOD

Expand one nonlinear term beyond the derivative, study a reciprocal power such as $1/x_n^2$, and apply Cesaro/Stolz.

7. Periodic continued-fraction fixed point**L1****RECOGNITION CUE**

The infinite tail reproduces the whole expression.

MATHEMATICAL SIGNATURE

$$x = a + \frac{1}{x} \implies x = \frac{a + \sqrt{a^2 + 4}}{2}$$

METHOD

Set $x = a + 1/x$, choose the admissible quadratic root, and justify convergence by monotonicity, contraction, or convergents.

8. Endpoint concentration / approximate identity**L1****RECOGNITION CUE**

$n x^n$ weights a continuous function near $x = 1$.

MATHEMATICAL SIGNATURE

$$n \int_0^1 x^n f(x) dx \longrightarrow f(1)$$

METHOD

Use integration by parts or the approximate-identity theorem; the limit is the endpoint value when the hypotheses hold.

Logic - Daily

Each problem is a theorem-recognition exercise disguised as a puzzle. The eight numbered entries correspond to the generator's eight-day rotation.

1. Vieta through strategic evaluations

P1

RECOGNITION CUE

A product over polynomial roots contains $r_i - c$ factors.

MATHEMATICAL SIGNATURE

$$p(c) = a \prod_i (c - r_i)$$

METHOD

Use $p(c)$ =leading times product($c - r_i$); split composite factors and track the parity sign.

2. Ballot/reflection counting

P1

RECOGNITION CUE

A lattice path must never cross a boundary.

MATHEMATICAL SIGNATURE

$$\#\{\text{admissible paths}\} = \#\{\text{all paths}\} - \#\{\text{reflected bad paths}\}$$

METHOD

Count unrestricted paths, reflect each bad path at its first forbidden step, and subtract the bijective bad count.

3. Parity invariant under absolute differences

P1

RECOGNITION CUE

Numbers are repeatedly replaced by $|u - v|$.

MATHEMATICAL SIGNATURE

$$\sum_i a_i \pmod{2} \text{ is preserved under the stated move}$$

METHOD

Reduce modulo 2, where subtraction equals addition; show the parity of the total is invariant until one number remains.

4. Biased gambler's ruin

P1

RECOGNITION CUE

A nearest-neighbour walk has two absorbing endpoints.

MATHEMATICAL SIGNATURE

$$h_i = p h_{i+1} + q h_{i-1}, \quad h_0 = 0, h_N = 1$$

METHOD

Solve the boundary recurrence with characteristic roots 1 and q/p , then impose $h_0 = 0$ and $h_N = 1$.

5. Sprague-Grundy Nim

P1

RECOGNITION CUE

Several heaps permit arbitrary reductions.

MATHEMATICAL SIGNATURE

$$G = h_1 \oplus \cdots \oplus h_m, \quad G = 0 \iff \text{losing position}$$

METHOD

XOR the heap sizes; zero is losing. For a winning move, reduce a heap h to $h \oplus G$ when that value is smaller.

6. Burnside rotational counting

P1

RECOGNITION CUE

Colourings are identified up to rotation.

MATHEMATICAL SIGNATURE

$$\#(X/G) = \frac{1}{|G|} \sum_{g \in G} |\text{Fix}(g)|$$

METHOD

Average the number fixed by each group element; a rotation by k fixes $\text{gcd}(k,n)$ colourings.

7. Multiplicative order and CRT

P1

RECOGNITION CUE

Huge powers are requested modulo 100 or another composite.

MATHEMATICAL SIGNATURE

$$a^n \pmod{p_i^{e_i}} \xrightarrow{\text{CRT}} a^n \pmod{N}$$

METHOD

Work modulo coprime prime powers, reduce exponents by multiplicative order, and recombine with CRT.

8. Bayesian beta posterior

P1

RECOGNITION CUE

An unknown coin bias has a uniform prior and repeated heads are observed.

MATHEMATICAL SIGNATURE

$$p \mid h, t \sim \text{Beta}(h+1, t+1), \quad \Pr(H_{\text{next}}) = \frac{h+1}{h+t+2}$$

METHOD

Update $\text{Beta}(1,1)$ to $\text{Beta}(h+1, t+1)$; the predictive probability is the posterior mean.

PART III

Screenshot-learned supplementary index

How to read the index

The entries in this part are the human-readable names of every learned calculus and logic pattern currently classified into an active generator slot. They supplement the 96 core families; close variants remain listed because their surface disguise is part of Daily Integral's difficulty. Retired mechanics, electromagnetism, and quantum slots are intentionally excluded from this calculus handbook.

Learned variants - Integral 01 - Beginner

14 distinct screenshot-learned variants are documented with their example, method, result when recorded, and distinguishing lesson.

1. Beginner exponential tangent with secant square

11

EXAMPLE

$$\int_0^{\pi/4} \left(\frac{5^{\tan x}}{\cos x} \right)^2 dx.$$

METHOD

Notice $\sec^2 x \, dx = d(\tan x)$, then integrate 25^u from 0 to 1.

RESULT

$$\frac{24}{\log 25}.$$

WHY IT MATTERS

Beginner integrals can still look visually dramatic; the entire problem collapses if the exponential and the hidden $\sec^2 x$ differential are seen as one substitution.

2. Change of base cancels logarithms to constant integrand

11

EXAMPLE

$$\int_0^{\ln^2(2e)} \frac{\log_{2e} x}{\log_2 x + \log_e x} dx.$$

METHOD

Convert every logarithm to natural logs. The factor $\ln x$ cancels, leaving the constant $\ln 2 / \ln^2(2e)$; the singularity at $x = 1$ is removable.

RESULT

$$\ln 2.$$

WHY IT MATTERS

A beginner integral may contain no genuine integration at all: uniform change of base reveals a constant, while the upper bound is engineered to cancel its denominator.

3. Cube sum denominator collapse (beginner integral)

11

EXAMPLE

$$\int_0^1 \frac{1+x}{1-x^{1/3}+x^{2/3}} dx$$

METHOD

$1+x = 1 + (x^{1/3})^3 = (1+x^{1/3})(1-x^{1/3}+x^{2/3})$, so the integrand collapses to $1+x^{1/3}$.

RESULT

$$\int_0^1 (1+x^{1/3}) dx = 1 + \frac{3}{4} = \frac{7}{4}.$$

WHY IT MATTERS

a beginner-level "the denominator is the quadratic factor of a sum of cubes" collapse; same spirit as the $(1-x^4)/(1-x)$ factorisation but with fractional powers.

4. Exponential king symmetry perfect square base

11

EXAMPLE

$$\int_0^1 \frac{45^x}{2025^x + 1} dx \text{ (note } 2025 = 45^2).$$

CONTINUED

METHOD

Since $2025^x = (45^x)^2$, substitute $u = 45^x$, $du = 45^x \ln 45 \, dx$, collapsing the integrand to $\frac{1}{\ln 45} \int_1^{45} \frac{du}{u^2+1}$ - a pure arctan.

RESULT

$$\frac{\arctan 45 - \pi/4}{\ln 45} \approx 0.200$$

WHY IT MATTERS

Year-themed perfect-square base ($2025 = 45^2$) disguises a one-step u -sub to arctan. Distinct from the plain $u = x^n$ arctan I1 - the disguise is an exponential whose base-square sits in the denominator. The handcrafted-today aesthetic (2025) is the tell.

5. Half angle denominator collapse exposing $\sec^2 + 2 \tan \sec^2$

11

EXAMPLE

$$\int_0^{\pi/4} \frac{2(1 + 2 \tan x)}{1 + \cos 2x} dx.$$

METHOD

$1 + \cos 2x = 2 \cos^2 x$, so the integrand becomes $\frac{1 + 2 \tan x}{\cos^2 x} = \sec^2 x + 2 \tan x \sec^2 x = (\tan x + \tan^2 x)'$.

RESULT

$$[\tan x + \tan^2 x]_0^{\pi/4} = 1 + 1 = 2 \text{ (verified).}$$

WHY IT MATTERS

A Beginner where the disguise is a half-angle identity in the denominator; once $1 + \cos 2x \rightarrow 2 \cos^2 x$, the whole integrand is a visible exact derivative. Good fresh I1 template (denominator-identity rather than u -sub).

6. Log base bounds make reciprocal exponential collapse

11

EXAMPLE

$$\int_{\log_8(e)}^{\log_2(e)} \frac{e^{1/x}}{x^2(e^{1/x} + 1)} dx.$$

METHOD

The integrand is exactly $-\frac{d}{dx} \ln(1 + e^{1/x})$ (since $\frac{d}{dx} e^{1/x} = -\frac{1}{x^2} e^{1/x}$). So the antiderivative is $-\ln(1 + e^{1/x})$. The log-base limits are engineered so $1/x$ hits $\ln 2$ and $\ln 8$, hence $e^{1/x} = 2$ and 8 , giving $-\ln(1 + 2) + \ln(1 + 8) = \ln(9/3) = \ln 3$.

RESULT

$$\ln 3$$

WHY IT MATTERS

A Beginner integral whose only trick is recognising the whole fraction as $-\frac{d}{dx} \ln(1 + e^{1/x})$, with the cosmetic twist that the bounds are written as $\log_8 e, \log_2 e$ so that $1/x = \ln 8, \ln 2$ and the exponential evaluates to clean integers $8, 2$. Distinct from existing I1 single- u -sub entries - here the disguise lives in the limits (logarithm-of- e in a non- e base) rather than the integrand.

7. Logarithmic rational substitution with engineered endpoint

11

EXAMPLE

$$\int_0^{\sqrt{e^2-1}} \frac{x \ln(1+x^2)}{(1+x^2)^2} dx.$$

METHOD

Put $u = 1 + x^2$, then integrate $\ln u/u^2$ by parts; the upper endpoint becomes $u = e^2$.

RESULT

$$\frac{1}{2} - \frac{3}{2e^2}.$$

WHY IT MATTERS

The endpoint is designed to simplify the logarithm after the structural substitution; always transform bounds immediately.

8. Nested trig quotient collapses to sin cos

11

EXAMPLE

$$\int_{\ln(\pi/4)}^{\ln(\pi/2)} \frac{\tan(e^x)}{\sec^2(e^x)} e^x dx$$

METHOD

$u = e^x$ turns it into $\int \tan u \cos^2 u du = \int \sin u \cos u du = \frac{1}{2} \int \sin 2u du$; bounds $u : \pi/4 \rightarrow \pi/2$.

RESULT

1/8

WHY IT MATTERS

I1 can hide a one-step $u = e^x$ under a $\tan / \sec^2$ costume that simplifies to $\sin u \cos u$.

9. Parity trap radical on a symmetric interval

11

EXAMPLE

$$\int_{-1}^1 \sqrt{x^{13} + x^8} dx.$$

METHOD

Factor x^8 out of the radical to get $x^4 \sqrt{x^5 + 1}$ (even prefactor; integrand is real on $[-1, 1]$ since $x^8(x^5 + 1) \geq 0$ there), then $u = x^5 + 1$.

RESULT

$$\frac{4\sqrt{2}}{15} \approx 0.3771.$$

WHY IT MATTERS

The high odd power inside the radical is a decoy - you must check the domain/parity before any u -sub. A sharper Beginner than a plain symmetric integrand.

10. Product expansion collapsing to elementary derivatives

11

EXAMPLE

$$\int_0^{\pi/4} (\cos x + \sec x)(\tan x + \sec x) dx$$

METHOD

Expand the product: $\cos x \tan x + \cos x \sec x + \sec x \tan x + \sec^2 x = \sin x + 1 + \sec x \tan x + \sec^2 x$. Each term integrates to a named elementary primitive. Result: $-\cos x + x + \sec x + \tan x$ evaluated at 0, $\pi/4$, giving $\pi/4 + \frac{\sqrt{2}}{2} + 1$.

RESULT

$$\pi/4 + \frac{\sqrt{2}}{2} + 1$$

WHY IT MATTERS

A beginner-friendly disguise: the product looks like it might call for substitution, but distributive expansion reveals four standard primitives in disguise. Distinct from existing I1 single-substitution and algebraic-collapse flavors - here the trick is "multiply it out and recognise each piece".

11. Rationalize constant surd exponent

11

EXAMPLE

$$\int_0^1 (1-x)^{\frac{1}{\sqrt{2}+1}} dx$$

METHOD

The exponent is a constant in disguise: rationalize $\frac{1}{\sqrt{2}+1} = \sqrt{2} - 1$, then plain power rule $\int_0^1 (1-x)^a dx = \frac{1}{a+1}$ with $a+1 = \sqrt{2}$.

RESULT

$$\frac{1}{\sqrt{2}} = \frac{\sqrt{2}}{2} \approx 0.7071$$

WHY IT MATTERS

A beginner integral can hide its one-step nature inside an irrational constant exponent - the "trick" is arithmetic (rationalize the surd), not calculus.

12. Same base power product pythagorean exponent collapse

11

EXAMPLE

$$\int_0^{\sqrt{2\pi}} x^{\sin^2(x)} x^{\cos^2(x)} dx$$

METHOD

Same base x raised to two different trig-function exponents multiplies via $x^a \cdot x^b = x^{a+b}$; since $\sin^2 x + \cos^2 x = 1$ identically, the integrand collapses to the trivial $x^1 = x$.

RESULT

π .

WHY IT MATTERS

Distinct from the logged "algebraic identity collapse" and "infinite-product-geometric-exponent" I1 families - here the disguise is purely the exponent law $x^a x^b = x^{a+b}$ combined with the Pythagorean identity, not a ratio simplification or a series sum. An intimidating-looking two-exponent integrand is secretly $\int x dx$.

13. Stacked nested integral notation collapse

11

EXAMPLE

A "Beginner Integral" rendered as a tall diagonal stack of nested integral signs (each wrapping an inner integral of dx with bounds alternating among $[0, 1]$, $[-1, 1]$, $[1, 0]$, etc.), visually resembling a fraction-tower rather than a normal single integral.

METHOD

Every layer's integrand is just the constant value returned by the integral inside it, so each nested integral reduces to (constant) $\times$ (signed interval length); work bottom-up from the innermost $\int dx$, multiplying by each successive interval's signed length (a reversed bound like $\int_1^0$ contributes a sign flip) until the whole stack collapses to one number - no real calculus is needed, only careful order-of-operations reading of the notation.

RESULT

A small rational/integer constant (exact value depends on precisely re-reading which of the ~ 8 stacked bounds is $[0, 1]$ vs. $[-1, 1]$ vs. $[1, 0]$ from the compressed screenshot render - the notational-reading technique is the salient, confidently-extracted point, not a specific reverified numeral).

WHY IT MATTERS

Distinct from every existing I1 "one-step disguise" and from the nested-radical / infinite-continued-fraction / power-tower fixed-point families - the gimmick here is purely typographic (stacking $\int \dots dx$ symbols like a fraction tower) rather than algebraic self-similarity; it teaches parsing nested integral notation carefully rather than any integration trick.

14. Usb arcsine with irrational amplitude

11

EXAMPLE

$$\int_1^e \frac{1}{x\sqrt{e - \ln^2(x)}} dx$$

METHOD

One-step u -sub $u = \ln x$ (with $du = dx/x$ already sitting in the integrand) lands on the standard arcsine form $\int du/\sqrt{a^2 - u^2}$ with the non-obvious amplitude $a = \sqrt{e}$; limits become $u \in [0, 1]$, giving $\arcsin(u/\sqrt{e})$.

RESULT

$\arcsin(1/\sqrt{e}) \approx 0.6403$

WHY IT MATTERS

The catalogued I1 arcsine-form warm-ups all have integer or rational a ; here the constant under the radical is e itself, so the payoff is $\arcsin(1/\sqrt{e})$ - an irrational amplitude that looks wrong at first glance and discourages pattern-matching on " $\sqrt{a^2 - u^2}$ means $a \in \mathbb{Z}$ ". Clean Beginner-tier anchor for "the disguise is the constant, not the structure".

Learned variants - Integral 02 - Easy

27 distinct screenshot-learned variants are documented with their example, method, result when recorded, and distinguishing lesson.

1. Arcsin half angle radical integral

12

EXAMPLE

$$\int_0^1 \sin^{-1}(x) \sqrt{\frac{1+x}{1-x}} dx.$$

METHOD

Use $x = \sin \theta$ and the identity $\sqrt{(1 + \sin \theta)/(1 - \sin \theta)} = \sec \theta + \tan \theta$, then integrate by parts or reduce to elementary endpoint terms.

RESULT

Tractable closed form after the half-angle collapse; use as a style template rather than repeat verbatim.

WHY IT MATTERS

Easy integrals often reward recognising a disguised half-angle expression after an inverse-trig substitution.

2. Arctan difference identity self substitution

12

EXAMPLE

$$\int_0^1 \frac{32}{1+x^2} \tan^{-1}\left(\frac{1-x}{1+x}\right) dx$$

METHOD

Identity $\arctan \frac{1-x}{1+x} = \frac{\pi}{4} - \arctan x$ on $[0, 1]$; then $u = \arctan x$, $du = dx/(1+x^2)$ gives $\int_0^{\pi/4} (\frac{\pi}{4} - u) du = \pi^2/32$.

RESULT

$$32 \cdot \pi^2/32 = \pi^2$$

WHY IT MATTERS

The subtraction-form companion to the existing "double-angle arctan self-substitution" ($\arctan \frac{2x}{1-x^2} = 2 \arctan x$). Here $\arctan \frac{1-x}{1+x} = \frac{\pi}{4} - \arctan x$ is the disguise, and the 32 prefactor is engineered so the answer is a clean π^2 .

3. Bessel J0 disguised zero

12

EXAMPLE

$$\int_0^\pi (\cos(\sin x) - \cos(\sin \frac{x}{2})) dx$$

METHOD

$\int_0^\pi \cos(\sin x) dx = \pi J_0(1)$. For the second term, $u = x/2$ gives $2 \int_0^{\pi/2} \cos(\sin u) du = \pi J_0(1)$ as well (since $\int_0^{\pi/2} \cos(\sin u) du = \frac{\pi}{2} J_0(1)$ by symmetry about $\pi/2$). The two Bessel forms cancel exactly.

RESULT

0 (verified exactly).

WHY IT MATTERS

A "looks like it needs J_0 , but the two J_0 pieces are engineered to cancel" disguised-zero. Sits alongside the $x \rightarrow 1/x$ hidden-parity I4 zero, but at Easy difficulty and via Bessel rather than parity.

4. Composite rational trig with symmetric numerator denominator

12

EXAMPLE

$$\int_{\pi/4}^{\pi/3} \frac{\cos^3 x + \cos^5 x}{\sin^2 x + \sin^4 x} dx$$

METHOD

Factor: $\cos^3 x(1 + \cos^2 x)/[\sin^2 x(1 + \sin^2 x)]$. Use $1 + \cos^2 x = 2 - \sin^2 x$. Substitute $u = \sin x$, $du = \cos x dx$ - one factor of $\cos x$ goes into du , leaving $\cos^2 x = 1 - u^2$ for the numerator. Integrand becomes $(1 - u^2)(2 - u^2)/[u^2(1 + u^2)]$ in u . Partial fractions in u^2 .

RESULT

Closed form in terms of u -bounds $\sin(\pi/4) = 1/\sqrt{2}$ and $\sin(\pi/3) = \sqrt{3}/2$.

WHY IT MATTERS

Distinct from existing I2 trig-sub menu - this rewards the "factor the numerator and denominator to match each other's structure, then do one substitution" instinct. The symmetric polynomial structure in $\sin^2 x$ and $\cos^2 x$ is the tell.

5. Cot power reduction of a csc heavy trig integrand

12

EXAMPLE

$$\int_{\pi/4}^{\pi/3} \cot x \csc^2 x (\cot^2 x + 2) dx$$

METHOD

substitute $u = \cot x$, $du = -\csc^2 x dx$; the integrand becomes the polynomial $-(u^3 + 2u)$, antiderivative $-(u^4/4 + u^2)$. Evaluate $u : 1 \rightarrow 1/\sqrt{3}$.

RESULT

8/9 (verified numerically).

WHY IT MATTERS

Easy-integral flavour where the whole job is recognising that a pile of $\csc^2 / \cot$ factors is a polynomial in $u = \cot x$. Distinct from the default $x = a \tan \theta$ trig-sub; reward is "rewrite in one variable, integrate a polynomial".

6. Cubic in $\exp x$ factors to three linear partial fractions

12

EXAMPLE

$$\int_0^\infty \frac{6}{e^{3x} + 6e^{2x} + 11e^x + 6} dx.$$

METHOD

With $u = e^x$ ($du = u dx$) the denominator is $u^3 + 6u^2 + 11u + 6 = (u + 1)(u + 2)(u + 3)$, so the integral becomes $\int_1^\infty \frac{6 du}{u(u+1)(u+2)(u+3)}$. Heaviside cover-up gives residues $\frac{6}{1 \cdot 2 \cdot 3} = 1$ (at $u=0$), -3 , $+3$, -1 - the signed binomial pattern $\binom{3}{0}, \binom{3}{1}, \binom{3}{2}, \binom{3}{3}$. Integrate the four logs and evaluate the telescoping boundary.

RESULT

$$5 \ln 2 - 3 \ln 3 = \ln \frac{32}{27}.$$

WHY IT MATTERS

A clean "exponential-polynomial factors into consecutive linears" I2, distinct from the trig-sub default. The reusable signal is coefficients 1, 6, 11, 6 (or any $\prod(u+k)$) and the tidy binomial-coefficient residues $(-1)^k \binom{n}{k}$ over $u, u+1, \dots$; the $u = e^x$ front-substitution converts an intimidating exponential sum into a routine rational partial-fraction problem.

7. Derivative of denominator disguised by sec tan

12

EXAMPLE

$$\int_0^\pi \frac{e^x \sec x + 1}{e^x \sec x + \tan x} dx$$

METHOD

Multiply numerator and denominator by $\cos x$: integrand becomes $\frac{e^x + \cos x}{e^x + \sin x}$. The numerator is exactly $\frac{d}{dx}(e^x + \sin x)$, so the integral is $[\ln(e^x + \sin x)]_0^\pi = \ln(e^\pi) - \ln(1) = \pi$.

RESULT

CONTINUED

 π

WHY IT MATTERS

"Easy Integral" register where the sec/tan dressing hides a clean f'/f form after one $\cos x$ multiply. The tell is that clearing the secant turns the numerator into the derivative of the denominator. Distinct from existing f'/f flavors because the disguise is trigonometric rather than algebraic.

8. Double angle arctan self substitution

12

EXAMPLE

$$\int_0^1 \frac{\tan^{-1}\left(\frac{2x}{1-x^2}\right)}{1+x^2} dx$$

METHOD

Recognize $\arctan \frac{2x}{1-x^2} = 2 \arctan x$ (on $[0, 1)$), turning the integrand into $2 \arctan x \cdot \frac{d}{dx}(\arctan x)$, i.e. $\int 2u du$ with $u = \arctan x$.

RESULT

$$\frac{\pi^2}{16}$$

WHY IT MATTERS

The disguise is the trig identity, not the integration. Once the double-angle collapse is seen it is a one-line $u du$. Distinct from a plain $1/(1+x^2)$ arctan integral because the numerator hides a multiple of the substitution variable itself.

9. Double angle substitution quartic trig denominator to arctan

12

EXAMPLE

$$\int_0^{\pi/2} \frac{\sin(2x)}{\sin^4(x) + \cos^4(x)} dx$$

METHOD

Use $\sin^4 x + \cos^4 x = 1 - \frac{1}{2} \sin^2(2x) = \frac{1}{2}(1 + \cos^2(2x))$, then substitute $t = \cos(2x)$ so that $\sin(2x) dx = -\frac{1}{2} dt$. The integral becomes $\int_{-1}^1 \frac{dt}{1+t^2} = [\arctan t]_{-1}^1$ - the two $\frac{1}{2}$'s cancel exactly and the limits land symmetrically on ± 1 .

RESULT

$$\frac{\pi}{2} \text{ (verified numerically)}$$

WHY IT MATTERS

The $\sin^4 + \cos^4 = 1 - \frac{1}{2} \sin^2 2x$ identity is already catalogued, but only as a numerator collapse ($\int_0^\pi (\sin^4 + \cos^4) dx = 3\pi/4$). Here it sits in the denominator and the payoff is a full arctan over $[-1, 1]$, i.e. $\arctan(1) - \arctan(-1) = \pi/2$ - a "π from nowhere" Easy-tier result. Distinct from the logged exponential-denominator-to-arctan entry because the vehicle is a double-angle trig identity rather than $u = e^x$.

10. Exponential sum denominator to arctan

12

EXAMPLE

$$\int_0^\infty \frac{dx}{16 \cdot 4^{-x} + 4^x}$$

METHOD

Multiply num/denom by 4^x and substitute $u = 4^x$; the exponential-plus-reciprocal-exponential denominator collapses to a plain $16 + u^2$, giving an arctan antiderivative directly (with a $1/\ln 4$ Jacobian prefactor).

RESULT

$$\frac{1}{8 \ln 2} \left(\frac{\pi}{2} - \arctan \frac{1}{4} \right) \approx 0.2391.$$

WHY IT MATTERS

The documented I2 menu is explicitly "trig substitution on $\sqrt{x^2 \pm a^2}$ is the dominant flavor" - this problem has neither a square root nor a trig function; the "trig-sub feel" is imitated purely by an exponential substitution landing on $1/(a^2 + u^2)$. A distinct disguise mechanism for the same arctan payoff.

11. Factor powers from twin radicals to common u sub**12****EXAMPLE**

$$\int_0^1 \left(\sqrt{2x^7 + 3x^6 + x^4} + \sqrt{2x^5 + 3x^4 + x^2} \right) dx.$$

METHOD

Pull perfect even powers out of each radical so both collapse to a common radicand: sum becomes $(x^2+x)\sqrt{2x^3+3x^2+1}$; the merged prefactor is exactly $u'/6$ for $u = 2x^3 + 3x^2 + 1$.

RESULT

$$\frac{1}{9} u^{3/2} \Big|_1^6 = (6\sqrt{6} - 1)/9.$$

WHY IT MATTERS

Two ugly radicals that differ by a factor of x^{2k} are one radical in disguise; merge first, then look for the derivative alongside.

12. Finite continuant fraction integrand linear split**12****EXAMPLE**

$$\int_0^3 \frac{1}{1 + \frac{1}{1 + \frac{1}{1 + \frac{1}{1+x}}}} dx$$

METHOD

Unlike the catalog's existing infinite self-referential continued fractions (solved via a fixed-point equation for golden/silver ratios), this compound fraction is finite (fixed depth, terminating in $1+x$). Collapse it bottom-up as ordinary algebra: $D_1 = 1+x$, $D_2 = 1 + 1/D_1 = \frac{2+x}{1+x}$, $D_3 = 1 + 1/D_2 = \frac{3+2x}{2+x}$, so the integrand is $1/D_3 = \frac{2+x}{2x+3}$ - a single Möbius transform of x . Then split $\frac{2+x}{2x+3} = \frac{1}{2} + \frac{1/2}{2x+3}$ and integrate to a log.

RESULT

$$\frac{3}{2} + \frac{1}{4} \ln 3 \text{ (verified by hand).}$$

WHY IT MATTERS

Distinct from the existing "nested continued fractions resolved to golden/silver ratios before integrating" I4 entry - that family is infinite/self-similar and needs a fixed-point equation, whereas this one is a finite-depth continuant that closes by plain bottom-up substitution into a rational Möbius function of x , then a routine partial-fraction-style split. A good Easy-tier counterpart to the existing Hard CF flavor.

13. Half angle denominator $1 + \cos^2(x/2)$ with weighted $x \sin x$ **12****EXAMPLE**

$$\int_0^\pi \frac{x \sin x}{1 + \cos^2(x/2)} dx.$$

METHOD

Half-angle in the denominator; $\sin x = 2 \sin(x/2) \cos(x/2)$, sub $u = \cos(x/2)$ on the symmetric part. (Note: a naive King reflection $x \rightarrow \pi - x$ does NOT close here - it sends $\cos^2(x/2) \rightarrow \sin^2(x/2)$, so the denominator is not reflection-invariant. Handle the x -weight carefully.)

RESULT

≈ 2.36532 (numeric; the tidy " $\pi \ln 2$ " guess is WRONG - re-derive before trusting a closed form).

WHY IT MATTERS

Cautionary template: weighted trig integrals with a half-angle denominator look King-friendly but aren't always. Logged mainly so the menu carries the surface form; verify any closed form numerically.

14. Half angle radicals plus parity

I2

EXAMPLE

$$\int_{-1}^1 \frac{x \sin \sqrt{(1+x)/2} \cos \sqrt{(1-x)/2}}{\sqrt{1-x^2}} dx.$$

METHOD

Pair the integrand at x and $-x$ before attempting substitutions; the swapped half-angle radicals expose the part that survives symmetric integration.

WHY IT MATTERS

A symmetric interval plus factors exchanged by $x \mapsto -x$ is a deliberate invitation to compute $f(x) + f(-x)$.

15. Integrand reflection on shifted interval

I2

EXAMPLE

$$\int_{-1}^0 \frac{x}{e^x + x + 1} dx$$

METHOD

King's-property-style reflection on a non-symmetric interval. The substitution $x \mapsto -1 - x$ maps $[-1, 0]$ to itself; adding the original and reflected integrands lets the awkward $e^x + x + 1$ denominators combine algebraically (sometimes telescoping; sometimes leaving an elementary residue).

RESULT

Not derivable directly from the image transcription — the combined integrand depends on the precise denominator (the screenshot may also have $e^x + e^{-x} + 1$ rather than $e^x + x + 1$). The flavor lesson is the substitution, not the closed form.

WHY IT MATTERS

The I2 / I3 menu has King's property on $[0, \pi/2]$ and $[0, 1]$, but not on the shifted interval $[-1, 0]$ with an exponential-plus-polynomial denominator. Worth adding as an "Easy" reflection: the move is identical to the King trick but on a translated symmetry axis $x = -1/2$.

16. Iterated nested radical collapses to constant

I2

EXAMPLE

$$\int_0^{\pi/3} \frac{\left(\frac{e^{2x}}{2} - \frac{e^{2x} \cos(2x)}{2} \right)^{34}}{\sqrt[67]{\left(\frac{e^{2x}}{2} - \frac{e^{2x} \cos(2x)}{2} \right)^{34} \dots}} dx$$

METHOD

Factor out $e^{2x}/2$: the core expression is $e^{2x}(1 - \cos(2x))/2 = e^{2x} \sin^2(x)$. The infinitely-iterated 67th root of a positive number A raised to the 34th power converges to a fixed point: the self-similar structure $X = A^{34/67}/X^?$ collapses via the infinite-depth radical to $X = A^{34/(67-34)} = A$ or, for an infinite tower of the same expression, $X = A^{34/67 \cdot \frac{1}{1-34/67}} = A^1$. In either case the iterated radical equals the base, and the ratio is 1. Integral $= \int_0^{\pi/3} 1 dx = \pi/3$.

RESULT

$\pi/3$

WHY IT MATTERS

An integral disguised by baroque nested n th-root notation; the key is recognising that an infinitely self-similar radical tower $A^{34/67}/\sqrt[67]{A^{34}/\dots}$ converges to the base A divided by itself $= 1$, so the integrand is identically 1. First "infinite nested radical $= 1$ " I2 flavor in the catalog. Related to but distinct from the power-tower fixed-point D3 entry - here it appears inside an integral.

17. King reflection with nested cosine

12

EXAMPLE

$$\int_0^\pi \frac{x \sin x}{1 + \cos(\cos x)} dx.$$

METHOD

King's property $x \rightarrow \pi - x$ kills the x factor: $2I = \pi \int_0^\pi \frac{\sin x}{1 + \cos(\cos x)} dx$, then $u = \cos x$ gives $\frac{\pi}{2} \int_{-1}^1 \frac{du}{1 + \cos u} = \frac{\pi}{2} \int_{-1}^1 \frac{1}{2} \sec^2(u/2) du = \pi \tan(1/2)$.

RESULT

$$\pi \tan\left(\frac{1}{2}\right)$$

WHY IT MATTERS

Classic King reflection but the inner $\cos(\cos x)$ camouflage makes the u -sub the real payoff; the denominator becomes an even function so the symmetric-interval integral is clean.

18. Log derivative disguised numerator

12

EXAMPLE

$$\int_{-1}^0 \frac{1 - e^x}{xe^x + 1} dx$$

METHOD

Let $g(x) = xe^x + 1$; note $g'(x) = e^x + xe^x = e^x(1 + x)$, and the numerator $1 - e^x$ isn't literally g' , but the integrand can be rewritten as $1 - g'(x)/g(x)$ after recognizing $g'(x) = xe^x + e^x$ relates to the given numerator via algebraic manipulation. The antiderivative collapses to $x - \ln g(x)$.

RESULT

$$\ln(e - 1) \approx 0.5413$$

WHY IT MATTERS

A "disguised logarithmic derivative" flavor distinct from the catalogued exact-derivative I1/I2 families - the integrand doesn't look like f'/f at first glance because of the constant term folded into the numerator; recognizing the split into a polynomial part plus $-g'/g$ is the trick.

19. Log of $1 + x^6$ derivative inside (reciprocal power tidy)

12

EXAMPLE

$$\int \frac{1}{(1/x^5 + x) \ln(1 + x^6)} dx \text{ with bounds making } x^6 = e - 1, e^2 - 1$$

METHOD

$$\frac{1}{1/x^5 + x} = \frac{x^5}{1 + x^6}; \text{ then } u = \ln(1 + x^6), du = \frac{6x^5}{1 + x^6} dx \Rightarrow \frac{1}{6} \int du/u.$$

RESULT

$$\frac{1}{6} \ln 2$$

WHY IT MATTERS

an "easy" integral whose denominator hides both $d(1 + x^6)$ and $\ln(1 + x^6)$; answer is $\frac{1}{6} \ln(\text{ratio of logs})$.

20. Logderiv disguised $2x$ minus $\tan x$

12

EXAMPLE

$$\int_{3\pi/4}^\pi \frac{\cos 2x}{\cos^2 x (2x - \tan x)} dx$$

METHOD

Whole integrand is u'/u with $u = 2x - \tan x$: since $\cos 2x = \cos^2 x(1 - \tan^2 x)$ and $u' = 1 - \tan^2 x$, the $\cos^2 x$ cancels and it is $\int u'/u dx = \ln |u|$.

RESULT

$$\ln \frac{2\pi}{3\pi/2 + 1} \approx 0.0952$$

WHY IT MATTERS

A logarithmic-derivative camouflage where the f'/f structure is split across the numerator AND the $\cos^2 x$ denominator factor - richer than the existing "derivative-of-denominator camouflage" because the $\cos 2x$ rewrite is the hidden step.

21. Nested co function inverse cancellation**12****EXAMPLE**

$$\int_0^{\pi/4} \tan(\cot^{-1}(\tan^{-1}(\cot x))) dx$$

METHOD

peel nested inverse co-functions via $\cot^{-1} z = \pi/2 - \tan^{-1} z$ and $\tan(\cot^{-1} w) = 1/w$; the tower collapses to $\tan x$.

RESULT

$$\frac{1}{2} \ln 2$$

WHY IT MATTERS

a deliberately intimidating stack of inverse trig that is pure cosmetics; reward is spotting the co-function identities, not integrating anything hard.

22. Nested cube root radical to beta function**12****EXAMPLE**

$$\int_0^1 \sqrt{1 - \sqrt[3]{1-x}} dx$$

METHOD

Substitute $t = 1 - x$, then $t = u^3$, turning the nested radical into a plain algebraic integrand: $\int_0^1 \sqrt{1 - t^{1/3}} dt = 3 \int_0^1 u^2 \sqrt{1-u} du = 3B(3, 3/2)$.

RESULT

$$3B(3, 3/2) = \frac{16}{35} \approx 0.4571 \text{ (verified numerically).}$$

WHY IT MATTERS

A different route to a Beta-function payoff than the catalogued "recognize $\sin^{2p-1} \cos^{2q-1}$ " I3 trig-power family - here the Beta form emerges from a chain of two algebraic substitutions (linear shift, then cube) applied to a nested radical, with no trig in sight. Good Easy-tier "Beta function in disguise" variant.

23. Nested radical absolute value split at half**12****EXAMPLE**

$$\int_0^1 \sqrt{x + \sqrt{|1-2x|}} dx$$

METHOD

Split at $x = 1/2$ (where $|1-2x|$ changes sign). On $[0, 1/2]$: $|1-2x| = 1-2x$, so the inner expression is $x + \sqrt{1-2x}$; on $[1/2, 1]$: $|1-2x| = 2x-1$, inner is $x + \sqrt{2x-1}$. Each half admits a rationalising substitution ($u = \sqrt{1-2x}$ or $u = \sqrt{2x-1}$) to eliminate the nested radical.

RESULT

Numeric (exact closed form requires evaluating two elementary integrals; ≈ 0.868).

WHY IT MATTERS

"Absolute value inside a nested radical forces a domain split" - distinct from the existing piecewise-min-of-trig-functions I4 entry and the cube-root-reciprocal-cubed I4 entry. The second-level radical $\sqrt{|1-2x|}$ is the disguise; once the absolute value is resolved on each half, the inner expression reduces to an algebraic sub.

24. Quartic $x^4 + x^2 + 1$ factor collapse integrand**12****EXAMPLE**

$$\int_0^1 \frac{\sqrt{x(x^2+x+1)}}{\sqrt{x+1}\sqrt{x^4+x^2+1}} dx.$$

METHOD

Factor $x^4 + x^2 + 1 = (x^2 + x + 1)(x^2 - x + 1)$ (Sophie-Germain-style). The $(x^2 + x + 1)$ cancels, leaving $\sqrt{x/((x+1)(x^2-x+1))} = \sqrt{x/(x^3+1)}$, a clean reducible integrand.

RESULT

$$\int_0^1 \sqrt{x/(x^3+1)} dx \approx 0.84$$

CONTINUED

WHY IT MATTERS

The intimidating nested radical collapses entirely because of one algebraic factorisation of a biquadratic. Companion to the existing "sophie-germain-telescoping-sum" entry but applied to an integrand rather than a sum. Rewards recognising $x^4 + x^2 + 1$ as a product of two quadratics.

25. Shifted argument integral king disguise

12

EXAMPLE

$$\int_0^1 \frac{\sin x}{\sin(x-67)} dx$$

METHOD

Write $\sin x = \sin((x-67)+67) = \sin(x-67)\cos 67 + \cos(x-67)\sin 67$. The integrand splits into $\cos 67 + \sin 67 \cot(x-67)$. Integrate: $\cos 67 + \sin 67 \cdot \ln |\sin(x-67)| \Big|_0^1$.

RESULT

$$\cos 67 + \sin 67 [\ln |\sin(1-67)| - \ln |\sin(-67)|] = \cos 67 + \sin 67 \ln |\sin(-66)/\sin(-67)|$$

WHY IT MATTERS

A trig integral whose trick is decomposing $\sin x$ via the angle-addition formula $\sin x = \sin((x-67)+67)$ to separate the shifted denominator. Distinct from King's-property on $[0, \pi]$ - here the constant 67 (in radians) is a red herring; the move is a pure algebraic split. The cosine term integrates cleanly and the remaining cot integral gives a log.

26. Square root sec plus tan half angle Weierstrass

12

EXAMPLE

$$\int_{-\pi/2}^{\pi/2} \sqrt{\sec x + \tan x} dx$$

METHOD

$\sec x + \tan x = \frac{1+\sin x}{\cos x} = \frac{(1+\sin x)^2}{\cos^2 x/(1+\sin x)}$... cleaner: use half-angle $t = \tan(x/2)$: $\sec x + \tan x = \frac{1+t}{1-t} \cdot \frac{2}{1+t^2}$ (simplify) or note $\sec x + \tan x = \tan(\pi/4 + x/2)$, so $\sqrt{\sec x + \tan x} = \sqrt{\tan(\pi/4 + x/2)}$, and $\sec x + \tan x = (1 + \sin x)/\cos x = (\cos(x/2) + \sin(x/2))^2 / \cos^2(x/2)$, giving $\sqrt{\sec x + \tan x} = (\cos(x/2) + \sin(x/2))/|\cos(x/2)|$. Integrate over the symmetric interval.

RESULT

The integral evaluates via the half-angle split; by the even/odd structure on $[-\pi/2, \pi/2]$ the $\sin(x/2)$ term contributes 0 and the $\cos(x/2)/\cos(x/2) = 1$ term gives π .

WHY IT MATTERS

The $\sqrt{\sec x + \tan x}$ integrand collapses via the half-angle factorization to $(\cos(x/2) + \sin(x/2))/\cos(x/2)$; the odd part integrates to zero over the symmetric interval and only 1 remains. A new I2 half-angle-and-symmetry flavor distinct from the existing half-angle Vieta and secant-sub entries.

27. Sum product sine cosine via sum to product

12

EXAMPLE

$$\int_{\pi}^{3\pi/2} \frac{\sin(6x) + \sin(7x)}{\cos(6x) - \cos(7x)} dx$$

METHOD

Sum-to-product: $\sin A + \sin B = 2 \sin \frac{A+B}{2} \cos \frac{A-B}{2}$ and $\cos A - \cos B = -2 \sin \frac{A+B}{2} \sin \frac{A-B}{2}$. With $A = 6x$, $B = 7x$: numerator = $2 \sin(13x/2) \cos(x/2)$, denominator = $-2 \sin(13x/2) \sin(x/2)$. Ratio = $-\cot(x/2)$. Integrate: $-2 \ln |\sin(x/2)| \Big|_{\pi}^{3\pi/2}$.

RESULT

$$-2 \ln |\sin(3\pi/4)| + 2 \ln |\sin(\pi/2)| = -2 \ln(1/\sqrt{2}) + 0 = \ln 2$$

WHY IT MATTERS

Sum-to-product collapses a ratio of two multi-frequency trig sums to a single cot, which then integrates to a log. The cancellation of the common $\sin(13x/2)$ factor is the key step. A new I2 application of sum-to-product beyond the usual $[0, \pi/2]$ identity integrals.

Learned variants - Integral 03 - Medium

35 distinct screenshot-learned variants are documented with their example, method, result when recorded, and distinguishing lesson.

1. Absolute value rescues odd integrand into gamma moment base two

13

EXAMPLE

$$\int_{-\infty}^{\infty} |x^3 2^{-x^2}| dx$$

METHOD

The bars convert an ODD integrand into an EVEN one, so the integral is $2 \int_0^{\infty} x^3 2^{-x^2} dx$ rather than 0; then $t = x^2$ gives $\int_0^{\infty} t e^{-t \ln 2} dt = \Gamma(2)/(\ln 2)^2$. The non- e exponential base is the flourish that produces $\ln^2 2$ downstairs.

RESULT

$$\frac{1}{\ln^2 2} \approx 2.081369$$

WHY IT MATTERS

Inverse of the catalogued absolute-value-symmetry-kills-odd-numerator trick — there $|\cdot|$ annihilates, here it RESCUES. The tell is an odd integrand under bars on a symmetric interval: never write 0 reflexively. Pair with base- a exponentials to move $\ln a$ into the answer.

2. Absolute value symmetry kills odd numerator

13

EXAMPLE

$$\int_{-\pi/6}^{\pi/6} \frac{\pi + 4x^{11}}{1 - \sin(|x| + \pi/6)} dx$$

METHOD

The denominator is even in x (because of $|x|$), so the $4x^{11}$ contribution to the numerator integrates to 0 over the symmetric interval $[-\pi/6, \pi/6]$. The remaining integral $\int \pi/(1 - \sin(|x| + \pi/6)) dx$ reduces via $u = x + \pi/6$ on $[0, \pi/6]$ (twice, by even symmetry) then conjugate-multiply $(1 + \sin u)/(1 + \sin u)$ to $\int (\sec^2 u + \sec u \tan u) du = \tan u + \sec u$, evaluated $\pi/6 \rightarrow \pi/3$ gives 2. Total: $2 \cdot \pi \cdot 2 = 4\pi$.

RESULT

$$4\pi$$

WHY IT MATTERS

Existing I3 symmetry pattern is " $f(x) - f(-x) = 0$ collapses one half", but here $|x|$ in the denominator deliberately makes it even to permit an odd polynomial of arbitrary degree in the numerator - a more elegant disguise. The conjugate-multiply $1/(1 - \sin)$ is the second move.

3. Arcosh disguise ibp to arcsec

13

EXAMPLE

$$\int_1^{\infty} \frac{\ln(x + \sqrt{x^2 - 1})}{x^2} dx$$

METHOD

Recognise $\ln(x + \sqrt{x^2 - 1}) = \operatorname{arcosh} x$; IBP with $dv = x^{-2} dx$ leaves $\int_1^{\infty} \frac{dx}{x\sqrt{x^2 - 1}} = [\operatorname{arcsec} x]_1^{\infty} = \pi/2$.

RESULT

$$\pi/2$$

WHY IT MATTERS

Couples an inverse-hyperbolic disguise with an IBP that lands on an arcsec antiderivative - a clean two-step recognition distinct from the plain arctan-self-substitution mediums.

4. Arctan of rational that telescopes via factor

I3

EXAMPLE

$$\int_0^\infty \arctan\left(\left(\frac{x^3 - x}{x^4 - 3x^2 + 1}\right)^2\right) dx$$

METHOD

Factor: $x^4 - 3x^2 + 1 = (x^2 - x - 1)(x^2 + x - 1)$ and $x^3 - x = x(x^2 - 1)$. The argument of arctan becomes a perfect square of a difference of arctan terms via the subtraction identity $\arctan a - \arctan b = \arctan \frac{a-b}{1+ab}$; in fact $\frac{x^3-x}{x^4-3x^2+1}$ is itself $\tan(\arctan(x + \frac{1}{x}) - \arctan(x - \frac{1}{x}))$ (or similar telescoping identity). The outer arctan of a square collapses via the identity $\arctan(t^2) = \text{something integrable}$ and the limits use a Glasser or $x \rightarrow 1/x$ symmetry.

RESULT

$\pi^2/4$ (canonical competition answer for this class of arctan-squared integrals over $(0, \infty)$).

WHY IT MATTERS

Hard integral where both the rational-function argument AND the outer arctan-of-square structure must be decoded. The rational $\frac{x^3-x}{x^4-3x^2+1}$ is a telescoping arctan difference in disguise; the square and outer arctan then yield a clean π^2 payoff. New I3 flavor: "arctan of a rational that is itself an arctan-difference".

5. Derivative of power tower as whole integrand

I3

EXAMPLE

$$\int_0^{\pi/3} (\sec x)^{\sin x} (\cos x \ln \sec x + \sec x - \cos x) dx$$

METHOD

Let $g(x) = (\sec x)^{\sin x} = e^{\sin x \ln \sec x}$. Then $g'(x) = g(x)[\cos x \ln \sec x + \sin x \tan x]$. Rewrite $\sin x \tan x = (1 - \cos^2 x)/\cos x = \sec x - \cos x$, so the bracket equals the bracket in the integrand. The entire integrand IS $g'(x)$, and the integral is just $g(\pi/3) - g(0)$.

RESULT

$$2^{\sqrt{3}/2} - 1$$

WHY IT MATTERS

Existing I3 menu has "derivative-of-denominator camouflage" and "factor is $d/dx[\arctan \text{ thing}]$ ", but here the WHOLE baroque-looking integrand is $d/dx[f^g]$ for a power-with-trig-exponent base. The cosmetic trick is the $\sin x \tan x = \sec x - \cos x$ rewrite that hides the product rule inside.

6. Determinant of matrix of antiderivatives

I3

EXAMPLE

$$\det \left[\int_0^x \begin{pmatrix} \cos t & \sin 2t \\ 1 & t \cos t + \sin t \end{pmatrix} dt \right].$$

METHOD

Integrate the matrix entrywise first: $\int_0^x \cos t dt = \sin x$, $\int_0^x \sin 2t dt = \frac{1}{2}(1 - \cos 2x) = \sin^2 x$, $\int_0^x 1 dt = x$, and $\int_0^x (t \cos t + \sin t) dt = \int_0^x (t \sin t)' dt = x \sin x$ (the integrand is an exact derivative). The matrix is $\begin{pmatrix} \sin x & \sin^2 x \\ x & x \sin x \end{pmatrix}$, whose determinant $x \sin^2 x - x \sin^2 x = 0$ vanishes identically.

RESULT

0 (for all x).

WHY IT MATTERS

An I3 dressed as linear algebra: the "integral" is a 2×2 matrix and you must integrate each entry, spot $t \cos t + \sin t = (t \sin t)'$, then notice the two columns become proportional so the determinant is identically zero. Distinct from existing single-scalar I3 camouflage entries - the cosmetic layer is the determinant wrapper, and the punchline is a hidden rank-deficiency.

7. Difference of power functions as perfect derivative

13

EXAMPLE

$$\int_1^e \left\{ \left(\frac{x}{e} \right)^{2x} - \left(\frac{e}{x} \right)^x \right\} \ln x \, dx$$

METHOD

Recognise that $\frac{d}{dx}[(x/e)^{2x}] = (x/e)^{2x} \cdot 2(\ln(x/e)+1) = (x/e)^{2x} \cdot 2 \ln x$. Similarly for $(e/x)^x$: derivative is $(e/x)^x \cdot (\ln(e/x)-1) = -(e/x)^x \ln x$. Combining: the integrand is exactly $\frac{1}{2} \frac{d}{dx}[(x/e)^{2x}] + \frac{d}{dx}[(e/x)^x]$. FTC collapses everything.

RESULT

Boundary evaluation: $\frac{1}{2}(1 - e^{-2}) + (1 - e^0) = \frac{1}{2}(1 - e^{-2})$.

WHY IT MATTERS

Flagship "the integrand IS a known derivative, just written ugly" trick. Cousin of the existing derivative-of-denominator-camouflage pattern but at exponential-tower level. Highly characteristic of dailyintegral I3 disguises.

8. Double angle collapse then perfect derivative of cubic under square root

13

EXAMPLE

$$\int_0^{\pi/2} \frac{\sin \theta \sin 2\theta (\sin^6 \theta + \sin^4 \theta + \sin^2 \theta) \sqrt{2 \sin^4 \theta + 3 \sin^2 \theta + 6}}{1 - \cos 2\theta} d\theta$$

METHOD

First collapse $\sin \theta \sin 2\theta / (1 - \cos 2\theta) \rightarrow \cos \theta$ via double-angle identities. Then substitute $p = \sin^2 \theta$ (not the more obvious $u = \sin \theta$) - the key insight, since it reveals $\sin^6 \theta + \sin^4 \theta + \sin^2 \theta = p(p^2 + p + 1) = Q'(p)/6$ where $Q(p) = 2p^3 + 3p^2 + 6p$. The integrand becomes $\propto Q'(p)\sqrt{Q(p)}$, integrating to $\frac{2}{3}Q(p)^{3/2}$.

RESULT

$$\frac{11\sqrt{11}}{18}$$

WHY IT MATTERS

A visually terrifying two-screen-wide trig/radical monster whose real difficulty is entirely cosmetic. Distinct from "derivative-of-denominator camouflage" and "derivative of power tower" - here the camouflage is a polynomial-times-sqrt-of-cubic perfect derivative.

9. Equal magnitude hyperbolic over cosecant ratio with reversed limits

13

EXAMPLE

$$\frac{\int_{-\ln(\sqrt{2}-1)}^{\ln(\sqrt{2}-1)} (e^x + e^{-x})^{10} dx}{\int_{\pi/4}^{\pi/2} (2 \csc x)^{11} dx}$$

METHOD

Numerator = $2^{10} \int \cosh^{10} x \, dx$ and denominator = $2^{11} \int_{\pi/4}^{\pi/2} \csc^{11} x \, dx$; the substitution $x = \operatorname{arcsinh}(\cot \theta)$ (i.e. $\sinh x = \cot \theta$, $\cosh x \, dx = -\csc^2 \theta \, d\theta$, $\cosh x = \csc \theta$) maps the θ -integral onto the cosh-integral, so the two integrals have equal magnitude. The catch: the numerator's limits are written backwards - lower = $-\ln(\sqrt{2}-1) > 0$, upper = $\ln(\sqrt{2}-1) < 0$ - so the numerator is the negative of the matching positive integral. The ratio is therefore exactly -1 .

RESULT

-1

WHY IT MATTERS

Two stacked tricks: (i) a hidden $\sinh \leftrightarrow \cot$ substitution that makes a $\cosh^{10}$ integral and a $\csc^{11}$ integral numerically identical, and (ii) deliberately swapped integration limits ($\ln(\sqrt{2}-1) < 0$) that flip the sign. The clean -1 payoff is the tell. Distinct from existing I3 King/reflection and Gudermannian entries - the punchline is "two scary integrals are secretly $\pm$ the same number".

10. Exact differential of the x to the $1/x$ family

13

EXAMPLE

$$\int_0^2 x^{\frac{1}{x}-2} (1 - \ln x) dx.$$

METHOD

Recognise the whole integrand as a perfect derivative. Let $F(x) = x^{\frac{1}{x}-1}$. Then $\ln F = (\frac{1}{x}-1) \ln x$, so $F'/F = -\frac{\ln x}{x^2} - \frac{1}{x^2} + \frac{1}{x}$. $\frac{1}{x} = \frac{1-\ln x}{x^2}$... more cleanly $F'(x) = x^{\frac{1}{x}-2} (1 - \ln x)$. Hence the integral is $[x^{1/x-1}]_0^2$; at $x \rightarrow 0^+$, $x^{1/x-1} = x^{1/x}/x \rightarrow 0$ (double-exponential decay beats the $1/x$).

RESULT

$$2^{1/2-1} = 2^{-1/2} = \frac{1}{\sqrt{2}}.$$

WHY IT MATTERS

A camouflaged FTC recognition beyond the logged x^x/x^{-x} family: here the antiderivative is the exotic $x^{1/x-1}$ and the trick is spotting that the messy factor $(1 - \ln x)$ is exactly what the quotient-in-the-exponent produces. Distinct from the existing "integrand = $\frac{d}{dx} x^{\pm x}$ " note because the exponent is $\frac{1}{x} - 1$, not $\pm x$, and the lower-limit vanishing needs the $x^{1/x} \rightarrow 0$ argument.

11. Exponential rational algebraic collapse via t eq e^x

13

EXAMPLE

$$\int_0^\infty \frac{(1 - e^x)(1 - e^{-x})}{(1 + e^x)(e^x + e^{-x})(e^{-x} + 1)} dx$$

METHOD

Multiply out the numerator: $(1 - e^x)(1 - e^{-x}) = 2 - e^x - e^{-x}$. Substituting $t = e^x$, this becomes $-(t - 1)^2/t$. The denominator similarly simplifies: $(1 + t)(t + 1/t)(1/t + 1) = (1 + t)^2(t^2 + 1)/t^2$. With $dx = dt/t$, the whole integrand collapses to $-(t - 1)^2/[(1 + t)^2(t^2 + 1)] dt$, a routine rational integral over $t \in [1, \infty)$.

RESULT

A clean closed form in $\ln 2$ and π (rational-function-of- e^x reduces to a standard partial-fractions form after the algebraic simplification).

WHY IT MATTERS

Distinct from the existing " $t = e^x$ turns denominator into a product of consecutive linears" I2 entry - here the front-loaded algebraic identity on the numerator and denominator both (via difference-of-squares-style exponential simplification) is what makes the substitution pay off, not a factoring trick alone. A genuinely different "hide the simplification before the substitution" I3 flavor.

12. Floor step trig mean integral

13

EXAMPLE

$$\int_0^3 \text{mean}\left(\sin\left(\frac{\pi}{2} - \frac{\pi}{6}[x]\right), \cos\left(\frac{\pi}{6} + \frac{\pi}{6}[x]\right)\right) dx$$

METHOD

$[x]$ takes values 0, 1, 2 on $[0, 1)$, $[1, 2)$, $[2, 3)$. For each integer n : $\sin(\pi/2 - n\pi/6) = \cos(n\pi/6)$ and $\cos(\pi/6 + n\pi/6) = \cos((n+1)\pi/6)$. The mean is $\frac{1}{2}[\cos(n\pi/6) + \cos((n+1)\pi/6)]$. Sum over $n = 0, 1, 2$, each contributing integral of the constant on its unit interval.

RESULT

$$\frac{1}{2} \sum_{n=0}^2 [\cos(n\pi/6) + \cos((n+1)\pi/6)] = \frac{1}{2} [\cos 0 + 2 \cos(\pi/6) + 2 \cos(\pi/3) + \cos(\pi/2)] = \frac{1}{2} [1 + \sqrt{3} + 1 + 0] = 1 + \frac{\sqrt{3}}{2}$$

WHY IT MATTERS

A "floor function turns the integrand into a step function" I3 flavor where the step values are trigonometric but the floor collapses everything to arithmetic. The co-function identity $\sin(\pi/2 - \theta) = \cos \theta$ is the first simplification. Distinct from the floor-of-bounded-radical I4 entry - here the floor is in the argument of the trig function, not of the whole integrand.

13. Infinite product binary telescoping $\prod(1+x^{2^n}) = 1/(1-x^2)$

13

EXAMPLE

$$\exp\left(\sum_{k=2}^{49} \int_0^{1/k} \prod_{n=1}^{\infty} (1+x^{2^n}) dx\right).$$

METHOD

The Cantor/binary product $\prod_{n=0}^{\infty} (1+x^{2^n}) = \frac{1}{1-x}$; starting at $n=1$ drops the $(1+x)$ factor, giving $\prod_{n \geq 1} (1+x^{2^n}) = \frac{1}{1-x^2}$. Then $\int_0^{1/k} \frac{dx}{1-x^2} = \operatorname{artanh} \frac{1}{k} = \frac{1}{2} \ln \frac{k+1}{k-1}$, and the sum telescopes: $\prod_{k=2}^{49} \frac{k+1}{k-1} = \frac{49 \cdot 50}{1 \cdot 2} = 1225$.

RESULT

$$\exp\left(\frac{1}{2} \ln 1225\right) = \exp(\ln 35) = 35 \text{ (verified).}$$

WHY IT MATTERS

Flagship Medium. Two stacked tricks (a binary infinite-product identity, then an artanh log-telescope) wrapped in an intimidating $\exp(\sum \int \prod \dots)$ shell that collapses to a clean integer. Strong new I3 anchor.

14. Integral as equation solve for parameter then Vieta

13

EXAMPLE

Let $a_1, \dots, a_n$ be all real a with $\int_1^e \frac{\sqrt{\ln x}}{x(a - (\ln x)^{3/2})^2} dx = 1$. Find $a_1^2 + \dots + a_n^2$.

METHOD

Substitute $t = (\ln x)^{3/2}$ (so $\sqrt{\ln x} \frac{dx}{x} = \frac{2}{3} dt$, $t: 0 \rightarrow 1$); the integral collapses to $\frac{2}{3} \cdot \frac{1}{a(a-1)}$. Setting it equal to 1 gives a quadratic $3a^2 - 3a - 2 = 0$; then Vieta: $\sum a_i^2 = (\sum a_i)^2 - 2 \prod a_i = 1 + \frac{4}{3} = \frac{7}{3}$.

RESULT

$$\frac{7}{3}.$$

WHY IT MATTERS

The integral is a shell. The actual reward is (i) spotting the substitution that linearises the denominator into $\int dt/(a-t)^2$, then (ii) recognising the resulting equation in a as a quadratic and finishing with a symmetric-function-of-roots (Vieta) step. Distinct from ordinary I3 evaluation: the answer is a property of the set of parameter values that make the integral hit a target, not the integral itself. A genuine two-layer (calculus then algebra) archetype.

15. Inverse function integral with engineered jacobian cancellation

13

EXAMPLE

Let $f(x) = x^{2^x}$ and g its inverse. Evaluate $\int_0^2 \frac{dx}{\log_2(e) + g(x)}$.

METHOD

Substitute $x = f(t) = t^{2^t}$ so $g(x) = t$ and $dx = 2^t(1+t \ln 2) dt$; limits $x: 0 \rightarrow 2$ become $t: 0 \rightarrow 1$ (since $t^{2^t} = 2 \Rightarrow t = 1$). Write $\log_2(e) = 1/\ln 2$, so $\log_2(e) + t = (1+t \ln 2)/\ln 2$. The awkward Jacobian factor $(1+t \ln 2)$ cancels exactly, leaving $\int_0^1 \ln 2 \cdot 2^t dt = [2^t]_0^1$.

RESULT

$$1.$$

WHY IT MATTERS

A "sub $x = f(t)$, and the denominator was reverse-engineered to annihilate $f'(t)$ " I3 trick. The tell is the odd constant $\log_2(e) = 1/\ln 2$ sitting next to the inverse function - it is precisely the piece that makes $f'(t)/(\text{denominator})$ collapse to a bare exponential. Distinct from the logged Gudermannian-inverse I3 and the derivative-of-denominator camouflage; here the inverse function never has to be found in closed form.

16. Inverse Gudermannian geometric series integral

13

EXAMPLE

$$I = \int_0^{\pi/2} \frac{(\operatorname{gd}^{-1}(x))^5 - 1}{\operatorname{gd}^{-1}(x) - 1} dx, \text{ where } \operatorname{gd}^{-1}(\phi) \text{ is the inverse Gudermannian.}$$

CONTINUED

METHOD

The fraction $\frac{u^5-1}{u-1} = u^4 + u^3 + u^2 + u + 1$ (geometric sum factorization) with $u = \text{gd}^{-1}(x)$. The inverse Gudermannian is $\text{gd}^{-1}(x) = \ln |\tan(x/2 + \pi/4)| = \ln |\sec x + \tan x|$. The substitution $u = \text{gd}^{-1}(x)$, $du = \sec x \, dx$ transforms the integral into a sum of power integrals $\int_0^\infty (u^4 + u^3 + u^2 + u + 1) du / \cosh u$ (a Fourier-type series via the Gudermannian $dx = \text{sech } u \, du$). Each $\int_0^\infty u^n \text{sech } u \, du$ gives an Euler-type sum involving β -function / Dirichlet β values.

RESULT

Integer closest to I (the problem asks for the nearest integer; from the structure $I \approx$ small integer).

WHY IT MATTERS

The key insight is the geometric-sum factorization $\frac{u^n-1}{u-1}$ applied to an inverse Gudermannian, then the substitution $u = \text{gd}^{-1}(x)$ to convert to a sech-weighted polynomial integral. A genuinely new I3/I4 flavor: "Gudermannian substitution + polynomial via geometric series + sech Fourier integrals". First appearance of the Gudermannian in the learned-patterns catalog.

17. Inverse trig composition collapse

13

EXAMPLE

$$\int_{5/4}^{5/3} \sin(\sec^{-1} x) \, dx$$

METHOD

Right-triangle identity collapses $\sin(\sec^{-1} x) = \sqrt{x^2 - 1}/x$, leaving an integrand whose antiderivative is $\sqrt{x^2 - 1} - \sec^{-1} x$ (verifiable by direct differentiation).

RESULT

$$\frac{4}{3} - \frac{3}{4} - \sec^{-1}(5/3) + \sec^{-1}(5/4)$$

WHY IT MATTERS

Tests recognition that $\sin \circ \sec^{-1}$ collapses algebraically; not the usual $x = \sec \theta$ Wallis-style. The trig sub is hidden inside the integrand itself.

18. Iterated polynomial reflection symmetry $f(x) + f(a-x) = c$

13

EXAMPLE

$P(x) = 2x^3 - 48x^2 + 256x + 8$; with P^n the n -fold composition, evaluate $\int_0^{16} P^{1729}(x) \, dx$.

METHOD

Verify $P(16-x) + P(x) = 16$ (constant). The reflection identity is inherited by every iterate, so $P^{1729}(x) + P^{1729}(16-x) = 16$; hence $\int_0^{16} P^{1729} = \frac{1}{2} \cdot 16 \cdot 16$.

RESULT

128.

WHY IT MATTERS

A polynomial built so that $f(a-x) + f(x) = c$ passes that symmetry to all compositional iterates; the scary "1729th iterate" is a red herring. New I3/I4 anchor: "iterate preserves midpoint reflection."

19. King property mismatched base exponent swap

13

EXAMPLE

$$\int_0^{\pi/2} \frac{\sin(x)^{\cos(x)}}{\sin(x)^{\cos(x)} + \cos(x)^{\sin(x)}} \, dx$$

METHOD

Under $x \mapsto \pi/2 - x$, $\sin \leftrightarrow \cos$, so the integrand maps to $\cos(x)^{\sin(x)} / (\cos(x)^{\sin(x)} + \sin(x)^{\cos(x)}) = 1 - (\text{original integrand})$. Adding the integral to its reflection gives $2I = \int_0^{\pi/2} 1 \, dx = \pi/2$.

RESULT

$I = \pi/4$ (verified numerically).

CONTINUED

WHY IT MATTERS

The existing I3 King's-property menu is all same-function reflections (e.g. $\sin^n / (\sin^n + \cos^n)$); here the base and exponent are swapped between the two terms ($\sin^{\cos}$ vs $\cos^{\sin}$) rather than just the trig function alone, so it's less obviously King-shaped at first glance even though the mechanism is identical.

20. Leibniz arctan quotient recombination

13

EXAMPLE

$$\int_1^{\sqrt{3}} x \frac{\int_0^1 \frac{dt}{1+t^2} - \int_0^x \frac{dt}{1+t^2}}{x^2} dx$$

METHOD

Recognize the embedded definite integrals as arctan (inner-from-0 = $\arctan x$, the constant one = $\arctan 1 = \pi/4$), so the integrand is $\frac{(\pi/4)x - \arctan x}{x^2}$, which is exactly $\frac{d}{dx} \left(-\frac{\arctan x}{x} + \frac{\pi}{4} \ln x \right)$ - integrate by reverse quotient rule.

RESULT

closed form involving $\pi/12$ and $\ln 3$ (evaluate $-\arctan x/x + \frac{\pi}{4} \ln x$ from 1 to $\sqrt{3}$).

WHY IT MATTERS

Distinct from "integral-inside" limits and from variable-limit Leibniz derivatives - here the disguised arctan's recombine so the whole rational-of-arctan integrand is an exact reverse-quotient-rule derivative. A recognise-the-antiderivative-of-a-quotient trick, not a substitution or Feynman parameter.

21. Log cosine over ($e^x + 1$) Dirichlet symmetry

13

EXAMPLE

$$\int_{-\pi/2}^{\pi/2} \frac{\ln(\cos x)}{e^x + 1} dx$$

METHOD

$x \rightarrow -x$ gives $\frac{1}{e^x + 1} + \frac{1}{e^{-x} + 1} = 1$; halve to $\int_0^{\pi/2} \ln \cos x dx = -\frac{\pi}{2} \ln 2$.

RESULT

$$-\frac{\pi}{2} \ln 2$$

WHY IT MATTERS

the $1/(e^x + 1)$ weight on a symmetric interval is the classic "add the reflection, weight collapses to 1/2" trick married to Euler's $\int_0^{\pi/2} \ln \cos$.

22. Log linear over quadratic tan sub cosine shift cancellation

13

EXAMPLE

$$\int_{-3}^1 \frac{\ln(3-x)}{3+x^2} dx$$

METHOD

Substitute $x = \sqrt{3} \tan \theta$, so $dx/(3+x^2) = d\theta/\sqrt{3}$ and the limits $x = -3, 1$ become $\theta = -\pi/3, \pi/6$ - an interval of length exactly $\pi/2$. Factor the log: $3 - \sqrt{3} \tan \theta = 2\sqrt{3} \cos(\theta + \pi/6)/\cos \theta$, so the integrand splits into $\ln(2\sqrt{3}) + \ln \cos(\theta + \pi/6) - \ln \cos \theta$. Shifting the first log-cosine integral by $\pi/6$ maps its range to $[-\pi/6, \pi/3]$, and reflecting the second ($\theta \rightarrow -\theta$) maps it to the same range - so the two transcendental pieces cancel identically and only the constant survives.

RESULT

$$\frac{\pi \ln(2\sqrt{3})}{2\sqrt{3}} = \frac{\sqrt{3} \pi \ln 12}{12} \approx 1.12678 \text{ (verified numerically to 25 digits)}$$

WHY IT MATTERS

A Serret/Coxeter-style "log of a linear function over an irreducible quadratic" integral where the engineered asymmetric limits $[-3, 1]$ are the whole trick: they are exactly the pair that makes the tan-substituted interval have length $\pi/2$ and makes the two $\int \ln \cos$ pieces coincide after a shift and a reflection. Distinct from the catalogued $\ln \cos$ / Poisson-kernel and King's-property entries, which use symmetric intervals and reflection within one integral; here the cancellation is between two different log-cosine integrals produced by factoring $a - b \tan \theta$ into a phase-shifted cosine.

23. Log reciprocal negation area collapse

13

EXAMPLE

Let $f(x) = \ln(x)$. Find the area bounded by $f^2(|x|)$ and $-|f(1/x)|$.

METHOD

Recognise the reciprocal-log identity $\ln(1/|x|) = -\ln|x|$, so $-|f(1/x)| = -|-\ln|x|| = -|\ln|x||$ - the second curve is secretly built from the same quantity $t = |\ln|x||$ as the first ($f^2(|x|) = t^2$). The problem reduces to comparing $y = t^2$ (always ≥ 0) against $y = -t$ (always ≤ 0), which meet only where $t = 0$, i.e. $|x| = 1$.

RESULT

The two curves touch only at $x = \pm 1$ and diverge in opposite directions as $x \rightarrow 0$ or $x \rightarrow \infty$, so the "bounded" region is degenerate - either the enclosed area is 0 (single tangency point) or the problem is testing that no finite bounded region exists. Flagged as a possible intentional gotcha; not fully resolved from the source capture.

WHY IT MATTERS

A new I3 disguise family built on the log's reciprocal-negation identity $f(1/x) = -f(x)$ to force two seemingly unrelated composite curves to collapse onto the same underlying variable $t = |\ln|x||$. Distinct from the catalogued abs-value-symmetry and odd-times-weight I3 entries because the payoff is about curve identity/degeneracy rather than parity cancellation inside a single integral.

24. Nested exp of log ratio collapses to minus four log

13

EXAMPLE

$$\int_0^{\sqrt{2}} \ln\left(\frac{1 + e^{-\ln \frac{1+x^4}{1-x^4}}}{1 - e^{-\ln \frac{1+x^4}{1-x^4}}}\right) dx$$

METHOD

$e^{-\ln A} = 1/A$ with $A = \frac{1+x^4}{1-x^4}$. Then $\frac{1+1/A}{1-1/A} = \frac{A+1}{A-1} = \frac{2/(1-x^4)}{2x^4/(1-x^4)} = x^{-4}$, so the integrand is $\ln(x^{-4}) = -4\ln x$. Hence

$$\int_0^{\sqrt{2}} -4\ln x \, dx = -4[x\ln x - x]_0^{\sqrt{2}}.$$

RESULT

$$-4(\sqrt{2}\ln\sqrt{2} - \sqrt{2}) = 2\sqrt{2}(2 - \ln 2) \approx 3.696$$

WHY IT MATTERS

"Medium Integral" maximal-camouflage flavor: a tower of $\ln$, $\exp$, and a rational argument all unwind to the trivial $-4\ln x$. Rewards recognising $e^{-\ln A} = 1/A$ first, then the $\frac{A+1}{A-1}$ algebra. Distinct from existing camouflage entries (those used continued fractions / golden ratios); this one uses log-exp inversion.

25. Nested radical fixed point in definite integral

13

EXAMPLE

$$\int_0^1 \sqrt{\frac{1}{x} + \sqrt{\frac{1}{x} + \sqrt{\frac{1}{x} + \dots}}} dx.$$

METHOD

Inner radical is a fixed point $y = \sqrt{1/x + y} \Rightarrow y = \frac{1+\sqrt{1+4/x}}{2}$; then integrate the closed form over $[0, 1]$ (the $\sqrt{1+4/x}$ piece gives a $\sqrt{x^2+4x}$ -type antiderivative).

RESULT

finite closed form ≈ 2.40 .

WHY IT MATTERS

dailyintegral pairs the "infinite-nesting fixed point" trope (usually a derivative/limit trick) with a definite INTEGRAL. The tell is ... inside a radical with a $1/x$ that survives integration. Distinct from existing Ramanujan-nested-radical (that was a limit) and Trojan-sub integrals.

26. Numerator sq minus denominator sq constant arc length form**13****EXAMPLE**

$$\int_0^e \frac{\sqrt{e^{2x} + e^{-2x} + e^2 + e^{-2}}}{e^x + e^{-x}} dx.$$

METHOD

Notice $N^2 - D^2$ is a constant $(e - e^{-1})^2$, so the integrand is $\sqrt{1 + \sinh^2(1) \operatorname{sech}^2 x}$ (arc-length style). Sub $s = \sinh x$ and split $\frac{\sqrt{s^2+b^2}}{s^2+1}$ into $\frac{1}{\sqrt{s^2+b^2}} + \frac{b^2-1}{(s^2+1)\sqrt{s^2+b^2}}$.

RESULT

$$\sinh^{-1}(\sinh e / \cosh 1) + \sinh(1) \arctan(\sinh 1 \sinh e / \sqrt{\sinh^2 e + \cosh^2 1}) \approx 3.29598.$$

WHY IT MATTERS

When a radical-over-sum integrand hides $N^2 = D^2 + c$, rewrite as $\sqrt{1 + c/D^2}$; hyperbolic substitution then standard-splits.

27. Odd times semicircle weight**13****EXAMPLE**

$$\int_{-2}^2 \left(x^3 \cos \frac{x}{2} + \frac{1}{2} \right) \sqrt{4 - x^2} dx$$

METHOD

Symmetric interval: $x^3 \cos(x/2)$ is odd and $\sqrt{4 - x^2}$ even, so that product integrates to zero; only $\frac{1}{2} \sqrt{4 - x^2}$ survives as half the area of a radius-2 semicircle.

RESULT

$$\pi$$

WHY IT MATTERS

A nasty-looking term $(x^3 \cos \frac{x}{2})$ is pure decoy that vanishes by parity against an even weight; the real content is a geometric semicircle-area read-off. Rewards parity inspection before any integration attempt.

28. Parameter threshold integral equation (count of real solutions)**13****EXAMPLE**

"Find the minimum positive integer k such that $\int_0^{2x} \frac{kt}{10^x + 10^t} dt = 1$ has exactly two distinct real solutions in x ."

METHOD

treat the LHS as a function of x ; analyse its shape (monotonicity / single hump) and find the k at which the horizontal line $= 1$ first cuts it twice.

RESULT

a small integer k (threshold value).

WHY IT MATTERS

an integral-equation framed as a "how many roots / minimum parameter" puzzle - a hybrid integral/logic register the site likes for Medium Integral.

29. Radical odd power cosine rationalization**13****EXAMPLE**

$$\int_0^{\pi/6} \frac{dx}{\sqrt{\cos^5 x (1 + \sin x)}}$$

METHOD

Pair the odd power of cosine under the radical with $(1 + \sin x)$ using $\cos^2 x = (1 - \sin x)(1 + \sin x)$ to rationalize, then $t = \sin x$ (or $t = \tan(x/2)$) gives an algebraic integral.

RESULT

elementary but messy (numeric ≈ 0.61).

WHY IT MATTERS

A rationalization that exploits $\cos^2 = (1 - \sin)(1 + \sin)$ to tame an odd cosine power under a radical - distinct from the existing half-angle and Weierstrass I3 entries.

30. Rational function nested radical

I3

EXAMPLE

$$\int_0^1 \sqrt{\frac{x}{1-x} + \sqrt{\frac{x}{1-x} + \sqrt{\frac{x}{1-x} + \cdots}}} dx$$

METHOD

The classic infinite-nested-radical fixed-point trick, but the repeating inner term is a rational function of x (namely $x/(1-x)$) rather than a constant, so the fixed point $y(x) = \frac{1}{2}(1 + \sqrt{1 + 4x/(1-x)})$ is itself a function to be integrated over $[0, 1]$.

RESULT

Requires solving the quadratic $y^2 - y - \frac{x}{1-x} = 0$ then integrating the closed form.

WHY IT MATTERS

A genuine twist on the standard "nested radical converges to a constant" I4/L1 flavor - here the self-similar block is x -dependent, turning a fixed-point puzzle into a genuine integration problem. Good for I3/I4 variety.

31. Reflection engineered log ratio integral

I3

EXAMPLE

$$\int_0^1 \ln \frac{1+(4x-4x^2)^2}{1+(1-x^2)^2} dx.$$

METHOD

King's property $x \rightarrow 1-x$; the numerator's $4x(1-x)$ and the denominator's $(1-x)(1+x)$ are reflection-related, forcing heavy cancellation.

WHY IT MATTERS

A template for medium log-ratio integrals engineered to collapse under $x \rightarrow 1-x$.

32. Secant sub half angle arctan self derivative

I3

EXAMPLE

$$\int_1^2 \arctan\left(\sqrt{\frac{1-1/x}{1+1/x}}\right) \left(\frac{1-1/x}{1+1/x}\right) \frac{dx}{x\sqrt{x^2-1}}.$$

METHOD

Sub $x = \sec \theta$ ($\theta \in [0, \pi/3]$), $dx = \sec \theta \tan \theta d\theta$, $\sqrt{x^2-1} = \tan \theta$, so $dx/(x\sqrt{x^2-1}) = d\theta$. The radical becomes $\sqrt{(1-\cos \theta)/(1+\cos \theta)} = \tan(\theta/2)$, and the integral reduces to $\int_0^{\pi/3} \arctan(\tan \frac{\theta}{2}) \tan^2 \frac{\theta}{2} d\theta = \int_0^{\pi/3} \frac{\theta}{2} \tan^2 \frac{\theta}{2} d\theta$.

RESULT

reduces to an elementary $\int (\theta/2)(\sec^2(\theta/2) - 1) d\theta$ over $[0, \pi/3]$ via IBP.

WHY IT MATTERS

A double payoff: the $dx/(x\sqrt{x^2-1}) = d\theta$ identity AND the half-angle simplification of the radical-inside-arctan. Distinct from the existing half-angle-Vieta entry - here the half-angle appears inside an arctan whose own argument is the integrating factor.

33. Self map closed form iterate then power substitution

I3

EXAMPLE

$$f(x) = \frac{x}{(1+x^4)^{1/4}}, g(x) = f(f(f(f(x))))); \text{ evaluate } 18 \int_0^{\sqrt[4]{20}} x^2 g(x) dx.$$

METHOD

Direct computation shows $f(f(x)) = x/(1+2x^4)^{1/4}$, and induction gives the closed form $f^{(n)}(x) = x/(1+nx^4)^{1/4}$, so $g = f^{(4)}(x) = x/(1+4x^4)^{1/4}$. The integral then falls to a single $u = 1+4x^4$ substitution.

RESULT

39.

CONTINUED

WHY IT MATTERS

A different iterated-function mechanism than the existing menu's fixed-point, reflection-symmetry, and root-orbit iteration entries - here the payoff is discovering (by direct algebra, not a fixed-point argument) a closed-form formula for the n -fold iterate itself, which is then plugged into a definite integral.

34. Surprising constant via half angle elliptic form

13

EXAMPLE

$$\int_0^{\pi/4} \sec x \sqrt{\sec 2x} dx$$

METHOD

Substitute $s = \sin x$, reducing the integral to $\int ds / [(1 - s^2)\sqrt{1 - 2s^2}]$, then $s = \frac{1}{\sqrt{2}} \sin \theta$ turns it into the standard form $\int d\theta / (a^2 \cos^2 \theta + b^2 \sin^2 \theta) = \pi / (2ab)$.

RESULT

$$\pi/2$$

WHY IT MATTERS

A "famous surprising integral" flavor where an intimidating $\sqrt{\sec 2x}$ factor collapses to a textbook $a^2 \cos^2 + b^2 \sin^2$ form after a chain of two substitutions. Distinct from the catalogued Weierstrass/complete-the-square I3 families.

35. Tan substitution with power expansion between special angles

13

EXAMPLE

$$\int_{\pi/6}^{\pi/4} \frac{dx}{\sqrt{(\sin x)^{11} (\cos x)^5}}, \text{ with the result encoded as } 2^a(2^5 3^{1/a} - b)/45 \text{ and asking for } a + b.$$

METHOD

Let $t = \tan x$. The powers of $\sin x$ and $\cos x$ turn into a finite sum of powers of t , and the bounds $1/\sqrt{3}$ to 1 produce powers of 2 and 3.

RESULT

Encoded-answer format; solve symbolically, then match natural-number parameters.

WHY IT MATTERS

Medium integrals may hide a long but clean algebraic power integral behind special-angle tangent bounds and a deliberately encoded final answer.

Learned variants - Integral 04 - Hard

36 distinct screenshot-learned variants are documented with their example, method, result when recorded, and distinguishing lesson.

1. Abel plana dilog sech arccot (expert template; value not certified)

I4

EXAMPLE

$$\int_0^\infty \frac{\cot^{-1}\left(\frac{1}{3t} - \frac{\text{Li}_2(e^{-\pi t})}{\pi^2 t} - \frac{t}{2}\right)}{\cosh(\pi t)} dt$$

METHOD

A $\text{sech}(\pi t)$ kernel against an arccot whose argument bundles $\text{Li}_2(e^{-\pi t})$ and $1/t$ terms - the signature of an Abel-Plana / Binet-type evaluation; expand Li_2 and swap sum/integral against the cosh kernel.

RESULT

not certified from the screenshot (expert-tier; expected clean constant).

WHY IT MATTERS

Logs a template (Abel-Plana + dilogarithm + sech kernel) as a flagship I4 disguise distinct from the Beta/Mellin/Binet/Feynman families already on the menu. No value asserted - flagged for later verification.

2. Airy modulus wronskian arctan collapse

I4

EXAMPLE

$$\int_0^\infty \frac{dx}{\text{Ai}^2(x) + \text{Bi}^2(x)}$$

METHOD

Use $\frac{d}{dx} \arctan \frac{\text{Bi}(x)}{\text{Ai}(x)} = \frac{\text{Ai Bi}' - \text{Bi Ai}'}{\text{Ai}^2 + \text{Bi}^2} = \frac{W}{\text{Ai}^2 + \text{Bi}^2}$ with the Airy Wronskian $W = 1/\pi$, turning the integral into a telescoping arctan jump over $[0, \infty)$.

RESULT

$$\frac{\pi^2}{6}.$$

WHY IT MATTERS

A special-function integral solved by a Wronskian-driven exact antiderivative (arctan of the ratio of two independent ODE solutions). Generalizes "derivative-of-denominator camouflage" and "double-angle arctan self-substitution" to ODE-solution pairs; not covered by the Beta/digamma/Mellin/Binet I4 families.

3. Barnes G log integral Glaisher Kinkelin

I4

EXAMPLE

$$\int_0^1 \ln(G(x)) dx, \text{ where } G \text{ is the Barnes G-function.}$$

METHOD

Reduce the log-Barnes-G integral over $[0, 1]$ via the Adamchik/Raabe-type $\ln G$ integral identity to the Glaisher-Kinkelin constant, expressing the result through $\zeta'(-1) = \frac{1}{12} - \ln A$. First appearance of Barnes $G / \zeta'(-1)$ in the menu (existing I4 special-function family covers only $\Gamma/\text{Beta}/\text{digamma}/\text{Mellin}/\text{Binet}/\text{polylog}/\text{Airy}$).

RESULT

$$2\zeta'(-1) - \frac{1}{4} \ln(2\pi) - \frac{1}{12} = \frac{1}{12} - 2 \ln A - \frac{1}{4} \ln(2\pi) \approx -0.873645.$$

WHY IT MATTERS

A bare $\int_0^1 \ln(\text{special function})$ usually wants the function's Raabe/Kummer integral identity; Barnes G lands on the Glaisher-Kinkelin constant.

4. Binet bose einstein integral with rational factor

14

EXAMPLE

$$100 \int_0^{\infty} \frac{x^3}{(e^{2\pi x} - 1)(x^4 + x^2 + 1)} dx$$

METHOD

Expand $1/(e^{2\pi x} - 1) = \sum_{n \geq 1} e^{-2\pi n x}$. Factor $x^4 + x^2 + 1 = (x^2 + x + 1)(x^2 - x + 1)$ via cyclotomic identity, then partial-fraction. Each resulting piece is a Binet-style integral $\int x^a e^{-bx}/(\text{quadratic}) dx$ that evaluates via $\ln \Gamma$ / digamma at rational arguments (cube-roots-of-unity poles). Connects to Hurwitz ζ .

RESULT

Closed form in $\ln \Gamma(1/3)$, $\ln \Gamma(2/3)$, $\ln 2$, $\ln 3$. The $\times 100$ prefactor is engineered to make the final number a clean integer/rational.

WHY IT MATTERS

Existing I4 menu has

Γ -B/ ψ reductions and Mellin pieces. The Binet-formula flavor ($e^{2\pi x} - 1$ in denominator $\Leftrightarrow$ log- Γ integrals) is its own flagship trick. The $x^4 + x^2 + 1$ cyclotomic factorisation is the cosmetic disguise.

5. Binomial reflection log derivative exact integrand

14

EXAMPLE

With $C(a)$ the coefficient of x^{2026} in $(1+x)^a$, find $\int_0^1 C(-y-1) \left(\sum_{k=1}^{2026} \frac{1}{y+k} \right) dy$.

METHOD

Reflection $C(-y-1) = \prod_{k=1}^{2026} \frac{y+k}{2026!} = P(y)$ (even degree kills the sign), and the harmonic sum is $(\ln P)'$, so the integrand is exactly $P'(y)$ - FTC finishes it.

RESULT

$$P(1) - P(0) = 2027 - 1 = 2026.$$

WHY IT MATTERS

A binomial coefficient at a negated upper argument is a rising-factorial product; pairing it with its own digamma-style harmonic sum manufactures an exact derivative.

6. Complementary fourth root arctan difference cosh sub

14

EXAMPLE

$$\int_1^{\infty} \frac{\tan^{-1}\left(\sqrt[4]{\frac{x+1}{x-1}}\right) - \tan^{-1}\left(\sqrt[4]{\frac{x-1}{x+1}}\right)}{\sqrt{x^2-1}} dx.$$

METHOD

The two radicands are reciprocals, so writing $a = \sqrt[4]{(x+1)/(x-1)}$ the numerator is $\arctan a - \arctan(1/a) = 2 \arctan a - \frac{\pi}{2}$. Substitute $x = \cosh s$ ($\sqrt{x^2-1} = \sinh s$, $dx/\sqrt{x^2-1} = ds$, $s : 0 \rightarrow \infty$), and $(x+1)/(x-1) = \coth^2(s/2)$ so $a = \sqrt{\coth(s/2)}$. The integrand becomes $\frac{\pi}{2} - 2 \arctan \sqrt{\tanh(s/2)}$, which integrates over $[0, \infty)$ to $\frac{\pi \ln 2}{2}$.

RESULT

$$\frac{\pi \ln 2}{2} \text{ (verified numerically to 40 digits).}$$

WHY IT MATTERS

A Hard integral built from the complementary identity $\arctan a - \arctan(1/a) = 2 \arctan a - \frac{\pi}{2}$ together with the $x = \cosh s$ reduction $dx/\sqrt{x^2-1} = ds$ and the half-angle simplification $\sqrt[4]{(x+1)/(x-1)} = \sqrt{\coth(s/2)}$. Distinct from the existing secant-sub half-angle I3 entry (that had a single arctan and an algebraic weight) - here it is a difference of complementary arctans over a bare $1/\sqrt{x^2-1}$, and the clean $\frac{\pi}{2} \ln 2$ value is the payoff.

7. Cube root reciprocal cubed odd identity

14

EXAMPLE

$$\int_{-1}^1 \frac{dx}{[x^{-1/3}]^3}.$$

METHOD

$(x^{-1/3})^3 = x^{-1}$ inverted gives integrand $= x$ - but on $x < 0$ the real cube root keeps the sign, so the integrand is genuinely the odd function x over a symmetric interval; the integral vanishes by parity.

RESULT

0

WHY IT MATTERS

The cosmetic "reciprocal-cube-root cubed" wrapper hides the odd identity $f(x) = x$. The real subtlety is the branch of the cube root for negative x (it stays real and odd, so no absolute-value trap). A pure "looks horrible, is secretly odd $\Rightarrow$ 0" camouflage, complementing the existing abs-value-symmetry entry.

8. Dilog times log squared over eisenstein denominator

14

EXAMPLE

$$\int_0^\infty \frac{\text{Li}_2(-x) \ln^2 x}{x^2 - x + 1} dx.$$

METHOD

A dilogarithm numerator with a $\ln^2 x$ weight over the Eisenstein denominator $x^2 - x + 1 = \frac{x^3+1}{x+1}$ (primitive 6th-roots of unity). Attack by the self-reciprocal fold $x \rightarrow 1/x$ (the kernel $\frac{\ln^2 x}{x^2-x+1}$ is $x \mapsto 1/x$ symmetric up to a Jacobian) together with $\text{Li}_2(-x) + \text{Li}_2(-1/x) = -\frac{\pi^2}{6} - \frac{1}{2} \ln^2 x$ (Landen/inversion), reducing to $\int_0^\infty \frac{\ln^k x}{x^2-x+1} dx$ moments evaluable by the digamma/Hurwitz reflection at the sixth roots.

RESULT

No elementary closed form; numerically ≈ -15.996 (a $\pi^4 / \zeta(3) \ln$ / polygamma-at-thirds combination).

WHY IT MATTERS

Flagship "hard integral is a scaffold of named identities" I4 genre: dilog-inversion + $x \rightarrow 1/x$ folding + $\frac{\ln^k x}{x^2-x+1}$ moments via sixth-root digamma reflection. Worth keeping as an aspirational template; when authoring, prefer a variant engineered to a clean closed form (e.g. drop the $\ln^2$ weight so the moment integrals collapse to π^2 -rationals).

9. Double integral arctan ratio symmetrization

14

EXAMPLE

$$\int_0^\infty \int_0^\infty e^{-(x+y)} \tan^{-1}\left(\frac{x}{y}\right) dx dy$$

METHOD

Swap the roles of x and y (the weight $e^{-(x+y)}$ is symmetric, the region is symmetric) to get the same integral with $\tan^{-1}(y/x)$. Add the two copies and use $\arctan(t) + \arctan(1/t) = \pi/2$ for $t > 0$: $2I = \frac{\pi}{2} \int_0^\infty \int_0^\infty e^{-(x+y)} dx dy = \frac{\pi}{2} \cdot 1$.

RESULT

$I = \frac{\pi}{4} \approx 0.785398$ (verified numerically)

WHY IT MATTERS

First double-integral entry in the I4 menu, and the first use of the complementary-arctan identity as a two-variable symmetrization (the logged reciprocal-arctan entries are one-dimensional Leibniz/limit problems). The lesson is that an intimidating iterated improper integral can be pure bookkeeping: symmetric weight + symmetric domain + an antisymmetric-under-swap integrand whose two orientations sum to a constant, so the answer is $\frac{1}{2} \cdot (\text{constant}) \cdot (\text{total mass})$. No integration of arctan is ever performed.

10. Double integral change of variables with jacobian triangle to square

I4

EXAMPLE

$$\int_0^1 \int_0^{1-x} e^{y/(x+y)} dy dx$$

METHOD

Substitute $u = x + y$, $v = \frac{y}{x+y}$, i.e. $x = u(1-v)$, $y = uv$, with Jacobian u . The triangle maps to the unit SQUARE and the integral factors completely: $\int_0^1 u du \int_0^1 e^v dv$.

RESULT

$$\frac{e-1}{2} \approx 0.859141$$

WHY IT MATTERS

First genuine Jacobian change-of-variables in the I4 slot — the catalogued double-integral I4 (2026–07–30) used swap-symmetry with no Jacobian. Two double integrals in two weeks means multivariable substitution is now an established I4 device, not a one-off. The tell is a homogeneous-degree-zero exponent like $y/(x+y)$ over a triangle: pick the ratio as one new variable and the sum as the other.

11. Double log half angle quotient

I4

EXAMPLE

$$\int_0^\pi \frac{\ln(1+\cos x) \ln(\sin x)}{\tan x + \sec x} dx$$

METHOD

Rewrite $\tan x + \sec x = \frac{1+\sin x}{\cos x}$ and $1 + \cos x = 2\cos^2(x/2)$, $\sin x = 2\sin(x/2)\cos(x/2)$ to collapse the denominator and expose a product of two logs (of half-angle trig factors); likely resolved via Fourier series for $\ln \sin / \ln \cos$ or Feynman parametrization.

RESULT

Not shown on screen.

WHY IT MATTERS

Combines two distinct log-of-trig factors divided by a half-angle-collapsing denominator - not in the documented I4 menu (Binet, Beta/digamma, Mellin, polylog-ladder, Feynman-arctan). Moderate confidence this is genuinely new rather than a relabeled Euler-log-sin variant.

12. Dual base fractional part fractal product integral

I4

EXAMPLE

$$\int_0^1 \left(\sum_{n=0}^{\infty} \frac{\{2^n x\}}{2^n} \right) \left(\sum_{m=0}^{\infty} \frac{\{3^m x\}}{3^m} \right) dx, \text{ where } \{\cdot\} \text{ is the fractional-part function.}$$

METHOD

Each factor is a Takagi-function-style fractal series built from fractional parts scaled by a different base (2 and 3). Termwise expansion of the product gives a double sum $\sum_{n,m} \frac{1}{2^n 3^m} \int_0^1 \{2^n x\} \{3^m x\} dx$; each inner integral is a periodic sawtooth-correlation that must be evaluated via the mean/covariance structure of fractional-part functions at commensurate/incommensurate frequencies $2^n, 3^m$.

RESULT

Reduces to a convergent double series in powers of $1/2$ and $1/3$; not fully closed-formed in the source capture, but structurally tractable via $\int_0^1 \{ax\} \{bx\} dx$ sawtooth-correlation formulas.

WHY IT MATTERS

First appearance of a genuine fractal/self-similar fractional-part construction (Takagi-function flavor) in the I4 menu, and the first to combine TWO independent bases (2 and 3) multiplicatively before integrating. Distinct from all catalogued Beta/digamma/Mellin/Binet/polylog I4 families - the difficulty is entirely in taming an infinite product of self-similar sawtooth series, not special-function recognition.

13. Feynman arctan weight logratio

14

EXAMPLE

$$\int_0^1 \ln\left(\frac{2+x}{2-x}\right) \frac{dx}{x\sqrt{1-x^2}}$$

METHOD

Parametrise $I(a) = \int_0^1 \ln \frac{a+x}{a-x} \frac{dx}{x\sqrt{1-x^2}}$. Then $I'(a) = -2 \int_0^1 \frac{dx}{(a^2-x^2)\sqrt{1-x^2}} = -\frac{\pi}{a\sqrt{a^2-1}}$, so $I(a) = \frac{\pi^2}{2} - \pi \arccos(1/a)$ (constant fixed by $I(\infty) = 0$). At $a = 2$: $\arccos(1/2) = \pi/3$.

RESULT

$$\boxed{\frac{\pi^2}{6}} \text{ (verified).}$$

WHY IT MATTERS

Feynman differentiation with the $\frac{1}{x\sqrt{1-x^2}}$ weight; the inner a -derivative kills the $1/x$ and produces the standard $\int_0^1 dx/((a^2-x^2)\sqrt{1-x^2}) = \pi/(2a\sqrt{a^2-1})$. Clean $\pi^2/6$ payoff - a new I4 Feynman template distinct from the $(x^a-1)/\ln x$ one.

14. Floor function telescoping integral with $\pi \sin(\pi x)$ weight

14

EXAMPLE

$$\int_1^\infty \pi \sin(\pi x) \frac{1}{|x||x+1|} dx.$$

METHOD

On each $[n, n+1]$, $|x| = n$ so $\int_n^{n+1} \pi \sin \pi x dx = 2(-1)^n$; the weight telescopes $\frac{1}{n(n+1)} = \frac{1}{n} - \frac{1}{n+1}$. Sum $\sum 2(-1)^n/(n(n+1)) = 2 - 4 \ln 2$.

RESULT

$2 - 4 \ln 2 \approx -0.7726$ (verified numerically).

WHY IT MATTERS

A hard integral whose "trick" is recognising the integrand is piecewise-constant-weighted and reduces to an alternating telescoping series. Distinct from analytic special-function hard integrals.

15. Floor of bounded radical

14

EXAMPLE

$$\int_0^\infty \left\lfloor \sqrt{\frac{20x}{x^2+1}} \right\rfloor dx$$

METHOD

The radicand $20x/(x^2+1)$ peaks at $x = 1$ with value 10, so the floor of $\sqrt{\cdot}$ takes values $\{0, 1, 2, 3\}$. Solve $20x/(x^2+1) = k^2$ for $k = 1, 2, 3$: quadratic $k^2x^2 - 20x + k^2 = 0$, roots $x = (10 \pm \sqrt{100 - k^4})/k^2$. Total length on which $\lfloor \cdot \rfloor \geq k$ is Δx_k ; integral becomes $\sum_{k=1}^3 \Delta x_k$. Telescoping/Vieta: sum of roots = $20/k^2$.

RESULT

$$\Delta x_k = 2\sqrt{100 - k^4}/k^2; \text{ total} = 2\sqrt{99} + \frac{1}{2}\sqrt{84} + \frac{2}{9}\sqrt{19}.$$

WHY IT MATTERS

New "floor on a bounded function" sub-genre for I4 - distinct from Wallis, *Gamma*-reduction, or symmetry tricks. The shape of the integrand (peak-and-decay rational radical) forces a finite sum, and Vieta on the level-set quadratic is the trick.

16. Floor of root exponent golden ratio series

14

EXAMPLE

Find the integer closest to $\int_0^\infty \left(\frac{1}{\phi}\right)^{\lfloor \sqrt[3]{x} \rfloor} dx$, where ϕ is the golden ratio.

CONTINUED

METHOD

The floor $\lfloor x^{1/3} \rfloor = n$ on each shell $x \in [n^3, (n+1)^3)$, of length $(n+1)^3 - n^3 = 3n^2 + 3n + 1$. So the integral $= \sum_{n \geq 0} (3n^2 + 3n + 1) r^n$ with $r = 1/\phi$. Using $\sum r^n = 1/(1-r)$, $\sum nr^n = r/(1-r)^2$, $\sum n^2 r^n = r(1+r)/(1-r)^3$ and the golden identities $r(1+r) = 1$, $1-r = 1/\phi^2$, the sum collapses to $3\phi^6 + 3\phi^3 + \phi^2$.

RESULT

$3\phi^6 + 3\phi^3 + \phi^2 \approx 69.16$, closest integer 69. (A first parse mis-estimated ~ 47 ; the correct value is ~ 69.16 .)

WHY IT MATTERS

"Floor of a root in the exponent turns an improper integral into an arithmetic-times-geometric series", with a golden-ratio base chosen so $r(1+r) = 1$ makes the second-moment sum collapse to a clean power of ϕ . Distinct from the existing I4 $\Gamma/B/\text{Mellin}/\text{Binet}$ families - the payoff is a finite-coefficient geometric series and the disguise is the cube-root floor, with a "nearest integer" answer convention.

17. Fractional part of log integral via exponential integral

14

EXAMPLE

$$\int_0^1 \ln(\{\ln x\}) dx$$

METHOD

Sub $t = -\ln x$ turns this into $\int_0^\infty \ln(\{-t\}) e^{-t} dt$. Use $\{-t\} = 1 - \{t\}$, decompose $[0, \infty)$ into unit cells, sum the geometric factor $\sum e^{-n} = e/(e-1)$, leaving a fixed integral that reduces to $\gamma - \text{Ei}(1)$.

RESULT

$$\frac{\gamma - \text{Ei}(1)}{e-1} \approx -0.76699.$$

WHY IT MATTERS

First fractional-part-of-a-transcendental-function integral in the corpus. Combines a floor/fractional-part periodic decomposition, a reflection identity for fractional parts of negatives, and a bridge to the Ei/γ family. Genuinely new mechanism for I4.

18. Gamma reflection to sinh power integral

14

EXAMPLE

$$\int_0^\infty x \Gamma^2\left(1 + \frac{ix^2}{\pi}\right) \Gamma^2\left(1 - \frac{ix^2}{\pi}\right) dx$$

METHOD

Use $|\Gamma(1+iy)|^2 = \Gamma(1+iy)\Gamma(1-iy) = \pi y / \sinh(\pi y)$ with $y = x^2/\pi$, collapsing the Gamma product to $x^2 / \sinh(x^2)$. Squaring and multiplying by x gives $x^5 / \sinh^2(x^2)$; substituting $u = x^2$ reduces this to $\frac{1}{2} \int_0^\infty u^2 / \sinh^2 u du$, which expands via $1/\sinh^2 u = 4 \sum_{n \geq 1} n e^{-2nu}$ into $\sum 1/n^2 = \pi^2/6$.

RESULT

$$\frac{\pi^2}{12} \approx 0.8225 \text{ (verified numerically).}$$

WHY IT MATTERS

A new I4 mechanism - complex-argument Gamma-product reflection collapsing to a hyperbolic function, then a second reduction of a $u^2 / \sinh^2 u$ integral via its exponential series to a Basel-sum zeta value. Distinct from the catalogued Beta/digamma/Mellin/Binet I4 families, which don't involve complex-argument Gamma reflection.

19. Gaussian times odd power cosine via product to sum

14

EXAMPLE

$$\int_0^\infty e^{-x^2} \cos^3(2x) dx$$

METHOD

$\cos^3(2x) = \frac{3}{4} \cos(2x) + \frac{1}{4} \cos(6x)$. Use $\int_0^\infty e^{-x^2} \cos(ax) dx = \frac{\sqrt{\pi}}{2} e^{-a^2/4}$ on each frequency ($a = 2, 6$).

RESULT

$$\frac{\sqrt{\pi}}{8} (3e^{-1} + e^{-9}) = \frac{\sqrt{\pi}(1+3e^8)e^{-9}}{8} \approx 0.2445$$

CONTINUED

WHY IT MATTERS

"Hard Integral" register combining the Gaussian-cosine Laplace-type formula with power-reduction. Distinct from existing Gaussian entries because the odd power forces a product-to-sum split into TWO frequencies, each evaluated by the same closed form.

20. Glasser master theorem x minus π over x

14

EXAMPLE

$$\int_0^\infty \frac{x^2 \cos\left(\frac{\pi}{x} - x\right) \cos\left(\sqrt{\left(x - \frac{\pi}{x}\right)^2 + 2}\right)}{x^4 - (2\pi - 1)x^2 + \pi^2} dx.$$

METHOD

Recognise the denominator $x^4 - (2\pi - 1)x^2 + \pi^2 = x^2\left((x - \frac{\pi}{x})^2 + 1\right)$ and that $\cos(\frac{\pi}{x} - x) = \cos u$ with $u = x - \frac{\pi}{x}$. Glasser's master theorem ($\int_0^\infty F(x - a/x) dx = \int_{-\infty}^\infty F(u) du$ for $a > 0$) folds the integrand onto $u \in (-\infty, \infty)$, reducing to $\frac{1}{2} \int_{-\infty}^\infty \frac{\cos u \cos \sqrt{u^2 + 2}}{u^2 + 1} du$ - an even, decaying Cauchy-type kernel evaluable by contour / Bessel methods.

RESULT

No clean elementary form; numerically ≈ 0.31297 (the Glasser-folded kernel integrates to ≈ 0.62311).

WHY IT MATTERS

Flagship I4 disguise: an intimidating quartic denominator is secretly $x^2((x - \frac{\pi}{x})^2 + 1)$, unlocking Glasser's master theorem. Cousin of the logged Cauchy-Schlömilch $\int \cos(x - \pi/x) e^{-(x-1/x)^2}$ entry but here the shift is π/x (not $1/x$) and the collapse comes from the master-theorem substitution rather than a Gaussian. The reusable signal is a rational denominator that factors as $x^2 \cdot$ (quadratic in $x - a/x$).

21. King reflection on antisymmetric square root reciprocal integrand

14

EXAMPLE

$$\int_0^1 \frac{\sqrt{1/x} - \sqrt{1/(1-x)}}{1-2x} dx$$

METHOD

under $x \rightarrow 1 - x$ the two sqrt terms swap and $1 - 2x \rightarrow -(1 - 2x)$, so the integrand is invariant; sub $x = \sin^2 \theta$ finishes it.

RESULT

a clean $\log/\pi$ -free constant

WHY IT MATTERS

King's property applied to a reciprocal-sqrt difference where the denominator sign-flip is exactly cancelled by the numerator swap.

22. Lambert W times log integral

14

EXAMPLE

$$\int_0^\infty W\left(\frac{1}{x^2}\right) \ln(x) dx$$

METHOD

substitute $w = W(1/x^2)$ (so $x = e^{-w/2}/\sqrt{w}$) or use the inverse $1/x^2 = we^w$; reduces to a Gamma/digamma-weighted integral in w .

RESULT

closed form via Γ', ψ (numeric $\sim$ small; verify before reuse)

WHY IT MATTERS

a Lambert-W integrand becomes elementary under $w = W(\cdot)$, turning dx into $e^{-w/2}$ -type weights.

23. Ln arctan over 1 minus u^2 Catalan**14****EXAMPLE**

$$\int_{-\pi/2}^0 \frac{\ln(\cos x) \arctan(\cos x)}{\sin x} dx$$

METHOD

Sub $u = \cos x$ giving $\int_0^1 \frac{\ln u \arctan u}{1-u^2} du$; expand $\arctan u = \sum (-1)^k u^{2k+1}/(2k+1)$ and $\ln u/(1-u^2) = -\sum u^{2n} \ln u$; double-sum recognises Catalan's constant $G = \sum (-1)^k/(2k+1)^2$.

RESULT

Closed form involves G , $\ln 2$, and π^2 (Catalan-constant target).

WHY IT MATTERS

Catalan's constant G joins the Hard-integral menu alongside Li_2 , $\zeta(3)$, γ . The "double-series leading to alternating sum of reciprocal squares" structure is the signature.

24. Lnsq arctansq double maclaurin on unit interval**14****EXAMPLE**

$$\int_0^1 \ln^2(x) (\arctan x)^2 dx$$

METHOD

Expand $(\arctan x)^2 = \sum_{n \geq 1} c_n x^{2n}$ as the Cauchy product of the arctan Maclaurin series. Integrate term-by-term using $\int_0^1 x^{2n} \ln^2 x dx = 2/(2n+1)^3$. The resulting double sum reduces to a combination of Catalan's constant G , π^3 , $\zeta(3)$, and $\ln 2$.

RESULT

Closed form involving G , π^3 , $\zeta(3)$, $\ln 2$ (numerically ≈ 0.146).

WHY IT MATTERS

Existing I4 menu has β -differentiation and Mellin reductions, but the product of two transcendently-disjoint factors ($\ln^2$ and $\arctan^2$) on $[0, 1]$ requires a different route - series-product then term-wise integration. Catalan's constant emerging from a "clean-looking" integrand is the satisfaction.

25. Log sine poisson kernel fourier integral**14****EXAMPLE**

$$\int_0^{2\pi} \frac{\ln(2 \sin(x/2))}{5/4 - \cos x} dx$$

METHOD

Expand $\ln(2 \sin(x/2)) = -\sum_{n \geq 1} \cos(nx)/n$ (classical Fourier series) and recognise $1/(5/4 - \cos x)$ as (a multiple of) the Poisson kernel with $r = 1/2$, whose own cosine series is geometric in r^n . Multiply the two series and integrate term-by-term over a full period so only matching-frequency products survive (orthogonality), collapsing the double sum to a single geometric series in r .

RESULT

reduces to a closed form in terms of $\ln 2$ and powers of $1/2$ (not fully hand-evaluated in the source capture).

WHY IT MATTERS

A genuinely new I4 family - combines a Fourier-series identity for $\ln \sin$ with Poisson-kernel orthogonality, distinct from the existing Euler/reflection, Beta-function, and Vardi/Malmsten I4 menus.

26. Log squared ratio of shifted sines**14****EXAMPLE**

$$\int_0^{\pi/3} \ln^2 \left(\frac{\sin x}{\sin(x + \pi/3)} \right) dx$$

METHOD

The substitution $x \rightarrow \pi/3 - x$ swaps numerator/denominator and negates the log, so $\ln^2$ is symmetric. Splits into log-sine / Clausen-function integrals with $\zeta(2)$ values.

CONTINUED

RESULT

$$\pi^3/54$$

WHY IT MATTERS

A log-squared-of-a-sine-ratio with $\pi/3$ self-similarity - distinct from the existing I4 Euler $\int \ln \sin$ reflection and $\ln^2 x \arctan^2 x$ Catalan patterns. The clean $\pi^3/54$ payoff is the tell.

27. Nascent delta lorentzian kernel zero sum

14

EXAMPLE

$$\lim_{n \rightarrow \infty} \int_{-\infty}^{\infty} \frac{n}{1 + n^2(x^4 - 1)^2} dx$$

METHOD

Recognize $\frac{n}{1+n^2u^2}$ as a nascent (Cauchy/Lorentzian) delta function with $\int \frac{n}{1+n^2u^2} du = \pi$ for every n , concentrating onto the zeros of $u = g(x)$ as $n \rightarrow \infty$; near each simple zero x_0 of g , the local contribution is $\pi/|g'(x_0)|$. Sum over the real zeros of $g(x) = x^4 - 1$ (namely $x = \pm 1$).

RESULT

$$\pi/2.$$

WHY IT MATTERS

First appearance of the "weakly-converging Lorentzian/Cauchy kernel = sum of weighted Dirac deltas at the zeros of an inner function" mechanism in the corpus - a distinct special-limit trick from the existing Beta/Gamma/Mellin/Binet/reflection I4 families, closer to a physics-style distributional argument than an algebraic identity.

28. Nested continued fractions resolved to golden/silver ratios before integrating

14

EXAMPLE

$$\int_0^1 \frac{\ln\left(1 + \frac{x}{1 + \frac{x}{1 + \dots}}\right) - \ln\left(x + \frac{1}{x + \frac{1}{x + \dots}}\right)}{x} dx.$$

METHOD

Solve each continued fraction by its fixed-point equation: $y = \frac{x}{1+y} \Rightarrow 1+y = \frac{1+\sqrt{1+4x}}{2}$; $z = x + \frac{1}{z} \Rightarrow z = \frac{x+\sqrt{x^2+4}}{2}$. The numerator becomes $\ln \frac{1+\sqrt{1+4x}}{2} - \ln \frac{x+\sqrt{x^2+4}}{2}$; integrate $/x$ over $[0, 1]$ (Feynman parameter or numeric).

RESULT

≈ -0.30 (numeric; not a tidy closed form).

WHY IT MATTERS

Continued-fraction sibling of the existing nested-radical fixed-point family - the insight is the SAME (treat the self-similar block as a fixed point) but the surface is a CF, and there are two of them differenced under one log. Good Hard "intimidating self-reference" template.

29. Oscillatory log integral by parameter differentiation

14

EXAMPLE

$$\int_0^\infty \frac{\sin^3 x \log x}{x} dx.$$

METHOD

Expand $\sin^3 x = (3 \sin x - \sin 3x)/4$, introduce a parameter integral only in a convergent difference, and use scaling/differentiation to isolate logarithmic constants.

RESULT

Conditional-convergence closed form; generate only with a careful convergence-aware solution.

WHY IT MATTERS

Hard integrals can be oscillatory and improper; the trick is not brute force but packaging divergent-looking pieces into a convergent parameter difference.

30. Piecewise min of trig functions

14

EXAMPLE

$$\int_{3\pi/4}^{5\pi/4} \min(\sin x, \cos x, \tan x) dx$$

METHOD

Partition the interval by the crossings $\sin = \cos$, $\sin = \tan$, $\cos = \tan$, identify which of the three is the minimum on each sub-interval (on $[3\pi/4, 5\pi/4]$, $\tan$ stays bounded and is typically the minimum on most of the range), then integrate piecewise.

RESULT

Not transcribed.

WHY IT MATTERS

Genuinely fresh I4 flavor. The existing Hard-integral menu has competition-style symmetry tricks, β -function reductions, and ugly-trig camouflage — but no min/max of trig functions piecewise. Rewards the analytical skill of identifying envelope segments before integrating.

31. Polylog ladder antiderivative

14

EXAMPLE

$$\int_0^1 \frac{\text{Li}_3(x) - \text{Li}_3(-x)}{x} dx$$

METHOD

Use $\frac{d}{dx} \text{Li}_{s+1}(x) = \text{Li}_s(x)/x$, so the antiderivative is $\text{Li}_4(x) - \text{Li}_4(-x)$; evaluate $\text{Li}_4(1) - \text{Li}_4(-1) = \zeta(4) + \frac{7}{8}\zeta(4) = \frac{15}{8}\zeta(4)$.

RESULT

$$\frac{\pi^4}{48} \text{ (i.e. } \frac{15}{8}\zeta(4)\text{)}$$

WHY IT MATTERS

The $1/x$ factor is precisely the polylogarithm ladder operator, turning the integral into a trivial step up from Li_3 to Li_4 . The odd combination then selects odd terms, giving the $\frac{15}{8}$ rational multiple of $\zeta(4)$. Rewards fluency with polylog calculus over series manipulation.

32. Product of shifted double factorials collapses to π via Gamma reflection (under $\ln^3$)

14

EXAMPLE

$$\int_0^1 \ln^3((-x-1)!!(x-2)!!(x-1)!!(-x)!!) dx.$$

METHOD

Write the analytic double factorial via Γ ; the four shifted factors pair through Euler reflection $\Gamma(z)\Gamma(1-z) = \pi/\sin \pi z$, so the product inside the log is the constant π , independent of x . Integrand becomes $\ln^3 \pi$.

RESULT

$$\ln^3 \pi \approx 1.500.$$

WHY IT MATTERS

"Ugly x -dependent argument is secretly constant" hard-integral flavour, powered by Gamma reflection on double-factorials. Distinct from the log-sine reflection family.

33. Reciprocal digamma improper symmetry

14

EXAMPLE

$$\int_{-\infty}^{\infty} ((\psi(1/z + 1) + \gamma)^{-2} + 1)^{-1} dz.$$

METHOD

Name the special-function core, pair z with $-z$, and use digamma recurrence/reflection identities before considering direct antiderivatives.

CONTINUED

WHY IT MATTERS

In a whole-line special-function integral, functional equations and involutive symmetry are the first tools, not symbolic integration.

34. Sophomores dream x to x minus x over $x \ln x$

I4

EXAMPLE

$$\int_0^1 \frac{x^x - x}{x \ln x} dx.$$

METHOD

$x^x = e^{x \ln x} = \sum_{k \geq 0} \frac{(x \ln x)^k}{k!}$, so integrand $= \sum_{n \geq 1} \frac{(x \ln x)^{n-1}}{n!}$; with $\int_0^1 x^a (\ln x)^b dx = \frac{(-1)^b b!}{(a+1)^{b+1}}$ the integral collapses to $\sum_{n \geq 1} \frac{(-1)^{n-1}}{n n^n}$.

RESULT

$$\sum_{n \geq 1} \frac{(-1)^{n-1}}{n n^n} \approx 0.8224.$$

WHY IT MATTERS

Sophomore's-dream relative - the $\frac{x^x - x}{x \ln x}$ shell turns the standard $\int_0^1 x^x$ series into a $1/(n n^n)$ series. Distinct from the plain $\int_0^1 x^x$ (06 - 22) and $\int_0^1 x^{-x}$ entries.

35. Trigamma of floor reciprocal telescoping integral

I4

EXAMPLE

$$\int_0^1 \psi_1([1/x] + 1) dx, \text{ where } \psi_1 \text{ is the trigamma function.}$$

METHOD

Substitute on dyadic-style shells: $[1/x] = n$ for $x \in (\frac{1}{n+1}, \frac{1}{n}]$, an interval of length $\frac{1}{n} - \frac{1}{n+1} = \frac{1}{n(n+1)}$. So the integral $= \sum_{n \geq 1} \psi_1(n+1) \cdot \frac{1}{n(n+1)}$. Use $\psi_1(n+1) = \zeta(2) - \sum_{k=1}^n 1/k^2$ and Abel summation / the series identity $\sum_{n \geq 1} \frac{\psi_1(n+1)}{n(n+1)}$, which collapses via telescoping of $1/(n(n+1)) = 1/n - 1/(n+1)$ against the partial sums of $1/k^2$. Reduces to a combination of $\zeta(2)$ and $\zeta(3)$.

RESULT

Closed form in $\zeta(2) = \pi^2/6$ and $\zeta(3)$ (specifically $\zeta(2) - \zeta(3)$ type combination after telescoping).

WHY IT MATTERS

A new I4 floor flavor where the floor sits inside a special function argument (trigamma), not inside the integrand value or exponent. The shell decomposition $[1/x] = n$ produces the harmonic weight $1/(n(n+1))$ which telescopes against the trigamma's defining tail. Distinct from the floor-of-bounded-radical and floor-in-exponent I4 entries.

36. Triple log nielsen Euler sum

I4

EXAMPLE

$$\int_0^1 \frac{\ln x \ln(1-x) \ln(ex - ex^2)}{x} dx.$$

METHOD

$\ln(ex(1-x)) = 1 + \ln x + \ln(1-x)$, splitting into Nielsen/Euler integrals $\int_0^1 \frac{\ln^a x \ln^b(1-x)}{x} dx$ evaluated by $\zeta(2), \zeta(3), \zeta(4)$.

RESULT

rational combination of $\zeta(2), \zeta(3), \zeta(4)$ (i.e. $\pi^2, \zeta(3), \pi^4$).

WHY IT MATTERS

Flagship hard-integral "decompose a product of logs into a basis of Euler sums" pattern; the e inside the log seeds the lone $\zeta(3)$ via the constant-1 term.

Learned variants - Derivative 01 - Easy

10 distinct screenshot-learned variants are documented with their example, method, result when recorded, and distinguishing lesson.

1. Book cover chord related rate (law of cosines)

D1

EXAMPLE

covers 25 cm, opening angle increasing 0.1 rad/s; rate the edge-to-edge distance grows at 90 deg.

METHOD

chord $d = 2 \cdot 25 \sin(\theta/2) = 50 \sin(\theta/2)$; $\dot{d} = 25 \cos(\theta/2)\dot{\theta}$; at $\theta = \pi/2$, $\cos(\pi/4) = \frac{\sqrt{2}}{2}$.

RESULT

$1.25\sqrt{2} \approx 1.77$ cm/s

WHY IT MATTERS

book/hinge related-rates reduce to the chord formula $2r \sin(\theta/2)$, not raw law-of-cosines grinding.

2. Derivative problem that is really simplification

D1

EXAMPLE

$f(x) = \frac{(\sin x + e^{\cos x})(\cos^2 x + \sin^2 x)}{\csc^2 x}$, find $f'(\pi/2)$.

METHOD

Simplify first: $\cos^2 x + \sin^2 x = 1$ and $1/\csc^2 x = \sin^2 x$, so $f(x) = (\sin x + e^{\cos x}) \sin^2 x$.

RESULT

Simple value after product/chain rule; exact computation is secondary to the simplification lesson.

WHY IT MATTERS

Easy derivative screenshots often contain noisy identities whose purpose is to reward simplification before differentiating.

3. Half angle ratio radical collapse

D1

EXAMPLE

$g(x) = \tan^{-1} \sqrt{\frac{1-\cos x}{1+\cos x}}$; find $g'(1)$.

METHOD

$\sqrt{(1-\cos x)/(1+\cos x)} = \tan(x/2)$ on $(0, \pi)$, so $g(x) = x/2$ and $g'(x) = 1/2$ identically.

RESULT

1/2

WHY IT MATTERS

D1 hidden-identity collapse using the tangent half-angle ratio identity; the arctan and radical are pure cosmetics.

4. Leibniz with reciprocal limits plus complementary arctan

D1

EXAMPLE

$f(x) = \int_{1/x}^x \frac{\arctan t}{t} dt$ for $x > 0$; find $f'(2026)$.

METHOD

Leibniz on BOTH moving limits gives $f'(x) = \frac{\arctan x}{x} + \frac{\arctan(1/x)}{x}$ (the lower limit contributes + because $\frac{d}{dx}(1/x) = -1/x^2$ cancels the sign against the $1/t$). The complementary identity $\arctan x + \arctan(1/x) = \frac{\pi}{2}$ then collapses everything to $f'(x) = \frac{\pi}{2x}$ identically.

CONTINUED

RESULT

$$\frac{\pi}{4052} \approx 0.000775319$$

WHY IT MATTERS

Distinct from the catalogued two-moving-limits Leibniz entry — the point here is the RECIPROCAL pairing $[1/x, x]$, engineered so the integrand's two contributions are complementary. Reciprocal structure is a live house motif (see also $t + 1/t$ in the same batch).

5. Manhattan (L^1) distance between reflected curves $\ln x$ and e^x

D1

EXAMPLE

minimise the L^1 distance between a point on $y = \ln x$ and a point on $y = e^x$.

METHOD

The curves are mirror images across $y = x$; the closest L^1 approach is governed by where the slope equals 1, giving the points $(0, 1)$ on e^x and $(1, 0)$ on $\ln x$, distance $|1| + |1| = 2$.

RESULT

2.

WHY IT MATTERS

Reflection-symmetry optimisation with a Manhattan metric; a geometric Easy-derivative anchor distinct from clean-eval cascades.

6. Nested chain rule at cosmetic point

D1

EXAMPLE

$f(x) = \sin(\cos(1 + x^2))$; find $f'(\pi - 1)$.

METHOD

Apply the chain rule from the outside inward, then evaluate; do not assume a complicated-looking point must force a named-angle simplification.

RESULT

$$-2(\pi - 1) \cos(\cos(1 + (\pi - 1)^2)) \sin(1 + (\pi - 1)^2).$$

WHY IT MATTERS

Some Daily Integral easy derivatives test disciplined nesting rather than a hidden cancellation.

7. Normal line intercept under value/derivative constraints (easy derivative)

D1

EXAMPLE

$f(x) = e^{ax^2+bx+1}$ with $f(1) = f(0) = f'(0)$; find the x -intercept of the line through $(1, f(1))$ perpendicular to the tangent there.

METHOD

use the three constraints to pin a, b ; tangent slope $f'(1)$; normal slope $-1/f'(1)$; solve the normal's x -intercept.

RESULT

a clean rational/decimal.

WHY IT MATTERS

an "easy" derivative whose work is really algebra (solve for parameters) plus the normal-line geometry, not differentiation difficulty.

8. Related rates geometric ratio chain

D1

EXAMPLE

The incircle of a square changes area at $3 \text{ m}^2/\text{s}$ when it is 10 m^2 . Find the rate of change of the volume of the circumsphere of the cube built on that square.

METHOD

Chain geometric ratios: incircle area $\frac{\pi a^2}{4} \rightarrow a$; cube side $a \rightarrow$ circumsphere radius $R = \frac{a\sqrt{3}}{2} \rightarrow V = \frac{\sqrt{3}}{2}\pi a^3$. Relate $\dot{V}$ to $\dot{A}_o$ through a .

RESULT

$$\dot{V} = 18\sqrt{30/\pi} \approx 55.6 \text{ m}^3/\text{s}.$$

WHY IT MATTERS

An easy-slot related-rates problem whose difficulty is the geometric-ratio chaining (incircle $\rightarrow$ circumsphere across square $\rightarrow$ cube), not the calculus.

CONTINUED

9. Sum of midpoints of sign intervals of a reciprocal combination

D1

EXAMPLE

$f(x) = 2x^2 - 2x + 4$; find the sum of the midpoints of all intervals on which $\frac{1}{f(x)} + \frac{1}{f'(x)} + \frac{1}{f''(x)} < 0$.

METHOD

$f > 0$ always (negative discriminant), $f' = 4x - 2$ (sign-change at $x = \frac{1}{2}$), $f'' = 4$. Combine over a common denominator, build the sign chart from the numerator roots and the $x = \frac{1}{2}$ point, then average interval endpoints.

RESULT

a small rational from the sign chart (work the cubic numerator; the special point sits at $x = \frac{1}{2}$).

WHY IT MATTERS

A Derivative slot that is really a sign-analysis puzzle - exercises f, f', f'' together rather than a single differentiation. Fresh Easy-Derivative flavor.

10. Tangent line meeting a second line at a fixed angle (angle between lines)

D1

EXAMPLE

$f(x) = \sin(\frac{\pi x}{2})\cos(\frac{\pi x}{4})$; the tangent at $x = 4$ meets a line ℓ at 60° with $m_\ell > m_{\tan}$. Find m_ℓ .

METHOD

compute $f'(4) = -\pi/2$ (product rule; clean because $\sin(2\pi) = 0, \cos(\pi) = -1$), then solve $\tan 60^\circ = \left| \frac{m_\ell - m_t}{1 + m_\ell m_t} \right|$ for the larger root.

RESULT

$$m_\ell = \frac{\sqrt{3} - \pi/2}{1 + (\pi/2)\sqrt{3}} \approx \mathbf{0.0433} \text{ (verified)}.$$

WHY IT MATTERS

Marries a clean-evaluation derivative with the analytic-geometry angle-between-lines formula - a new Easy-derivative surface form where the calculus is one step and the twist is the geometry.

Learned variants - Derivative 02 - Medium

26 distinct screenshot-learned variants are documented with their example, method, result when recorded, and distinguishing lesson.

1. Abel integral equation (medium derivative)

D2

EXAMPLE

$y(x)$ continuous with $\int_0^x \frac{y(t)}{\sqrt{x-t}} dt = x^2$ for all $x > 0$; find $\pi y'(4)$.

METHOD

this is an Abel (half-order) integral equation; the convolution with $x^{-1/2}$ is a fractional integral, inverted via another fractional/Abel operator (or Laplace transform: $\hat{y}(s)\sqrt{\pi/s} = 2/s^3$).

RESULT

a clean closed form (involves $\sqrt{\cdot}$ / rational).

WHY IT MATTERS

the site files convolution/Abel integral equations under "Derivative"; recognising the $1/\sqrt{x-t}$ kernel as a fractional integral is the whole trick. Strong DE3/D3 crossover candidate too.

2. Anger function disguised Bessel j0 high order derivative

D2

EXAMPLE

The Anger function $J_u(x) = \frac{1}{\pi} \int_0^\pi \cos(u\theta - x \sin \theta) d\theta$. Compute $J_0'''(0)$.

METHOD

With $u = 0$, $J_0(x) = \frac{1}{\pi} \int_0^\pi \cos(x \sin \theta) d\theta$ is exactly the standard integral representation of the Bessel function $J_0(x)$. Differentiating under the integral sign four times gives $J_0'''(0) = \frac{1}{\pi} \int_0^\pi \sin^4 \theta d\theta = \frac{1}{\pi} \cdot \frac{3\pi}{8}$ (Wallis), matching the x^4 Maclaurin coefficient of J_0 .

RESULT

$J_0'''(0) = \frac{3}{8}$ (verified by direct differentiation under the integral).

WHY IT MATTERS

Distinct from the already-logged "bessel-J0-disguised-zero" I2 entry (which is a cancellation trick) - here the Anger-function integral representation is the disguise for J_0 itself, and the payoff is extracting a specific high-order derivative at 0 via Wallis integrals / the Maclaurin series, not a cancellation.

3. Antiderivative times derivative sign integer count

D2

EXAMPLE

For $f(x) = 2 \sin^2 x - 4 \cos x - 1$, with antiderivative $F(x) = \int f dx$ taken with $C = 0$, count (and/or sum) the integer points x at which $F(x) \cdot f'(x) > 0$.

METHOD

Build $F(x) = x - \sin x \cos x - 4 \sin x$ and $f'(x) = 4 \sin x \cos x + 4 \sin x = 4 \sin x(1 + \cos x)$; analyse the sign of the product on the integers in the stated window. Combines an explicit antiderivative (with a fixed constant) against the derivative of the same function - a sign-pattern hunt, not a closed-form evaluation.

RESULT

depends on the stated integer window (problem as screenshotted was partly underspecified).

WHY IT MATTERS

Distinctive hybrid - most dailyintegral derivative slots ask for a single $f^{(n)}(a)$ value or a stationary point; this one couples the antiderivative and the derivative of one function through an inequality and then quantifies over integers. Worth keeping as a "sign-analysis over a lattice" derivative flavour.

4. Derivative expression sign inequality, sum of integer solutions

D2

EXAMPLE

$f = x^3/3 + 2x^2 - 9x$; sum all integers x with $\frac{(f'(x)-1)(f''(x)+f'(x))}{f(x)f''(x)} < 0$.

METHOD

compute f', f'' (here $f' = x^2 + 4x - 9$, $f'' = 2x + 4$); build a sign chart of the rational expression across all real roots/poles; tally integer x in the negative regions.

RESULT

(sum of integers in the solution set - compute the sign chart)

WHY IT MATTERS

a derivative problem that is really a rational-inequality sign-chart over the critical numbers of f, f', f'' .

5. Derivative range bounds endpoint values by mean value theorem

D2

EXAMPLE

Let $f : [0, 3] \rightarrow \mathbb{R}$ be differentiable with $\max f' = 8$, $\min f' = -6.5$, and $f(0) = 10$. How many integer values are possible for $f(3)$?

METHOD

The mean value theorem gives $(f(3) - f(0))/3 = f'(c)$ for some $c \in (0, 3)$, so $-6.5 \leq (f(3) - 10)/3 \leq 8$. Convert this slope interval into an endpoint interval and count its integers; Darboux-compatible derivatives can realise each intermediate average.

RESULT

$-9.5 \leq f(3) \leq 34$, containing the 44 integers $-9, -8, \dots, 34$.

WHY IT MATTERS

This is a derivative problem whose core is global endpoint control, not differentiation: extrema of f' plus the mean value theorem bound every secant slope.

6. Floor of telescoping decimal sum to 99th power

D2

EXAMPLE

$S_n = \frac{1}{1+10^{-2}} + \frac{2 \cdot 10^{-2}}{1+10^{-4}} + \frac{4 \cdot 10^{-6}}{1+10^{-8}} + \dots$ (n terms); find $\lfloor (\lim_{n \rightarrow \infty} S_n)^{99} \rfloor$.

METHOD

With $x = 10^{-1}$ the k -th term is $\frac{2^{k-1}x^{2^{k-2}}}{1+x^{2^k}}$, a telescoping doubling product whose partial sums march to $\frac{1}{1-x^2} = \frac{100}{99} = 1.\overline{01}$. Then $(100/99)^{99} = (1 + 1/99)^{99} \rightarrow e^-$ from below, so the floor is 2.

RESULT

2

WHY IT MATTERS

"Medium Derivative" register that is really a series-limit-then-floor problem. The two hooks: (i) recognising the binary/doubling telescoping that gives $1/(1-x^2)$, and (ii) the $(1 + 1/99)^{99} \approx e \approx 2.718$ punchline that makes the floor 2, not 1. Distinct from existing telescoping-doubling-product entries because of the 99th-power floor twist.

7. Function equals the infinite sum of its own higher derivatives

D2

EXAMPLE

$f(x) = e^{ax}$ with $f(x) = \sum_{n=1}^{\infty} f^{(n)}(x)$; find $f(10)$.

METHOD

$f^{(n)} = a^n e^{ax}$, so the condition is $\sum_{n \geq 1} a^n = a/(1-a) = 1 \Rightarrow a = \frac{1}{2}$ (geometric series, $|a| < 1$). Then $f(10) = e^5$.

RESULT

$e^5 \approx 148.413$ (verified).

WHY IT MATTERS

A self-referential derivative condition solved by a geometric series in the growth rate - clean Medium-derivative flavour, payoff lands on e^5 .

8. Gamma functional equation log derivative telescope

D2

EXAMPLE

Differentiable f satisfies $f(x+1) = xf(x)$ for $x \geq 1$. Let $g(x) = f'(x)/f(x)$. Find $12(g(5) - g(1))$.

METHOD

Recognize the recurrence as the Gamma function's functional equation (no explicit formula for f needed); log-differentiate both sides to get $g(x+1) - g(x) = 1/x$ (the digamma recurrence), then telescope $g(5) - g(1) = \sum_{k=1}^4 1/k$.

RESULT

25.

WHY IT MATTERS

A new D2 mechanism distinct from the logged implicit-differentiation, self-referential-definition, and Leibniz families - the function is never given explicitly, only through a Gamma-like functional equation, and the derivative question is answered purely by differencing logarithmic derivatives into a harmonic telescoping sum.

9. Gauss continued fraction for tan recognition

D2

EXAMPLE

$f(x) = \frac{x}{1 - \frac{x^2}{3 - \frac{x^2}{5 - \frac{x^2}{7 - \dots}}}}$. Find $f'(\pi/4)$.

METHOD

Recognise as the Gauss/Euler continued fraction for $\tan x$ (the canonical CF expansion). So $f(x) = \tan x$, $f'(\pi/4) = \sec^2(\pi/4) = 2$.

RESULT

2

WHY IT MATTERS

Pattern-recognition of a classical continued-fraction identity. Same family as Maclaurin-series-as-product recognition (recognise the closed form, then differentiate trivially) but with CFs instead of power series. Likely sibling problems: CF for e^x , for $\sqrt{1+x}$, etc.

10. Leibniz product nth derivative at zero

D2

EXAMPLE

Let $f(x) = e^x g(x)$. Given $g(0) + g'''(0) = 5$ and $g'(0) + g''(0) = 2$, find $f'''(0)$.

METHOD

Leibniz rule for the n -th derivative of a product: $f^{(n)}(x) = \sum_{k=0}^n \binom{n}{k} (e^x)^{(k)} g^{(n-k)}(x) = \sum_{k=0}^n \binom{n}{k} e^x g^{(n-k)}(x)$. At $x = 0$: $f'''(0) = \sum_{k=0}^3 \binom{3}{k} g^{(3-k)}(0) = g'''(0) + 3g''(0) + 3g'(0) + g(0) = [g(0) + g'''(0)] + 3[g'(0) + g''(0)] = 5 + 3 \cdot 2 = 11$.

RESULT

11

WHY IT MATTERS

A "Leibniz product rule for the n -th derivative, evaluated at a point where only certain combinations of the factor's derivatives are given" D2 flavor. The binomial coefficients $\binom{3}{k} = 1, 3, 3, 1$ allow the given sums $g(0) + g'''(0)$ and $g'(0) + g''(0)$ to collapse the full expression. Distinct from existing D2 implicit-diff and log-diff entries.

11. Limit derivative via log Taylor

D2

EXAMPLE

$f(x) = \left(\frac{\ln x}{x-1}\right)^{1/(x-1)}$ with $f(1) = e^{-1/2}$; find $f'(1)$.

METHOD

f is an indeterminate 1^∞ form at $x = 1$; take $\ln f$ and Taylor-expand about $x = 1$: $\ln f = -\frac{1}{2} + \frac{5}{24}(x-1) + \dots$, so $f'(1) = f(1) \cdot \frac{5}{24}$.

RESULT

$$\frac{5}{24}e^{-1/2}$$

WHY IT MATTERS

The derivative exists only as a limit at the defining singular point; you cannot differentiate directly. The slope is read off the linear Taylor coefficient of $\ln f$, requiring expansion of $\ln(\ln x/(x-1))$ to second order. A "derivative at a point defined by continuous extension" twist.

CONTINUED

12. Log quotient split then differentiate at transformed point

D2

EXAMPLE

$f(x) = \ln\left(\frac{x^x \tan x}{e^{x^2/\ln x}}\right)$; find $f'(e^x)$.

METHOD

Split the log before differentiating: $f(x) = x \ln x + \ln \tan x - \frac{x^2}{\ln x}$. Differentiate termwise: $\frac{d}{dx}(x \ln x) = \ln x + 1$; $\frac{d}{dx} \ln \tan x = \sec x \csc x = \frac{2}{\sin 2x}$; and $\frac{d}{dx}(x^2/\ln x) = \frac{2x \ln x - x}{(\ln x)^2}$. Then evaluate the resulting f' at the point $x = e^x$ (i.e. the literal argument e^x), where the $\ln$'s collapse cleanly.

RESULT

Closed form obtained by substituting the argument into f' ; the $\ln(e^x)$ terms simplify the log-pieces to elementary expressions.

WHY IT MATTERS

A Medium-derivative "logarithmic differentiation" anchor whose disguise is a three-factor product/quotient inside a single $\ln$ (a power-tower base x^x , a trig factor, and an exponential with a rational exponent). The skill is to split-then-differentiate rather than differentiate the quotient directly, plus the cosmetic trick of evaluating at a e^x -type argument so the logs telescope. Distinct from existing D2 self-referential and Leibniz entries.

13. Monstrous mostly constant camouflage

D2

EXAMPLE

$f(x) = \frac{9 [\sin(\gamma)]^{99999 \cos(1/x)}}{e^2 + 2\pi - 2026e + \cos^{1000}(\pi/e)} + \sum_{n=1}^{1000} n$. Find $f'(x)$.

METHOD

Nearly everything is an elaborate constant: γ , the whole denominator, and $\sum_1^{1000} n = 500500$ are constants/distractors. The only x -dependent piece is the exponent $99999 \cos(1/x)$ on the constant base $\sin \gamma$, so $f'(x) = \frac{9}{(\text{const})} (\sin \gamma)^{99999 \cos(1/x)} \ln(\sin \gamma) \cdot 99999 \frac{\sin(1/x)}{x^2}$. (Common variant: if the exponent is also constant, f is wholly constant and $f' \equiv 0$.)

RESULT

nonzero log-derivative of the single live term (or 0 in the fully-constant variant). [Parse uncertain - the redesigned card clipped/obscured the exponent; recorded as a flavor, not a verified value.]

WHY IT MATTERS

A "wall of intimidating constants hiding at most one x -dependent factor" trap. Rewards scanning every sub-expression for x before differentiating, discarding all constants (named constants γ, e, π , finite sums, $\cos$ of a constant), and log-differentiating only the surviving term. Extends the D1 "secretly constant" entry to the partially-constant case.

14. Nested trig stationary points sum of y **D2****EXAMPLE**

$f(x) = \tan(\cos(\sin x))$; find the sum of the y -values at the stationary points on $[-\frac{\pi}{2}, \frac{\pi}{2}]$.

METHOD

$f'(x) = \sec^2(\cos(\sin x)) \cdot (-\sin(\sin x)) \cdot \cos x$. Zeros: $\cos x = 0$ ($x = \pm \frac{\pi}{2}$) or $\sin(\sin x) = 0$ ($x = 0$). Evaluate f at each:
 $f(0) = \tan(\cos 0) = \tan 1$; $f(\pm \frac{\pi}{2}) = \tan(\cos 1)$.

RESULT

$\tan 1 + 2 \tan(\cos 1) \approx 2.78$

WHY IT MATTERS

A "find the critical points of a 3-deep nested composition, then sum the function values" D2 - distinct from the high-order-derivative and self-referential entries. Trick is reading off which factor of f' vanishes where.

15. Nth derivative via series coefficient gamma normalization**D2****EXAMPLE**

$f(u, x) = \frac{u - ux - x}{(1-x)^2}$; with $g = \partial_x^{15} f$, find $g(\pi, 0)/\Gamma(15)$.

METHOD

Decompose $f = \frac{u}{1-x} - \frac{x}{(1-x)^2}$; use $\frac{1}{1-x} = \sum x^n$, $\frac{1}{(1-x)^2} = \sum (n+1)x^n$; the x^{15} coefficient $\times 15!$ gives $g(\pi, 0)$, and dividing by $\Gamma(15) = 14!$ yields a clean linear-in- π value.

RESULT

$15\pi - 16$ (i.e. $\Gamma(15)$ -normalized coefficient).

WHY IT MATTERS

A "high-order derivative" framed so $/\Gamma(n)$ cancels the factorial and exposes the generating-function coefficient. Distinct from the Fibonacci-GF (06 - 18 - v2) and partial-fractions (06 - 14) nth-derivative entries.

16. Optimization labelled as derivative monomial constraint**D2****EXAMPLE**

If $x^2 y^3 = 729$ and $x, y > 0$, find $x + y$ for the point (x, y) closest to the origin.

METHOD

Minimize $x^2 + y^2$ subject to $x^2 y^3 = 729$ using logarithms/Lagrange multipliers or a one-variable reduction.

RESULT

Clean algebraic values after balancing derivative conditions.

WHY IT MATTERS

Medium "Derivative" can mean optimization, not merely computing $f'(a)$. Include occasional constrained-optimization problems under the derivative slot.

17. Parametric area bounded by curve tangent axis**D2****EXAMPLE**

f defined on $[-1, 1]$ has $f''(x) = \tan^{-1}(3^x)$. Point P on $y = f(x)$ has x -coordinate $\sin t$. Let A be the area bounded by the curve, the tangent at P , and the y -axis. Find dA/dt at $t = \pi/6$.

METHOD

Write $A = \int_0^{x_P} [f(x_P) + f'(x_P)(x - x_P) - f(x)] dx$. Differentiate w.r.t. x_P : top-limit term cancels (tangent meets curve at $x = x_P$); leaves $f''(x_P) \int_0^{x_P} (x - x_P) dx$... cleaner - $dA/dx_P = -\frac{1}{2} x_P^2 f''(x_P)$ (standard result for tangent-line area to the axis). Chain by $dx_P/dt = \cos t$. At $t = \pi/6$: $x_P = 1/2$, $\cos t = \sqrt{3}/2$, $f''(1/2) = \arctan(\sqrt{3}) = \pi/3$. Answer $-\frac{1}{2} \cdot \frac{1}{4} \cdot \frac{\pi}{3}$.

$$\frac{\sqrt{3}}{2} = -\frac{\pi\sqrt{3}}{48}.$$

RESULT

$$-\frac{\pi\sqrt{3}}{48}$$

WHY IT MATTERS

CONTINUED

Compound parametric construction - a t -parameterised tangent-area function. Builds on existing D2 implicit-diff and self-referential menus, but adds the "tangent-line area cleanly differentiates to $-\frac{1}{2}x^2 f'''$ " structural lemma.

18. Related rates isosceles triangle vertex angle to circumradius

D2

EXAMPLE

An isosceles triangle has two equal sides of length $a = 2.4$. The angle θ between these sides increases at a rate of 0.015 rad/s. Find the rate of change of the circumcircle radius R when [given θ].

METHOD

Law of Sines gives $R = \frac{a}{2 \sin(\theta/2)}$ (relating R to the vertex angle via the base, which subtends angle $\pi - \theta$ isn't needed directly - the base side $b = 2a \sin(\theta/2)$ opposite... the key relation is $R = \text{base}/(2 \sin(\text{opposite angle}))$). Differentiate implicitly with respect to t using the chain rule on $\theta(t)$ to get dR/dt in terms of $d\theta/dt$.

RESULT

Numeric value depending on the (partially captured) θ snapshot; formula is $dR/dt = -\frac{a \cos(\theta/2)}{4 \sin^2(\theta/2)} \cdot \frac{d\theta}{dt}$.

WHY IT MATTERS

A new related-rates D2 sub-flavor distinct from the catalogued "incircle-of-square to circumsphere-of-cube" geometric-ratio-chain and the "book-cover chord, law of cosines" entries - here the driver is the vertex angle of an isosceles triangle and the target is the circumradius via the Law of Sines, a different geometric relation entirely.

19. Series coeff via mixed high partials

D2

EXAMPLE

$\frac{\partial^{134} f}{\partial x^{67} \partial y^{67}} \Big|_{(0,0)} = (b!)^2$ for $f = \frac{1}{1-xy}$; find b .

METHOD

$f = \sum_n (xy)^n$; only the $n = 67$ term survives the symmetric mixed partial at the origin, giving $(67!)^2$.

RESULT

$b = 67$

WHY IT MATTERS

Treats an absurdly high mixed partial as pure Taylor-coefficient extraction - the multivariable cousin of the single-variable "Maclaurin coefficient = derivative" trick. New for the derivative menu.

20. Sum of ALL derivatives at 0 as a geometric series

D2

EXAMPLE

$f(x) = e^{x/3}$; find $\sum_{n=0}^{\infty} f^{(n)}(0)$.

METHOD

$f^{(n)}(x) = (1/3)^n e^{x/3}$ so $f^{(n)}(0) = (1/3)^n$; sum the geometric series $\sum_{n \geq 0} (1/3)^n$. Generalises: for $f = e^{ax}$ with $|a| < 1$, $\sum_n f^{(n)}(0) = \frac{1}{1-a}$.

RESULT

$\frac{1}{1-1/3} = \frac{3}{2}$ (verified).

WHY IT MATTERS

Clean "sum every derivative" framing - reframes a derivative question as a series sum. Nice Medium-Derivative template distinct from high-order-at-a-point.

21. Sum all derivatives polynomial finite sum

D2

EXAMPLE

Let $f(x) = x^3 + 3x^2 - x - 1$. Find $\sum_{n=1}^{\infty} \frac{d^n}{dx^n} f(x)$ at $x = 1.2$.

CONTINUED

METHOD

The series truncates after $n = 3$ since $f^{(n)} = 0$ for $n \geq 4$. Compute $f'(x) = 3x^2 + 6x - 1$, $f''(x) = 6x + 6$, $f'''(x) = 6$. Evaluate at $x = 1.2$: $f'(1.2) = 3(1.44) + 7.2 - 1 = 10.52$, $f''(1.2) = 7.2 + 6 = 13.2$, $f'''(1.2) = 6$. Sum = $10.52 + 13.2 + 6 = 29.72$.

RESULT

29.72

WHY IT MATTERS

An "infinite sum of derivatives" that is secretly a finite sum because the function is a polynomial. The trick is recognising that the series terminates, then computing three derivatives numerically. A new D2 sub-flavor: "sum over all orders, but the function is of finite degree so the series is finite."

22. Two level power tower at a transcendental evaluation point

D2

EXAMPLE

$f(x) = x^{x^{\sin x}}$, find $f'(e)$.

METHOD

Nested logarithmic differentiation; $\ln f = x^{\sin x} \ln x$ with inner $g = x^{\sin x}$, $\ln g = \sin x \ln x$.

RESULT

≈ -2.67 (no clean closed form; messy evaluation point by design).

WHY IT MATTERS

Two stacked variable exponents plus a non-clean eval point - a sharper Medium derivative than the single-tower x^x family.

23. Variable base logarithm change of base

D2

EXAMPLE

$f(x) = \log_{(x^2+3x)}(x^3 + \sin x)$; find $f'(3)$.

METHOD

Change of base $f = \frac{\ln(x^3 + \sin x)}{\ln(x^2 + 3x)}$, then quotient rule. Clean-ish evaluation point $x = 3$ gives base-arg 18, top-arg $27 + \sin 3$.

RESULT

numeric ≈ 0.134 .

WHY IT MATTERS

Differentiating a logarithm whose BASE is also a function of x - a change-of-base-then-quotient-rule flavor not previously in the D2 menu.

24. Variable limits plus parameter Leibniz

D2

EXAMPLE

$f(x) = \int_x^{x^2} \frac{e^{xt}}{t} dt$; find $f'(1)$.

METHOD

Full Leibniz rule with three contributions: (i) upper-limit term $(d/dx)(x^2) \cdot e^{x \cdot x^2} / x^2 = (2/x)e^{x^3}$; (ii) lower-limit term $-1 \cdot e^{x \cdot x} / x = -e^{x^2} / x$; (iii) parameter term $\int_x^{x^2} \partial_x [e^{xt} / t] dt = \int_x^{x^2} e^{xt} dt = (e^{x^3} - e^{x^2}) / x$. At $x = 1$ the three pieces sum to e .

RESULT

$f'(1) = e$

WHY IT MATTERS

Existing D2 has implicit diff and self-referential definitions but not the "both limits depend on x AND the integrand has x as a parameter" trifecta. Forces clean Leibniz bookkeeping; the answer collapsing to e is a satisfying tell.

25. Variable upper limit product collapses via triangular exponent**D2****EXAMPLE**

$f(x) = \left(\prod_{i=1}^x x^i\right)^2$; find $f'(2)$.

METHOD

Collapse via the arithmetic-series exponent $\sum_{i=1}^x i = x(x+1)/2$, giving $f(x) = x^{x^2+x}$, then apply ordinary logarithmic differentiation.

RESULT

$$f'(2) = 64(3 + 5 \ln 2) \approx 413.81.$$

WHY IT MATTERS

Distinct from the D3 "power tower" family and the I2 "infinite product with geometric-series exponent" entry - a finite PRODUCT (not a tower) collapses via a triangular number, a different combinatorial mechanism, then feeds ordinary log-differentiation.

26. Wire cut square circle min area**D2****EXAMPLE**

A wire of length 16 is cut into two pieces forming a square and a circle. For the cut that minimises total area, find the length of the larger piece.

METHOD

Circle piece c , square piece $16 - c$; $A(c) = \frac{c^2}{4\pi} + \left(\frac{16-c}{4}\right)^2$. $A'(c) = \frac{c}{2\pi} - \frac{16-c}{8} = 0 \Rightarrow c = \frac{16\pi}{4+\pi}$ (circle), square piece = $\frac{64}{4+\pi}$.

RESULT

$$\text{larger piece} = \frac{64}{4+\pi} \approx 8.96 \text{ (the square)}$$

WHY IT MATTERS

The canonical "wire into square+circle, minimise area" optimization - the menu lists optimization for D2 but not this archetype. Clean closed form $64/(4+\pi)$; at the minimum the square gets the larger share.

Learned variants - Derivative 03 - Hard

41 distinct screenshot-learned variants are documented with their example, method, result when recorded, and distinguishing lesson.

1. $\{x, -x, 1/x, -1/x\}$ substitution group functional equation D3

EXAMPLE

f real with $\frac{1}{x}f(-x) + f(1/x) = x$ for all $x \neq 0$; find $f'(e)$.

METHOD

Apply the substitutions $x \rightarrow -x, 1/x, -1/x$ to generate a linear system in $f(\pm x), f(\pm 1/x)$; solve for $f(x)$ in closed form, then differentiate.

RESULT

closed-form f then $f'(e)$ (clean target; full solve required).

WHY IT MATTERS

The substitution orbit mixes negation AND reciprocal (order-4 group), unlike the usual single $x \rightarrow 1/x$ trick. New Hard-Derivative template.

2. Alternating odd order derivative operator series D3

EXAMPLE

Let $y(x)$ satisfy $y = y' - y''' + y^{(5)} - y^{(7)} + \dots$ (alternating odd-order derivatives), with $\lim_{n \rightarrow \infty} y^{(n)} = 0$ and $y(0) = 0$. Find $\frac{y'(0)}{y(2)}$.

METHOD

For a trial $y = e^{rx}$ the RHS is $y \sum_{k \geq 0} (-1)^k r^{2k+1} = y \cdot \frac{r}{1+r^2}$ (geometric in r^2), so the operator equation forces $\frac{r}{1+r^2} = 1 \Rightarrow r^2 - r + 1 = 0 \Rightarrow r = e^{\pm i\pi/3}$. The decay condition selects the appropriate combination; with $y(0) = 0$ the solution is $y(x) = Ce^{x/2} \sin(\frac{\sqrt{3}}{2}x)$. Then $y'(0) = C \cdot \frac{\sqrt{3}}{2}$ and $y(2) = Ce \sin(\sqrt{3})$, so the ratio is $\frac{\sqrt{3}}{2e \sin(\sqrt{3})}$.

RESULT

$$\frac{y'(0)}{y(2)} = \frac{\sqrt{3}}{2e \sin \sqrt{3}}.$$

WHY IT MATTERS

A Hard-derivative operator puzzle: an infinite alternating series of odd-order derivatives is secretly a geometric series in the symbol r , giving a quadratic $r^2 - r + 1 = 0$ with complex roots $\rightarrow$ damped sinusoid. Distinct from the "function equals sum of ALL its derivatives" D2 entry (that one used $\sum a^n = a/(1-a)$ with all orders); here only odd orders appear with alternating sign, producing $r/(1+r^2)$ and a 60° complex root.

3. Complex analytic derivative real part D3

EXAMPLE

On the principal branch ($-\pi \leq \arg z \leq \pi$), $f(z) = z \ln z - z - \cos(z^2)$. Find $\Re[f'(1 + \sqrt{3}i)]$.

METHOD

Differentiate holomorphically: $\frac{d}{dz}[z \ln z - z] = \ln z$, so $f'(z) = \ln z + 2z \sin(z^2)$. Evaluate at $z = 1 + \sqrt{3}i$ (clean polar form $|z| = 2$, $\arg = \pi/3$, $z^2 = -2 + 2\sqrt{3}i$) and take the real part.

RESULT

$\Re[f'] = \ln 2 + \Re[2z \sin(z^2)]$ with $\ln z = \ln 2 + i\pi/3$.

CONTINUED

WHY IT MATTERS

The existing derivative menu is entirely real-variable; this is genuine complex differentiation on a stated principal branch with a "take the real part at a polar-clean point" payoff. The $z \ln z - z \rightarrow \ln z$ cancellation is a hidden simplify-before-differentiate, but the complex/branch framing is new.

4. Conjugate radical sum inverse derivative preimage

D3

EXAMPLE

$f(x) = \sqrt{x} + \sqrt{x+1}$; find $(f^{-1}((1 + \sqrt{2})^2))'$.

METHOD

The inverse-function theorem gives $(f^{-1})'(c) = 1/f'(a)$ where $f(a) = c$, so the real work is finding a . Rationalise: $\sqrt{x+1} - \sqrt{x} = 1/f(x)$, so with $s = f(a) = (1 + \sqrt{2})^2 = 3 + 2\sqrt{2}$ one gets $2\sqrt{a} = s - 1/s$; since $1/s = 3 - 2\sqrt{2}$, $s - 1/s = 4\sqrt{2}$, hence $\sqrt{a} = 2\sqrt{2}$ and $a = 8$ exactly. Then $f'(8) = \frac{1}{2\sqrt{8}} + \frac{1}{2 \cdot 3} = \frac{3+2\sqrt{2}}{12\sqrt{2}}$ and invert.

RESULT

$$\frac{12\sqrt{2}}{3 + 2\sqrt{2}} = 36\sqrt{2} - 48 \approx 2.91169 \text{ (verified numerically)}$$

WHY IT MATTERS

The catalogued inverse-derivative D3 entries all have a preimage you can guess ($f(0) = 1$, etc.); here the evaluation point $(1 + \sqrt{2})^2$ is deliberately written in squared form to hide that the preimage is the integer 8, and the only clean route is the conjugate identity $\sqrt{x+1} - \sqrt{x} = 1/f(x)$, which turns "solve $\sqrt{a} + \sqrt{a+1} = s$ " into the one-line $2\sqrt{a} = s - 1/s$. Bonus: the surd $3 + 2\sqrt{2}$ has $1/s = 3 - 2\sqrt{2}$ (unit norm), so every step stays exact and the answer is the tidy $36\sqrt{2} - 48$.

5. Constant arctan pair plus complex log

D3

EXAMPLE

$\frac{\partial^{100}}{\partial x^{50} \partial y^{50}} [\tan^{-1} \frac{x}{y} + \tan^{-1} \frac{y}{x} + \ln(x^2 + y^2)]_{(1,1)} = -\frac{a!}{2^b}$; find $a + b$.

METHOD

$\tan^{-1} \frac{x}{y} + \tan^{-1} \frac{y}{x} = \pi/2$ (constant, drops out); split $\ln(x^2 + y^2) = \ln(x + iy) + \ln(x - iy)$ and collapse to $99! [(1 + i)^{-100} + (1 - i)^{-100}]$ with $(1 \pm i)^{100} = -2^{50}$.

RESULT

$-99!/2^{49}$, so $a + b = 148$

WHY IT MATTERS

Two stacked tricks - annihilate a constant arctan pair, then complexify the remaining log for a 100th-order mixed partial. Distinct from existing complex-exponential high-order-derivative entries (those are single-variable).

6. Cotangent doubling telescope infinite tan series derivative

D3

EXAMPLE

Let $f(x) = \sum_{n=1}^{\infty} \frac{1}{2^n} \tan\left(\frac{x}{2^n}\right)$. Find $f'(\pi/3)$.

METHOD

Use the cotangent double-angle identity $\cot \theta - 2 \cot 2\theta = \tan \theta$, i.e. $\frac{1}{2^n} \tan(x/2^n) = \frac{1}{2^n} \cot(x/2^n) - \frac{1}{2^{n-1}} \cot(x/2^{n-1})$. The series telescopes: $\sum_{n=1}^N \rightarrow \frac{1}{2^N} \cot(x/2^N) - \cot x$, and as $N \rightarrow \infty$, $\frac{1}{2^N} \cot(x/2^N) \rightarrow 1/x$. So $f(x) = \frac{1}{x} - \cot x$ in closed form, and $f'(x) = -\frac{1}{x^2} + \csc^2 x$.

RESULT

$$f'(\pi/3) = -\frac{9}{\pi^2} + \frac{4}{3}.$$

WHY IT MATTERS

A new D3 family - an infinite series that looks like it needs termwise differentiation is instead first collapsed via a cotangent-doubling telescoping identity into a two-term closed form, and only then differentiated. Distinct from the catalogued ODE-to-series and fixed-point-iteration D3 flavors; the telescoping trick here is trigonometric identity-driven rather than algebraic/partial-fraction-driven.

7. Cyclic order three functional equation derivative

D3

EXAMPLE

f differentiable for $x \notin \{0, 1\}$ with $f(x) + 2f\left(\frac{x-1}{x}\right) = x$ for all valid x . Evaluate $f'(2)$.

METHOD

The map $x \mapsto \frac{x-1}{x}$ has order 3: $x \rightarrow \frac{x-1}{x} \rightarrow -\frac{1}{x-1} \rightarrow x$. Write the equation at all three orbit points to get a 3×3 linear system in $f(x), f\left(\frac{x-1}{x}\right), f\left(-\frac{1}{x-1}\right)$; solve for $f(x) = \frac{x^3 - 3x^2 - 2}{9x(x-1)}$, then differentiate.

RESULT

$$f'(2) = \frac{1}{2}.$$

WHY IT MATTERS

"Functional equation whose argument-map is a finite-order Möbius transformation" D3 flavor - the payoff is spotting the period-3 orbit and solving the resulting linear system, then a routine quotient-rule derivative. Distinct from the logged $\{x, -x, 1/x, -1/x\}$ substitution-group entry: that group is order 4 and the target is an integral; here it is an order-3 orbit of $\frac{x-1}{x}$ solved for a closed-form f before differentiating.

8. Cyclotomic product identity collapses to geometric polynomial

D3

EXAMPLE

$$f(x) = \prod_{k=1}^{2026} \left(x^2 - 2x \cos \frac{k\pi}{2027} + 1 \right); \text{ find } f'(1).$$

METHOD

The identity $x^{2n} - 1 = (x^2 - 1) \prod_{k=1}^{n-1} \left(x^2 - 2x \cos \frac{k\pi}{n} + 1 \right)$ with $n = 2027$ turns the 2026-factor product into $\frac{x^{4054} - 1}{x^2 - 1} = \sum_{j=0}^{2026} x^{2j}$. Then $f'(1) = 2 \sum_{j=0}^{2026} j = 2026 \cdot 2027$.

RESULT

4106702

WHY IT MATTERS

The year does STRUCTURAL work here — 2026 factors paired with modulus 2027 is exactly the n that makes the identity close. Do not assume a large constant is decorative. First entry using the $x^{2n} - 1$ product factorisation; distinct from the catalogued Chebyshev/root-of-unity counting entries.

9. Dawson function via defining ODE

D3

EXAMPLE

The Dawson function $F(x) = e^{-x^2} \int_0^x e^{t^2} dt$. Compute $F^{(5)}(0)$.

METHOD

F satisfies $F'(x) = 1 - 2xF(x)$. Differentiate repeatedly at 0 (odd function $\Rightarrow$ only odd Taylor terms): $F(x) = x - \frac{2}{3}x^3 + \frac{4}{15}x^5 - \dots$, so $F^{(5)}(0) = 5! \cdot \frac{4}{15}$.

RESULT

32

WHY IT MATTERS

Special-function-by-its-own-ODE-recursion - distinct from existing D3 Γ/ψ /Lambert-W flavors. The trick is converting the integral definition into a first-order ODE and reading off a Maclaurin coefficient.

10. Derivative of a Feynman parametric integral

D3

EXAMPLE

$$f(a) = \frac{d}{da} \int_0^\infty \frac{\ln(1 + a^2 x^2)}{1 + x^2} dx; \text{ evaluate } f(2).$$

METHOD

recognise the classical parametric integral $\int_0^\infty \frac{\ln(1 + a^2 x^2)}{1 + x^2} dx = \pi \ln(1 + a)$ (itself a Feynman/differentiate-under-the-sign result), so $f(a) = \pi/(1 + a)$.

RESULT

CONTINUED

$f(2) = \boxed{\pi/3}$ (verified: integral = $\pi \ln 3$, central-difference derivative = $\pi/3$).

WHY IT MATTERS

A Hard-derivative whose difficulty is knowing (or re-deriving) a named parametric-integral identity, then differentiating it - bridges the integral and derivative menus.

11. Determinant of log matrix mixed partial annihilation

D3

EXAMPLE

$f(x, y) = \det \begin{pmatrix} \ln x & \ln y & 1 \\ \ln y & \ln x & 1 \\ x & y & 1 \end{pmatrix}$; find $\frac{\partial^{110} f}{\partial x^{90} \partial y^{20}}$ evaluated at (67, 69).

METHOD

Expand the determinant: $f = (\ln x)^2 - (\ln y)^2 - x \ln x + x \ln y - y \ln x + y \ln y$. The pure terms $(\ln x)^2, (\ln y)^2$ die under the mixed partial (they depend on only one variable). Of the cross terms, each is at most linear in the variable being differentiated many times: $x \ln y$ and $-x \ln x$ are linear in x (killed by ∂_x^{90} , $90 \geq 2$), and $-y \ln x$, $y \ln y$ are linear in y (killed by ∂_y^{20} , $20 \geq 2$). Every term is annihilated, so the result is identically 0 - the point (67, 69) is a red herring.

RESULT

0

WHY IT MATTERS

A Hard-derivative trap where a 3×3 determinant of logs and linear entries expands to a function that is at most degree-one in x (and in y) within every cross term, so a high mixed partial wipes it out. Distinct from the existing "tan $^{-1} \frac{x}{y} + \tan^{-1} \frac{y}{x} + \ln(x^2 + y^2)$ " 100th-mixed-partial entry (which gave a nonzero answer via complexifying the log) and from the det-of-antiderivatives I3 entry (rank-deficiency $\rightarrow 0$): here the zero comes from low per-variable degree after the determinant expansion, not from complexification or proportional columns.

12. Differential operator power rolle zero count

D3

EXAMPLE

Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be thrice differentiable with at least 5 distinct real zeros. Define $F(x) = f(x) + 6f'(x) + 12f''(x) + 8f'''(x)$. Find the minimum number of distinct real zeros of F .

METHOD

Recognise the coefficients $1, 6, 12, 8 = \binom{3}{k} 2^k$, so $F = (1 + 2D)^3 f$ where $D = d/dx$. The operator $(1 + 2D)$ has integrating factor $e^{x/2}$: $(1 + 2D)g = 2e^{-x/2}(e^{x/2}g/2)'$, so if g has n zeros, $(1 + 2D)g$ has at least $n - 1$ zeros (Rolle on $e^{x/2}g$, which has the same zeros as g). Apply three times: $5 \rightarrow 4 \rightarrow 3 \rightarrow 2$.

RESULT

2

WHY IT MATTERS

Flagship "binomial coefficients reveal a differential operator power" trick. Existing D3 menu has special-function and power-tower flavors but no Rolle-with-integrating-factor zero count. Rewards two-stage spotting: binomial pattern + integrating factor.

13. Directional derivative multivariable with mixed exponent

D3

EXAMPLE

Find the directional derivative of $f(x, y, z) = x^3z + 1 + \sin(xy) - x^{zy}$ at $(3, 0, 1)$ along $\mathbf{v} = 2\mathbf{i} + 4\mathbf{j} + 3\mathbf{k}$.

METHOD

Compute partial derivatives at $(3, 0, 1)$: $f_x = 3x^2z + y \cos(xy) - zy x^{zy-1}$, $f_y = x \cos(xy) - x^{zy} \ln x \cdot z$, $f_z = x^3 - x^{zy} \ln x \cdot y$. At $(3, 0, 1)$: $f_x = 27 + 0 - 0 = 27$, $f_y = 3 \cos 0 - 1 \cdot \ln 3 \cdot 0 = 3$, wait: $f_y|_{(3,0,1)} = 3 \cdot 1 - 3^0 \ln 3 \cdot 1 = 3 - \ln 3$. $f_z|_{(3,0,1)} = 27 - 0 = 27$. Normalize $\hat{\mathbf{v}} = (2, 4, 3)/\sqrt{29}$. $D_{\mathbf{v}}f = \nabla f \cdot \hat{\mathbf{v}} = (27(2) + (3 - \ln 3)(4) + 27(3))/\sqrt{29}$.

CONTINUED

RESULT

$$(54 + 12 - 4 \ln 3 + 81)/\sqrt{29} = (147 - 4 \ln 3)/\sqrt{29}$$

WHY IT MATTERS

A D3 multivariable directional derivative with the tricky term x^{zy} (a function of all three variables). At $y = 0$ many terms vanish but f_y requires care with the $\ln x$ from the exponential. New D3 sub-flavor: "directional derivative of a function containing x^{zy} -type mixed exponentials".

14. Disguised identity via double angle and geometric series

D3

EXAMPLE

$$f(x) = x \cdot \left(\frac{1 + \cos 2x}{2} \right) \sum_{n=0}^{\infty} \left(\frac{\tan^2 x}{1 + \tan^2 x} \right)^n. \text{ Find } f'(\pi/5).$$

METHOD

$(1 + \cos 2x)/2 = \cos^2 x$ and $\tan^2 x/(1 + \tan^2 x) = \sin^2 x$, so the geometric series sums to $\sec^2 x$; the whole expression collapses to $f(x) = x \cos^2 x \sec^2 x = x$ identically, making $f'(x) \equiv 1$.

RESULT

$$f'(\pi/5) = 1.$$

WHY IT MATTERS

A "the baroque expression is secretly just x " gotcha - distinct from the existing D3 menu (self-referential functions, Lambert W, Schwarzian, power towers). Rewards recognizing a geometric-series-plus-double-angle collapse rather than grinding through the sum.

15. Extrema counting of $\cos(\text{exponential argument})$

D3

EXAMPLE

$$f(x) = \cos(3^{x+1}) \text{ on } [0, 3]; \text{ ratio of \#local extrema on } [0, 2) \text{ vs } [2, 3].$$

METHOD

extrema where $f' = -3^{x+1} \ln 3 \sin(3^{x+1}) = 0$, i.e. $3^{x+1} = k\pi$; count integer k in each image-interval ($u \in [3, 27)$ vs $[27, 81]$).

RESULT

$$8/17$$

WHY IT MATTERS

oscillation-counting where the phase grows exponentially; the count is a lattice-point tally in $k\pi$, not a derivative grind.

16. Fixed point exponent iteration

D3

EXAMPLE

$$f(x) = 1.3^x, \xi(f, x, n) = n\text{-th iterate of } f; \text{ find } \frac{d}{dx}(x^{\xi(f, 3, \infty)}) \text{ at } x = 2.$$

METHOD

The infinite iteration $\xi(f, 3, \infty)$ is the attracting fixed point L of $x = 1.3^x$ ($L \approx 1.471$); the exponent is then a constant and $\frac{d}{dx} x^L = L x^{L-1}$.

RESULT

$$L \cdot 2^{L-1} \approx 2.039, \text{ where } L \approx 1.4710 \text{ solves } L = 1.3^L.$$

WHY IT MATTERS

The exponent is defined by an infinite functional iteration that must first be resolved to a fixed point before any differentiation; the calculus step is trivial once you realize the exponent is just a transcendental constant. Rewards recognizing iterated-map convergence, not derivative rules.

17. Functional inequality forces constant

D3

EXAMPLE

f real, well-defined, with $f(x+y) + f(y+z) + f(z+x) \geq 3f(x+2y+3z)$ for all $x, y, z \in \mathbb{R}$. Find $f'(67)$.

METHOD

Reparametrize $u = x+y, v = y+z, w = z+x$ (free over $\mathbb{R}^3$); note $x+2y+3z = 2v+w$. Choosing parameters to compare arbitrary $f(s)$ vs $f(t)$ in both directions forces f constant, so the derivative is 0 everywhere.

RESULT

0

WHY IT MATTERS

A "derivative" problem whose hypothesis is an inequality valid for ALL reals usually collapses f to a constant - the calculus content is zero by design; the insight is the free reparametrization.

18. Geometric series under square root then differentiate

D3

EXAMPLE

$f(x) = \sqrt{x + 1 + \frac{1}{x} + \frac{1}{x^2} + \frac{1}{x^3} + \dots}$ for $x > 1$. Find $f'(2)$.

METHOD

The series under the radical is $x + 1 + \sum_{k=1}^{\infty} x^{-k} = x + 1 + \frac{1/x}{1-1/x} = x + 1 + \frac{1}{x-1} = \frac{x^2}{x-1}$ (for $x > 1$). So $f(x) = \sqrt{\frac{x^2}{x-1}} = \frac{x}{\sqrt{x-1}}$.

Differentiate: $f'(x) = \frac{\sqrt{x-1} - x/(2\sqrt{x-1})}{x-1} = \frac{2(x-1)-x}{2(x-1)^{3/2}} = \frac{x-2}{2(x-1)^{3/2}}$.

RESULT

$f'(2) = 0$

WHY IT MATTERS

An "infinite series under a radical" that first requires summing the geometric part, then simplifying the closed form before differentiating. At $x = 2$ the numerator vanishes, giving a clean 0. Distinct from the "fixed-point-exponent-iteration" and "lambert-W-of-inner-trig" D3 entries - here the key is a geometric series collapse inside a square root.

19. Gradient reconstruction with mixed PDE constraint

D3

EXAMPLE

Given $\nabla f = (3x^2 + by^2, 2bxy + a)$ with the extra constraint $\partial^3 f / \partial x^3 + (1/x) \partial^2 f / \partial y^2 = 10$, and $f(0,0) = 0, f(1,1) = 2$, find $f(1,2)$.

METHOD

Reconstruct the potential $f = x^3 + bxy^2 + ay$ from the given gradient (exactness/potential-function idea, same family as DE1's exact-equation trick but applied to a plain multivariable function rather than an ODE), then use the extra mixed-derivative constraint plus the two point conditions as a 2-equation system to pin down the free parameters a, b before evaluating at a third point.

RESULT

$f(1,2) = 7$ (with $a = -1, b = 2$).

WHY IT MATTERS

New D3 family - treats gradient reconstruction as a derivative problem, then layers on a differential-constraint equation to fix constants instead of simple boundary matching. Distinct from the catalogued implicit-differentiation and Faà di Bruno D3 families.

20. Half angle Vieta with complementary angle

D3

EXAMPLE

In triangle PQR , $\angle R = \pi/2$. $\tan(P/2)$ and $\tan(Q/2)$ are the roots of a quadratic $Q(x)$. Find $Q'(0.5)/Q(0)$.

METHOD

Since $P + Q = \pi/2$, $(P + Q)/2 = \pi/4$, so $\tan((P + Q)/2) = 1 = s/(1 - p)$ where $s = \text{sum of roots}$, $p = \text{product}$. Hence $s = 1 - p$, i.e. $s + p = 1$. Write $Q(x) = a(x^2 - sx + p)$; $Q(0) = ap$, $Q'(0.5) = a(1 - s) = ap$. Ratio collapses to 1.

RESULT

1

WHY IT MATTERS

The complementary-angle constraint forces $s + p = 1$, which by the clever evaluation point $x = 0.5$ in Q' produces an algebraic cancellation. A geometric Vieta flavor - distinct from existing Vieta-on-polynomial-coefficient flavors.

21. Herglotz coth csch parameter differentiation series

D3

EXAMPLE

$$f(x) = \sum_{n=1}^{\infty} \frac{x + n(-1)^n}{x^2 + n^2}; \text{ find } f'(-1) + f'(1).$$

METHOD

Split $f'(x)$ termwise into $\sum (n^2 - x^2 - 2xn(-1)^n)/(x^2 + n^2)^2$; the sum $f'(1) + f'(-1)$ collapses the alternating pieces and leaves $\sum 2(n^2 - 1)/(n^2 + 1)^2 = 2S_1(1) - 4S_2(1)$ where $S_1(a) = \sum 1/(n^2 + a^2)$ and $S_2(a) = \sum 1/(n^2 + a^2)^2$ are the standard Herglotz/Mittag-Leffler sums built from $\pi \coth(\pi a)$; S_2 is obtained by differentiating S_1 with respect to a^2 .

RESULT

$$f'(1) + f'(-1) = 1 - \frac{\pi^2}{\sinh^2 \pi} \approx 0.926 \text{ (verified numerically).}$$

WHY IT MATTERS

A new D3 family using the $\coth(\pi a)$ Mittag-Leffler expansion (and its parameter-derivative for the squared-denominator sum) to evaluate an infinite series of derivatives - distinct from the catalogued self-referential, Lambert-W, and Schwarzian D3 flavors; the mechanism is closer to a Basel-type special-sum technique applied to a derivative sum rather than a single function evaluation.

22. Hidden hyperbolic identity $y + y' = e^x$ with derivative valued exponent

D3

EXAMPLE

with $y = \sinh x$ and $p = x \ln(2x) - x + 1$, evaluate $(y + y')^{p''}$ at $x = 1$.

METHOD

$y + y' = \sinh x + \cosh x = e^x$; exponent $p' = \ln(2x)$, $p'' = 1/x$, so $p''(1) = 1$; result $= (e^1)^1 = e$.

RESULT

e.

WHY IT MATTERS

Two independent collapses (hyperbolic sum to e^x , and a log-derivative exponent to 1) compose into a clean payoff. Hard-derivative "two cosmetic layers" template.

23. Hyperbolic segment area ratio via tangent parallel chord

D3

EXAMPLE

A, B are fixed points on the same branch of $xy = 1$. A point P on the arc between them is chosen so that the area of triangle APB is maximised. Let S_1 = area of the region bounded by the hyperbola and chord AP , and S_2 = area of the region bounded by the hyperbola and chord PB . Find S_1/S_2 .

METHOD

At the area-maximising P , the tangent to the curve at P is parallel to chord AB (Archimedes' parabolic-segment lemma, generalising to any smooth curve: $\frac{d}{dt} \text{Area}(APB) = 0$ forces tangent $\parallel AB$). Combined with the affine symmetry of $xy = 1$ exchanging the two segments, the two flanking hyperbolic caps then have equal area.

RESULT

$$S_1/S_2 = 1$$

CONTINUED

WHY IT MATTERS

Whenever a point on a smooth curve is chosen to maximise the area of a triangle with two fixed points on that curve, the tangent at the optimal point is parallel to the chord - Archimedes' "parabolic segment" lemma in disguise. For conics with an affine/projective symmetry exchanging the two flanking caps (e.g. $xy = 1$ via $(x, y) \leftrightarrow (1/y, 1/x)$), the two segment areas are then forced equal. Distinct from existing D2/P1 inscribed-figure entries because the payoff is a ratio of segment areas (not the triangle area), and the answer collapses to a clean 1 via symmetry rather than calculus.

24. Infinite product geometric exponent collapse

D3

EXAMPLE

$$f(x) = \prod_{k=0}^{\infty} x^{x^k/(-1)^k} \text{ on } (0, 1); \text{ find } f'(\frac{1}{2}).$$

METHOD

The exponent sum is $\sum_{k \geq 0} (-x)^k = \frac{1}{1+x}$, so $\ln f = \frac{\ln x}{1+x}$ and $f(x) = x^{1/(1+x)}$. Then $f'/f = \frac{(1+x)/x - \ln x}{(1+x)^2}$.

RESULT

$$f'(\frac{1}{2}) = 2^{-2/3} \left(\frac{4}{3} + \frac{4 \ln 2}{9} \right) \approx 1.03$$

WHY IT MATTERS

An infinite product (not a power tower) whose exponents form a geometric series collapses to a single $x^{1/(1+x)}$. Distinct from the Lambert-W power-tower D3 family - the trick is recognising $x^k/(-1)^k = (-x)^k$ and summing the geometric series.

25. Inverse derivative of self similar logarithmic tower

D3

EXAMPLE

For $f(x) = \ln(x \ln(x \ln(\dots)))$, evaluate $(f^{-1})'(3/2)$.

METHOD

Self-similarity gives $f = \ln(xf)$, so $e^f = xf$ and the inverse is explicit: $f^{-1}(y) = e^y/y$. Differentiate the inverse directly instead of differentiating the infinite nesting.

RESULT

$$(f^{-1})'(3/2) = \frac{2e^{3/2}}{9}.$$

WHY IT MATTERS

An infinite logarithmic tower becomes elementary after solving its fixed-point equation for the original variable; inverse differentiation is then the easy final step.

26. Lagrange Burmann Catalan generating function

D3

EXAMPLE

With $f(x) = x - \frac{1}{3}x^2$ and Lagrange-Bürmann inversion, evaluate $\sum_{n=0}^{\infty} \frac{(2n)!}{n!^2 (n+1) 3^{2n+1}}$.

METHOD

$\frac{(2n)!}{n!^2 (n+1)} = C_n$ (Catalan), so the sum is $\frac{1}{3} \sum_n C_n 9^{-n} = \frac{1}{3} \cdot \frac{1 - \sqrt{1-4/9}}{2/9} = \frac{3-\sqrt{5}}{2}$, i.e. the inverse-function series of f evaluated at the appropriate point.

RESULT

$$\frac{3-\sqrt{5}}{2} \approx 0.3820$$

WHY IT MATTERS

"Hard Derivative" register dressing the Catalan generating function as a Lagrange-Bürmann series inversion. The tell is the $\frac{(2n)!}{n!^2 (n+1)}$ Catalan signature; answer is a clean golden-ratio-flavored surd $\frac{3-\sqrt{5}}{2} = \varphi^{-2}$. Distinct from existing Catalan/series entries because the framing is explicit series reversion of a quadratic.

27. Lambert w of inner trig gamma stationary**D3****EXAMPLE**

$f(x) = W(\sin(\Gamma(x)))$ on $[2, 4]$; find the sum of the x -coordinates of the stationary points.

METHOD

Since $W'(u) \neq 0$ wherever defined, $f'(x) = 0 \iff \frac{d}{dx} \sin(\Gamma(x)) = \cos(\Gamma(x)) \Gamma'(x) = 0$. So stationary points are the Gamma-minimum root $\Gamma'(x) = 0$ ($x \approx 2.4877$, the ψ root) together with roots of $\Gamma(x) = \pi/2$ on $[2, 4]$. The outer W is a red herring.

RESULT

numeric (sum ≈ 6.28); the insight (outer monotone function $\Rightarrow$ ignore it) is the point.

WHY IT MATTERS

Extends the special-function derivative family (Γ^W , power towers) with a "the outermost wrapper is strictly monotone so its derivative never vanishes - differentiate only the inside" trick. Distinct from log-diff towers; the payoff is recognising the wrapper is inert.

28. Multivariable partials combination at a point (hard derivative)**D3****EXAMPLE**

$f(x, y) = 10x^4y^3 - e^{3xy} + 1$; evaluate $(f_x + f_y + xy^2)^2 - (f_{xy} + f_{yy} + f_{xx})$ at $x = 2, y = 0$.

METHOD

compute first/second partials, evaluate at $y = 0$ (most polynomial terms die; $e^{3xy} \rightarrow 1$), then assemble.

RESULT

a clean integer.

WHY IT MATTERS

a multivariable-calculus derivative problem (partials + a contrived combination) where the eval point $y = 0$ kills most terms - the dailyintegral "evaluation point carries the joke" aesthetic in 2 variables.

29. Named slot is a shell (derivative slot whose core is complex region distance, open infimum trap)**D3****EXAMPLE**

Let $z \in \mathbb{C}$ satisfy $\operatorname{Re}(z^2) > 0$. Find the smallest possible value of $|z + i|$.

METHOD

$\operatorname{Re}(z^2) = x^2 - y^2 > 0 \iff |x| > |y|$ - the open horizontal double-wedge about the real axis (boundaries $y = \pm x$). Minimise the distance from $-i = (0, -1)$ (which sits in the opposite vertical wedge) to that region: nearest boundary line is $x + y = 0$, distance $\frac{|-1|}{\sqrt{2}} = \frac{1}{\sqrt{2}}$. Because the region is open, the value $\frac{\sqrt{2}}{2}$ is an infimum that is not attained.

RESULT

$\frac{\sqrt{2}}{2}$ (greatest lower bound; no minimiser).

WHY IT MATTERS

Two distinctive features. (1) The slot label ("derivative") is a decoy - the real skill is region geometry + point-to-line distance. (2) The open-set subtlety: the natural answer is an infimum, not a minimum, so a careful solver flags "not attained." Pair this with the I3 entry above: dailyintegral is increasingly shipping wrapper problems where the named technique is a shell around a different core skill (Vieta, complex geometry). Generator implication: occasionally seed an integral/derivative slot whose true reward is algebra/geometry, and when the natural extremum lives on an open boundary, state the infimum and note non-attainment.

30. ODE curve fitting then director circle distance**D3****EXAMPLE**

Curve $c_1: y = f(x)$ passes through $(2, 2)$ and $(8, 1/2)$ and satisfies $y y'' = 2(y')^2$. Curve c_2 is the director circle of $x^2 + y^2 = 2$. The shortest distance between c_1 and c_2 is $\sqrt{p} - q$ with $p, q \in \mathbb{N}$. Find $p^2 - q$.

CONTINUED

METHOD

ODE substitution $u = y'/y$: $u' = (y''/y) - (y'/y)^2 = 2u^2 - u^2 = u^2$, so $u = -1/(x - c)$, $y = A/(x - c)$. Fit through $(2, 2)$, $(8, 1/2)$ gives $y = 4/x$ (so $c = 0$, $A = 4$). Director circle of conic $x^2 + y^2 = 2$ is $x^2 + y^2 = 4$ (radius 2). Closest point on $xy = 4$ to origin is $(2, 2)$ at distance $2\sqrt{2}$, so min distance to circle $= 2\sqrt{2} - 2 = \sqrt{8} - 2$. Hence $p = 8$, $q = 2$, $p^2 - q = 62$.

RESULT

62

WHY IT MATTERS

Multi-domain composite - ODE + conic-section director circle + optimization. Rewards the " $u = y'/y$ separates the ODE" structural move AND the geometric director-circle definition. Distinct from existing D3 special-function and power-tower menus.

31. ODE solution as series coefficient

D3

EXAMPLE

Compute $f'(1)$ where $f(x) = \sum_{n=1}^{\infty} \frac{x^n}{n^2} y_n$ and y_n is the solution to $y'' - 2y' + y = e^t$, $y(0) = y'(0) = 0$, evaluated at $t = n$.

METHOD

Solve the ODE: repeated root $r = 1$ in the characteristic equation, resonant forcing gives $y(t) = \frac{t^2}{2} e^t$. Then $y_n = \frac{n^2}{2} e^n$. The n^2 cancels the n^2 in the denominator and the series becomes geometric: $f(x) = \frac{1}{2} \sum (ex)^n = ex/(2(1 - ex))$. Differentiate.

RESULT

$$f'(1) = e/(2(1 - e)^2)$$

WHY IT MATTERS

Two-stage engineering: (1) solve an ODE; (2) recognise that its values at integers conspire with the series prefactor to give a clean closed-form generating function. Distinct from existing D3 special-function and power-tower flavors - the trick is the cross-domain bridge from ODE to generating function.

32. Optimization labelled as derivative implicit curve extremum

D3

EXAMPLE

If $x^2 + y^3 = 64$ and $-8 < x < 8$, find the largest possible value of $(x + y)^2$.

METHOD

Treat $y = (64 - x^2)^{1/3}$, differentiate $x + y$, check interior critical points and boundary behavior under the domain restriction.

RESULT

Exact maximum after solving a low-degree algebraic critical equation.

WHY IT MATTERS

Hard derivative can be a global optimization problem on an implicit curve; always include endpoint/domain checks.

33. Power constraint xy minimization via substitution

D3

EXAMPLE

Let $x, y > 0$. If $x^y = 2026$, find the minimum possible value of the product xy .

METHOD

Write $y = \ln 2026 / \ln x$, so $xy = x \ln 2026 / \ln x$. Minimize by setting $d/dx[x/\ln x] = 0$: $(\ln x - 1)/(\ln x)^2 = 0 \Rightarrow \ln x = 1 \Rightarrow x = e$. At $x = e$: $y = \ln 2026$, $xy = e \ln 2026$.

RESULT

$$e \ln 2026$$

WHY IT MATTERS

"Minimize a product xy subject to $x^y = C$ " - the disguise is that you must convert the power constraint to a logarithmic relation, then optimise $x/\ln x$. The answer always has the form $e \ln C$ regardless of C . Distinct from existing D3 inverse-function and tower entries; the technique is a smooth calculus optimisation on the reduced one-variable function.

34. Power tower exponent arithmetic, $f'(1)$ by log diff**D3****EXAMPLE**

$$f(x) = \frac{x^{10x^2}}{\sqrt{x^{x^3}}} \cdot \frac{\sqrt[3]{x^{x^3}}}{x^{5x^3/2}} \cdot \frac{x^{x^3/13}}{\sqrt[720]{x^{x^3}}} \cdots, \text{ find } f'(1).$$

METHOD

$\ln f = \sum (\text{exponent}) \ln x$; collect the exponent as a single power series in x ; $f'(1) = (\text{exponent at } 1)$ since $f(1) = 1$ and the $\ln x$ factor's derivative is $1/x \rightarrow 1$.

RESULT

(exponent evaluated at $x = 1$; reduce the telescoping/geometric exponent sum)

WHY IT MATTERS

stacked radical-of-power-tower factors are pure exponent bookkeeping once you take $\ln$; only the constant term of the exponent series survives at $x = 1$.

35. Reciprocal limits Leibniz with complementary arctan**D3****EXAMPLE**

$$f(x) = \int_{1/x}^x \frac{\tan^{-1} t}{t} dt, \quad x > 0; \text{ find } f'(3).$$

METHOD

Two-term Leibniz rule (both limits variable, reciprocal of each other, no parameter in the integrand) reduces to $\frac{1}{x} [\tan^{-1} x + \tan^{-1}(1/x)]$, which collapses via $\tan^{-1} x + \tan^{-1}(1/x) = \pi/2$.

RESULT

$$f'(3) = \pi/6.$$

WHY IT MATTERS

A cleaner, two-contribution sibling of the 05 – 16 "variable-limits-plus-parameter-leibniz" D2 entry (which needed three contributions). The reciprocal-limits setup specifically to trigger the arctan complementary-angle constant is the new wrinkle.

36. Self referential nested radical derivative**D3****EXAMPLE**

$$f(x) = \sqrt{x + \sqrt{x^2 + \sqrt{x^3 + \sqrt{x^4 + f(x)}}}}; \text{ find } f'(1).$$

METHOD

At $x = 1$ the tower collapses to $f(1)^2 = 1 + f(1)$ along a self-similar branch giving $f(1) = \varphi$; implicit-differentiate the full four-level radical and substitute $x = 1$, $f = \varphi$.

RESULT

rational expression in $\varphi = (1 + \sqrt{5})/2$ (golden ratio).

WHY IT MATTERS

Combines the infinite/self-referential-nesting motif with implicit differentiation; the golden ratio fixed point is the recognisable hook.

37. Self similar trig derivative infinite sum**D3****EXAMPLE**

$$f(x) = \sqrt{1 + \cos x}; \text{ compute } \sum_{n=0}^{\infty} f^{(n)}\left(\frac{\pi}{2}\right).$$

METHOD

Near $x = \pi/2$, simplify $f(x) = \sqrt{2} \cos(x/2)$ (valid locally), so every derivative is again a cosine of a shifted/scaled argument: $f^{(n)}(\pi/2) = \sqrt{2} (1/2)^n \cos(\pi/4 + n\pi/2)$. Summing over n is then a complex geometric series (treat $\cos(\pi/4 + n\pi/2)$ as $\text{Re}[e^{i\pi/4} i^n]$ and sum the resulting geometric series in $i/2$).

RESULT

$$2/5$$

WHY IT MATTERS

CONTINUED

A new D3 family combining "simplify before differentiating" with an infinite sum of all derivatives at a point (rather than a single high-order derivative) - the self-similarity under differentiation (each derivative is a scaled, phase-shifted copy of the original) is what makes the infinite sum tractable via a geometric series.

38. Sum of inverse branch derivatives via Vieta

D3

EXAMPLE

Let $f(x) = x^3 - 3x + 1$. For a real y such that $f(x) = y$ has three local inverse branches $g \in A$, find $\sum_{g \in A} [g^{-1}(y)]'$.

METHOD

By the inverse function theorem $g'(y) = 1/f'(r_i)$ where the r_i are roots of $f(x) = y$. Then $\sum 1/f'(r_i) = \frac{1}{3} \sum 1/(r_i^2 - 1)$. Split via partial fractions; each piece is $\sum 1/(r_i - c) = -p'(c)/p(c)$ for $p(x) = x^3 - 3x + (1 - y)$. Both terms vanish because ± 1 are critical points of f , i.e. $p'(\pm 1) = 0$.

RESULT

0

WHY IT MATTERS

"The answer is zero because of a hidden symmetry" trick - the critical-point structure of f forces $p'(\pm 1) = 0$ identically. Distinct from existing D3 inverse-function and Vieta patterns because it aggregates over ALL inverse branches simultaneously.

39. Telescoping doubling product as log derivative

D3

EXAMPLE

For fixed integer $n \geq 1$, $f(x) = \frac{1}{1+x} + \frac{2x}{1+x^2} + \frac{4x^3}{1+x^4} + \dots + \frac{2^n x^{2^n-1}}{1+x^{2^n}}$; find $f'(0)$.

METHOD

Each term is $\frac{d}{dx} \ln(1+x^{2^k})$, so $f = \frac{d}{dx} \ln \prod_{k=0}^n (1+x^{2^k})$. The doubling product telescopes $\prod_{k=0}^n (1+x^{2^k}) = \frac{1-x^{2^{n+1}}}{1-x}$, hence $f(x) = \frac{d}{dx} \ln \frac{1-x^{2^{n+1}}}{1-x} = 1 + x + x^2 + \dots$ near 0; answer independent of n .

RESULT $f'(0) = 1$.**WHY IT MATTERS**

A sum of $\frac{2^k x^{2^k-1}}{1+x^{2^k}}$ is the log-derivative of a self-doubling product that telescopes to $\frac{1-x^{2^{n+1}}}{1-x}$; read off the Maclaurin coefficient.

40. Tied parameter limit Leibniz decoupling substitution

D3

EXAMPLE

$g(x) = \int_0^{x^2} \ln(1+tx^2) dt$; find $g'(1)$.

METHOD

Rather than grinding through the full two-term Leibniz rule, substitute $s = tx^2$ to decouple the parameter from the dummy variable: $g(x) = \frac{1}{x^2} \int_0^{x^4} \ln(1+s) ds = \frac{1}{x^2} [(1+x^4) \ln(1+x^4) - x^4]$, which is now an ordinary single-variable quotient-rule/FTC problem; the resulting $\ln(1+x^4)$ term carries a factor of $(x^4 - 1)$ that vanishes exactly at $x = 1$.

RESULT $g'(1) = 2$.**WHY IT MATTERS**

Distinct from both logged Leibniz D2/D3 entries (variable-limits-plus-parameter, and reciprocal-limits) - here the upper limit and the integrand's parameter are algebraically tied together (tx^2 under a limit of x^2), and the real insight is a substitution that collapses the two-variable Leibniz problem into single-variable calculus before differentiating, with the evaluation point chosen to null the transcendental term.

41. Variable rational exponent logarithmic differentiation**D3****EXAMPLE**

$f(x) = (3 \sin x)^{1/(x^2+1)}$ on $(0, \pi)$; find $f'(2)$.

METHOD

$\ln f = \frac{\ln(3 \sin x)}{x^2+1}$; $\frac{f'}{f} = \frac{-2x}{(x^2+1)^2} \ln(3 \sin x) + \frac{\cot x}{x^2+1}$; multiply by $f(2) = (3 \sin 2)^{1/5}$.

RESULT

$f'(2) = (3 \sin 2)^{1/5} \left[\frac{\cot 2}{5} - \frac{4 \ln(3 \sin 2)}{25} \right] \approx -0.27$ (numeric).

WHY IT MATTERS

Variant of the $(\sin x)^{\tan x}$ log-diff family, but the exponent is a rational function $1/(x^2 + 1)$ and the eval point $x = 2$ is deliberately non-clean - sharper Hard-Derivative execution test.

Learned variants - Differential 01 - Easy

8 distinct screenshot-learned variants are documented with their example, method, result when recorded, and distinguishing lesson.

1. Daily Differential Easy tier (branded applied first order ODE)

DE1

EXAMPLE

A whimsically branded scenario ("conflict-resolution time R in the Modulo workspace over network load spike t ") governed by $t \frac{dR}{dt} + 3R = 5t^2$, with $R(1) = 5$; find $R(2)$.

METHOD

first-order LINEAR (or Cauchy-Euler) ODE, integrating factor $\mu = t^3 \Rightarrow R = t^2 + 4/t^3$.

RESULT

$R(2) = 4.5$ (verified).

WHY IT MATTERS

the site's Easy Differential is ALWAYS a playful tech vignette (often self-referential to dailyintegral) wrapping a clean first-order ODE with a tidy IC and a single scalar ask. Register = 3 – 4 sentences of story, then the displayed ODE, then "Given ..., find ...".

2. Differential equation slot now appears on dailyintegral (RL circuit, first order linear)

DE1

EXAMPLE

RL circuit $L \frac{dI}{dt} + RI = E(t)$ with $R = 12 \Omega$, $L = 4 \text{ H}$, constant $E = 60 \text{ V}$, switch closed at $t = 0$; find $\lim_{t \rightarrow \infty} I(t)$.

METHOD

First-order linear constant-coefficient ODE via integrating factor; step response $I(t) = \frac{E}{R}(1 - e^{-Rt/L})$.

RESULT

$I(\infty) = E/R = 5 \text{ A}$; full solution $I(t) = 5(1 - e^{-3t})$.

WHY IT MATTERS

Confirms dailyintegral.com is now featuring differential-equation problems. Canonical first-order-linear / integrating-factor anchor for the new DE1 (Easy) tier - RL/RC circuits, Newton cooling, and mixing-tank problems all live here.

3. Dx th root notation disguising dy/dx

DE1

EXAMPLE

Solve $xe^{e^x} = \sqrt[dx]{e^{dy}}$ with $y(1) = e$; find the exact value of $y(e)$.

METHOD

Read the notation: $\sqrt[dx]{e^{dy}} = (e^{dy})^{1/dx} = e^{dy/dx}$. Taking logs gives the completely ordinary separable ODE $\frac{dy}{dx} = \ln(xe^{e^x}) = \ln x + e^x$. Integrate: $y = x \ln x - x + e^x + C$; the IC $y(1) = -1 + e + C = e$ forces $C = 1$; then $y(e) = e - e + e^e + 1$.

RESULT

$y(e) = e^e + 1 \approx 16.1543$

WHY IT MATTERS

A typographic disguise rather than a mathematical one - the derivative is hidden inside a radical index (dx -th root of e^{dy}), so the entire difficulty is recognising that $\sqrt[dx]{e^{dy}}$ means $e^{y'}$ and that taking $\ln$ of both sides linearises it. Sibling to the logged "stacked-nested-integral-notation-collapse" I2 entry (same "the gimmick is the notation" register) but on the DE side, and the payoff is a clean $e^e + 1$ from the e^x term surviving while $x \ln x - x$ cancels at $x = e$.

4. Factorable linear ODE separable after grouping

DE1

EXAMPLE

Solve $\frac{dy}{dx}(x + x^2) + y(x^3 + x^4) = x^3 + x^4$ with $y(0) = 2$; find the exact value of $y(1)$.

METHOD

Factor each coefficient: $x + x^2 = x(1 + x)$ and $x^3 + x^4 = x^3(1 + x)$. Divide through by $x(1 + x)$: $y' + x^2y = x^2$, a clean first-order linear ODE. Integrating factor $\mu = e^{x^3/3}$ gives $(e^{x^3/3}y)' = x^2e^{x^3/3}$, so $y = 1 + Ce^{-x^3/3}$. The IC $y(0) = 2 \Rightarrow C = 1$, hence $y(x) = 1 + e^{-x^3/3}$ and $y(1) = 1 + e^{-1/3}$.

RESULT

$$y(1) = 1 + e^{-1/3} \approx 1.7165.$$

WHY IT MATTERS

An Easy-Differential anchor where the disguise is bloated polynomial coefficients ($x + x^2$, $x^3 + x^4$) that share the common factor $x(1 + x)$; cancelling it exposes a textbook linear ODE $y' + x^2y = x^2$ whose particular solution is the constant 1. Rewards factoring before reaching for machinery. Distinct from the standard constant-coefficient DE2 entry below.

5. Newton cooling two equal time ratios

DE1

EXAMPLE

Coffee cools from 90°C toward a 20°C room and is 70°C after 10 minutes; find its temperature after 20 minutes.

METHOD

For $T' = -k(T - T_a)$, equal time intervals multiply the excess temperature by the same factor.

RESULT

$$T(20) = 20 + 50(5/7) = 390/7^\circ\text{C}.$$

WHY IT MATTERS

Working with the excess $T - T_a$ avoids solving explicitly for k .

6. Perfect square under radical easy ODE

DE1

EXAMPLE

Solve for $x > 1$: $\frac{dy}{dx} = \sqrt{x + 2\sqrt{x-1}}$, with $y(1) = 3$; find $y(5)$.

METHOD

$x + 2\sqrt{x-1} = (x-1) + 2\sqrt{x-1} + 1 = (\sqrt{x-1} + 1)^2$, so $y' = \sqrt{x-1} + 1$ and $y = \frac{2}{3}(x-1)^{3/2} + x + C$. Apply $y(1) = 3 \Rightarrow C = 2$; then $y(5) = \frac{2}{3} \cdot 8 + 7 = \frac{37}{3}$.

RESULT

$$\frac{37}{3} \approx 12.33$$

WHY IT MATTERS

First documented "Easy Differential" tier on dailyintegral that is a direct-integration ODE whose only trick is spotting a perfect square under the nested radical. The hook is $a + 2\sqrt{a-1} = (\sqrt{a-1} + 1)^2$. Pairs well with the new DE slot family.

7. Repeated root parameter Vieta product

DE1

EXAMPLE

Determine all k for which a parameterised second-order constant-coefficient ODE has general solution $(C_1 + C_2x)e^{\lambda x}$, then evaluate a product of the corresponding initial-value solutions.

METHOD

Set the characteristic discriminant to zero, use Vieta on the admissible parameter values, and postpone expanding the requested product until the IVP constants are expressed algebraically in k .

WHY IT MATTERS

The repeated-root form is a condition on the characteristic polynomial; the final product is engineered for symmetric-polynomial cancellation.

8. Reverse quotient rule with doubly exponential antiderivative log log ask**DE1****EXAMPLE**

Branded vignette: antimatter A from t tesseracts obeys $A = t \frac{dA}{dt} - (t^2 - 1)e^{e^{t+1/t} + t+1/t}$, with $A(1) = e^{e^2}$; find x such that $t = 40$ yields 10^{10^x} .

METHOD

Divide by t^2 so the left side becomes $\left(\frac{A}{t}\right)'$ (reverse QUOTIENT rule, not the usual reverse product rule); with $w = t + \frac{1}{t}$ the right side is exactly $\frac{d}{dt}e^w$. Hence $A = t e^{e^{t+1/t}}$, the IC forcing $C = 0$. Then take $\log_{10}$ twice.

RESULT

$x \approx 17.020421$

WHY IT MATTERS

Three things are new: the reverse-quotient-rule packaging, a doubly-exponential antiderivative keyed to the reciprocal combination $t + 1/t$, and MAGNITUDE BOOKKEEPING (10^{10^x}) as the ask — previously a P1-only device, now migrating into DE1. Also note the Easy Differential answer is genuinely messy, breaking the usual clean-payoff expectation for that tier.

Learned variants - Differential 02 - Medium

8 distinct screenshot-learned variants are documented with their example, method, result when recorded, and distinguishing lesson.

1. Daily Differential Medium tier (terse Bernoulli)

DE2

EXAMPLE

"Solve for $x > 0$: $\frac{dy}{dx} + \frac{y}{x} = xy^2$, $y(1) = \frac{1}{3}$; find $y(3)$."

METHOD

BERNOULLI $n = 2$, sub $v = 1/y \Rightarrow v = -x^2 + 4x \Rightarrow y = 1/(4x - x^2)$.

RESULT

$y(3) = \frac{1}{3}$ (verified).

WHY IT MATTERS

the Medium Differential is a TERSE, story-free statement rewarding ONE genuine technique (most often Bernoulli, otherwise a clean substitution / separable / exact). One line of setup, displayed ODE, an IC, "find $y(x_0)$ ".

2. Exponential linearization substitution

DE2

EXAMPLE

$y' = 2t^{-1} + e^{-y}$, $y(1) = 0$; find $y(1.5)$.

METHOD

Substitute $v = e^y$, so $v' = e^y y' = v(2/t + e^{-y}) = 2v/t + 1$, turning the nonlinear ODE into the linear equation $v' - \frac{2}{t}v = 1$. Solve via integrating factor $\mu = t^{-2}$ to get $v = -t + 2t^2$, then invert $y = \ln v$.

RESULT

$y(1.5) = \ln 3$

WHY IT MATTERS

A DE2 substitution family not in the current menu - "exponentiate the dependent variable" to linearize an equation with e^{-y} forcing, sibling to Bernoulli ($v = y^{1-n}$) but for exponential rather than power nonlinearity.

3. Golden ratio characteristic roots convergence forced constant

DE2

EXAMPLE

$y'' + y' - y = x$, given $\int_0^\infty (y + x + 1) dx = \varphi - 1$, find $y(1 - \varphi)$.

METHOD

The homogeneous characteristic roots of $r^2 + r - 1 = 0$ are exactly $\varphi - 1$ and $-\varphi$ (golden ratio identities). The growing-mode coefficient must vanish for the improper integral to converge, and the golden-ratio identity $\varphi^2 = \varphi + 1$ is used to simplify the remaining constant of integration.

RESULT

$y(1 - \varphi) = e + \varphi - 2$.

WHY IT MATTERS

New DE2 family - ties the characteristic-equation roots directly to φ and uses convergence-of-an-improper-integral (rather than a plain initial condition) as the second constraint. Distinct from the catalogued distinct-real-root, repeated-root, and complex-root DE2 families.

4. Homogeneous substitution to arctan with engineered zero

DE2

EXAMPLE

$x \frac{dy}{dx} - y = x^2 + y^2$, $y(\pi/2) = \pi/2$; find $y(\pi/4)$.

METHOD

Homogeneous of degree zero, so $v = y/x$ collapses everything to $v' = 1 + v^2$, giving $v = \tan(x + C)$ and $y = x \tan\left(x - \frac{\pi}{4}\right)$ once the IC pins C . The evaluation point is chosen to sit exactly on the zero of the tangent.

RESULT

0

WHY IT MATTERS

The $v = y/x \Rightarrow v' = 1 + v^2 \Rightarrow \arctan$ route is new to the log (catalogued homogeneous material used a first-integral trick instead). The packaging is the real lesson: IC and evaluation point conspire so the headline answer is exactly 0 — an engineered-cancellation tell that also appeared in P1 this batch (answer 1).

5. Linear first order ODE tan integrating factor

DE2

EXAMPLE

Solve $\frac{dy}{dx} + y \tan x = \cos^2 x$ on $(-\pi/2, \pi/2)$, $y(0) = 5$; find $y(\pi)$ (by continuation of the general solution).

METHOD

Integrating factor $\mu = e^{\int \tan x dx} = \sec x$. $(y \sec x)' = \cos^2 x \sec x = \cos x$, so $y \sec x = \sin x + C$, $y = (\sin x + C) \cos x$. $y(0) = C = 5$; evaluating the general solution at $x = \pi$ gives $(0 + 5)(-1) = -5$.

RESULT

-5

WHY IT MATTERS

"Medium Differential" tier - textbook linear first-order ODE with a $\tan x$ integrating factor, but with the deliberately out-of-domain evaluation point $x = \pi$ (problem instructs to use the formula's analytic continuation). Distinct from existing Bernoulli/RL-circuit ODE entries; the $\sec x$ integrating factor is the signature.

6. Logistic growth carrying capacity word problem

DE2

EXAMPLE

Deer population N in a park obeys $dN/dt = kN(L - N)$, $L = 500$; $N(0) = 100$, $N(4) = 200$; find t when $N = 350$.

METHOD

Textbook logistic equation, solved via partial fractions $\frac{1}{N(L-N)} = \frac{1}{L} \left(\frac{1}{N} + \frac{1}{L-N} \right)$ giving $N/(L-N) = Ae^{kLt}$; two data points calibrate A and kL , then invert for t .

RESULT

$t \approx 9.11$ years.

WHY IT MATTERS

Logistic growth / carrying-capacity ODE not previously logged in the DE menu - a clean addition to the DE1/DE2 separable-ODE word-problem rotation. NOTE: in the source screenshot this appeared as an elaborate branded word problem in the MEDIUM tier, whereas every previously logged DE2 example was a terse, story-free statement (Bernoulli, homogeneous, exact, etc.) - flagged as a possible register observation, not adopted as a rule change; the v7 three-register design (Easy=branded/Medium=terse/Hard=story-with-insight) remains the locked calibration unless Ben says otherwise.

7. Perfect square coefficient hidden Euler ODE

DE2

EXAMPLE

Solve for $x > -\frac{1}{2}$: $(4x^2 + 4x + 1)y'' + 2(2x + 1)y' + y = 0$, given $y(0) = 1$, $y'(0) = 1$. Find $y\left(\frac{e-1}{2}\right)$.

CONTINUED

METHOD

Recognize $4x^2 + 4x + 1 = (2x + 1)^2$, turning the ODE into a Cauchy-Euler equation in disguise via $u = 2x + 1$; repeated-root structure gives $y = (2x + 1)^{-1/2}(A + B \ln(2x + 1))$ -type solution. The evaluation point is engineered so $2x + 1 = e$, landing on a clean log value.

RESULT

Not fully shown; requires solving for A, B from the ICs.

WHY IT MATTERS

A fresh "quadratic coefficient factors as a perfect square of a linear term, revealing a hidden Cauchy-Euler equation" DE2 family, with the eval point chosen so $2x + 1 = e$ cleans up logs. Complements the existing Cauchy-Euler menu entries (which use $x^2y'' + bxy' + cy = 0$ directly rather than disguising it).

8. Tsiolkovsky rocket fuel fraction word problem

DE2

EXAMPLE

A rocket of initial mass m_0 burns fuel at steady rate α , ejecting it at speed u relative to the rocket (no gravity/drag). Starting from rest, find the minimum initial fuel percentage so the rocket reaches speed $\geq u$ by the time all fuel is used.

METHOD

Variable-mass Newton's-law-in-disguise ODE $m dv = -u dm$ separates directly to $v = u \ln(m_0/m)$; setting $v = u$ at burnout mass m_f gives $m_f = m_0/e$, so the fuel fraction is $1 - 1/e$.

RESULT

$1 - \frac{1}{e} \approx 63.21\%$.

WHY IT MATTERS

The classic Tsiolkovsky rocket equation, not previously logged in the DE menu - a "physics-flavored but explicitly DE-slotted" branded word problem in the Medium tier, continuing the precedent (already flagged once, with the logistic-growth deer-population entry) that story-word-problem DE2's do occur and should be logged as legitimate DE patterns rather than skipped as physics.

Learned variants - Differential 03 - Hard

17 distinct screenshot-learned variants are documented with their example, method, result when recorded, and distinguishing lesson.

1. Cauchy Euler singular point regularity condition

DE3

EXAMPLE

$x^2 y'' + 2xy' - 2y = 1$, given $y(1) = 1$, $y'(0) = 2$, and $y(2) = \Xi^3$, find $[\Xi^2]$.

METHOD

The Cauchy-Euler equation has characteristic roots $r = 1, -2$ and a constant particular solution $y_p = -1/2$, so the general solution is $y = C_1 x + C_2 x^{-2} - \frac{1}{2}$. Because $x = 0$ is a singular point of the ODE, the $C_2 x^{-2}$ branch makes y' blow up at $x = 0$ unless $C_2 = 0$; requiring $y'(0)$ to exist as a finite value therefore forces $C_2 = 0$ before the numeric value of $y'(0)$ is even used to pin C_1 . The remaining condition(s) then solve for C_1 and back out the implicit constant Ξ from $y(2)$.

RESULT

Not fully resolved from the source capture (the three stated conditions look over-determined for the two free constants unless $y'(0) = 2$ is read as the regularity-forced value of C_1 directly); the technique - using finiteness at a singular point as a boundary condition rather than a plain numeric one - is the notable new element.

WHY IT MATTERS

First DE3 entry where a boundary condition is a regularity/existence condition at a singular point of the ODE rather than a plain evaluated value - forces recognizing that one homogeneous branch must vanish identically before doing any algebra with the stated numeric conditions. Distinct from the catalogued double-resonance and Clairaut-envelope DE3 families.

2. Clairaut implicit ODE singular solution via envelope

DE3

EXAMPLE

A curve $\mathcal{C}$ in the upper half-plane ($y > 0$) satisfies $y^2 + (x - y/y')^2 = K^2$ where $K = \int_0^\infty \frac{\ln(1+x^4)}{1+x^2} dx$ (a fixed constant). Write parametric equations $x = x(t)$, $y = y(t)$ for the singular-solution curve, where $t = y'$ is the tangent slope, and find $x(1) + y(1)$.

METHOD

This is a disguised Clairaut/implicit first-order ODE where $p = y'$ appears algebraically. Treating p as a free parameter, differentiate the defining relation with respect to p (holding the implicit constraint) to find the envelope of the family of solution curves - the singular solution not obtainable from the general solution - yielding explicit parametric formulas $x(t), y(t)$ in terms of $t = p$ and the constant K .

RESULT

Depends on the evaluated constant K (a Feynman-trick / Beta-function-style improper integral); the final $x(1) + y(1)$ is a clean expression in K .

WHY IT MATTERS

First appearance of a genuine Clairaut-style envelope/singular-solution construction in the DE3 menu - distinct from all catalogued resonance, Cauchy-Euler, and homogeneous-first-integral DE3 families. The extra twist of burying a nontrivial definite integral inside the ODE's constant is a novel "two-stage" DE3 mechanism (evaluate hidden constant, then do envelope calculus).

3. Daily Differential Hard tier (classic integral identity hidden inside an ODE)

DE3

EXAMPLE

"Peak memory overhead M over $\theta \in [0, \pi/2]$ has relative growth $\frac{1}{M} \frac{dM}{d\theta} = \frac{4 \ln 7}{\pi} \cdot \frac{1}{1 + \tan^{\sqrt{13}} \theta}$, $M(0) = 12$; find $M(\pi/2)$."

METHOD

separable $\Rightarrow \ln \frac{M(\pi/2)}{12} = \frac{4 \ln 7}{\pi} \int_0^{\pi/2} \frac{d\theta}{1 + \tan^a \theta}$, and the symmetry identity $\int_0^{\pi/2} \frac{d\theta}{1 + \tan^a \theta} = \frac{\pi}{4}$ for ANY a collapses it to $\ln 7$.

RESULT

$M(\pi/2) = 12 \cdot 7 = 84$ (verified).

WHY IT MATTERS

the Hard Differential is an elaborate branded story whose PAYOFF is a famous definite-integral identity buried in a separable/linear ODE; the DE is just a shell. Reusable embeds: $\int_0^{\pi/2} \sin^a / (\sin^a + \cos^a) = \pi/4$, Frullani, Dirichlet $\int \sin x/x = \pi/2$, a Gaussian, a Beta/ Γ value.

4. Double resonance Cauchy Euler

DE3

EXAMPLE

$x^2 T'' - 5xT' + 9T = x^3 \ln x$, $T(1) = 4$, $T'(1) = 7$, find $T(e)$.

METHOD

Cauchy-Euler with a repeated indicial root ($r = 3$) that ALSO collides with the exponent structure of the forcing term after the $x = e^t$ substitution, requiring a doubly-resonant particular-solution ansatz (an extra factor of t^2 beyond the usual repeated-root t -multiplier).

RESULT

$T(e) = -5e^3/6$.

WHY IT MATTERS

A sharper, "double resonance" variant of the already-catalogued Cauchy-Euler and undetermined-coefficients-resonance DE3 families - worth rotating in occasionally for an extra bite of difficulty on the Hard tier.

5. Euler Cauchy double root resonant cube log

DE3

EXAMPLE

Solve $x^2 y'' - 3xy' + 4y = x^2 \ln x$ for $x > 0$ with $y(1) = 0$, $y'(1) = 0$; find $6y(e)e^{-2}$.

METHOD

Euler-Cauchy with indicial $m^2 - 4m + 4 = 0$ (double root $m = 2$), so homogeneous solutions x^2 , $x^2 \ln x$. The forcing $x^2 \ln x$ is doubly resonant, giving particular part $y_p = \frac{1}{6} x^2 \ln^3 x$. The initial conditions kill both constants, so $y = \frac{1}{6} x^2 \ln^3 x$, $y(e) = \frac{e^2}{6}$, and $6y(e)e^{-2} = 1$.

RESULT

1

WHY IT MATTERS

"Hard Differential" tier - Euler-Cauchy equation whose double indicial root makes a $\ln x$ -forcing triply resonant, producing a $\ln^3 x$ particular solution. The clean $6y(e)e^{-2} = 1$ packaging is the tell. Distinct from the existing homogeneous-ode-conserved-ratio DE3 entry (that one was a first integral; this is a second-order resonance computation).

6. Euler equidimensional ODE double root alternating Catalan sum

DE3

EXAMPLE

$y(x, n)$ satisfies $4y'' + \frac{y}{x^2} = x^n$ with $\partial_x y|_{x=0} = 0$. Find $\sum_{n=0}^{\infty} (-1)^n y(1, n)$.

METHOD

Homogeneous part is Euler/equidimensional: indicial $4r(r-1) + 1 = (2r-1)^2 = 0$ gives the double root $r = \frac{1}{2}$, so $y_h = \sqrt{x}(A+B \ln x)$; the regularity condition at $x = 0$ kills both ($A = B = 0$). Particular $y_p = k x^{n+2}$ with $k[4(n+2)(n+1)+1] = 1$ and $4(n+2)(n+1)+1 = (2n+3)^2$, so $y(1, n) = 1/(2n+3)^2$. Then $\sum (-1)^n/(2n+3)^2 = 1-G$ where $G = \sum_{k \geq 0} (-1)^k/(2k+1)^2$ is Catalan's constant.

RESULT

$$\sum_{n=0}^{\infty} (-1)^n y(1, n) = 1 - G \approx 0.084034.$$

WHY IT MATTERS

Two-stage DE3: (1) an Euler equation whose forcing x^n makes $y(1, n) = 1/(2n+3)^2$ a clean function of the parameter n ; (2) the alternating series reindexes to Catalan's constant G (offset from the standard $\sum (-1)^k/(2k+1)^2 = G$ by the missing $k=0$ term). The tell is the $(2r-1)^2$ double indicial root and the parametrised forcing engineered so the amplitude is $1/(2n+3)^2$. First DE-slot appearance of Catalan's constant.

CONTINUED

7. Euler method exact fraction numerical ODE

DE3

EXAMPLE

$dy/dx = 1/(3+3x+x^2)$, $y(0) = 1$; use Euler's method with $h = 0.1$ to approximate $y(0.2)$ as a coprime fraction a/b ; find b .

METHOD

Since the RHS depends only on x (not y), each Euler step $y_{k+1} = y_k + h f(x_k)$ is just adding an exact rational increment. Carry the iteration in exact fractions rather than decimals: $y_1 = 1 + \frac{1}{10} \cdot \frac{1}{3} = \frac{31}{30}$, then $y_2 = y_1 + \frac{1}{10} \cdot \frac{100}{331} = \frac{10561}{9930}$ (already coprime).

RESULT

$y(0.2) \approx \frac{10561}{9930}$, so $b = 9930$ (verified via exact rational arithmetic).

WHY IT MATTERS

First DE3 entry built around a numerical method (forward Euler) rather than a closed-form ODE technique - the twist is that the "approximation" is demanded as an exact coprime fraction, forcing careful rational bookkeeping through each step rather than floating-point iteration.

8. Homogeneous ODE conserved ratio

DE3

EXAMPLE

Solve $y' = \frac{2x^3y-y^4}{x^4-2xy^3}$, $y(1) = 2$; evaluate $\frac{x^3+y^3}{xy}$ at $x = 2026$.

METHOD

The equation is exact / homogeneous with first integral $x^3 + y^3 = Cxy$; the requested ratio is the conserved constant $C = (1+8)/2 = 9/2$, independent of x .

RESULT

$9/2$

WHY IT MATTERS

"Hard Differential" register where the differential equation is a shell hiding a conserved quantity; the year-themed evaluation point $x = 2026$ is a red herring because the ratio is invariant along the trajectory.

9. Normal mgf all orders derivative operator inverts to 1 minus D

DE3

EXAMPLE

Let Z be a standard normal variable with MGF $M_Z(t)$. Solve $y(t) + y'(t) + y''(t) + \dots = M_Z(t)$ (sum of all orders of derivative) and evaluate $y(2)$.

METHOD

First, $M_Z(t) = e^{t^2/2}$. The operator $1 + D + D^2 + \dots = (1-D)^{-1}$, so $(1-D)^{-1}y = e^{t^2/2} \Rightarrow y = (1-D)e^{t^2/2} = e^{t^2/2} - \frac{d}{dt}e^{t^2/2} = (1-t)e^{t^2/2}$. Hence $y(2) = (1-2)e^2 = -e^2$.

RESULT

$-e^2$

CONTINUED

WHY IT MATTERS

Two recognitions: (i) the standard-normal MGF is $e^{t^2/2}$, and (ii) the infinite sum-of-all-derivatives operator inverts to the finite first-order operator $(1 - D)$, which is then applied directly to a non-exponential RHS. Distinct from both existing operator-series entries: the D2 "sum of ALL derivatives at 0" entry sums $f^{(n)}(0)$ for a known e^{ax} , and the D3 "alternating-odd-order" entry solves a homogeneous operator equation by a characteristic quadratic. Here the RHS is a given Gaussian and the trick is operator inversion $(1 - D)^{-1} \rightarrow (1 - D)$ applied to it.

10. Orthogonal trajectories

DE3

EXAMPLE

Given the family of curves $e^y \tan(x/2 + \pi/4) = C$, find the orthogonal trajectory through a given point and evaluate it at $x = \pi/2$.

METHOD

Implicitly log-differentiate the family to find its slope $y' = -\sec x$; the orthogonal family has slope $y' = \cos x$ (negative reciprocal); integrate with the given initial condition to get the orthogonal curve, then evaluate.

RESULT

(site example) π at the specified evaluation point

WHY IT MATTERS

A classical ODE application absent from the current DE3 menu (which covers PDEs, eigenvalue/BVP, Frobenius, Laplace-transform IVPs, etc.) - orthogonal trajectories is a clean, self-contained first-order technique that fits the "elaborate story, clean separable payoff" Hard-Differential register even though it's technically first-order, not the usual PDE/eigenvalue Hard fare.

11. Ratio encoded ICs annihilate homogeneous

DE3

EXAMPLE

$f''' + f'' - f' - f = e^x \sin x$, with $\frac{3}{f(0)} = \frac{7}{f'(0)} = \frac{8}{f''(0)} = -25$; find $f(2\pi)$.

METHOD

Characteristic $(r - 1)(r + 1)^2$; undetermined coefficients give $f_p = -\frac{(4 \sin x + 3 \cos x)e^x}{25}$. The ratio-encoded ICs are rigged so every homogeneous term (e^x, e^{-x}, xe^{-x}) cancels, leaving only f_p .

RESULT

$$f(2\pi) = -\frac{3e^{2\pi}}{25} \approx -64.26$$

WHY IT MATTERS

Initial data presented as equal ratios specifically engineered to zero out the complementary solution - an elegant ICs gimmick worth reusing on the DE Hard tier.

12. Resonant undetermined coefficients second order

DE3

EXAMPLE

$y'' + 4y = 16 \sin 2x$, $y(0) = 3$, $y'(0) = 5$; find $y(\pi/4)$.

METHOD

Forcing frequency = natural frequency $\Rightarrow$ resonance; particular $y_p = -4x \cos 2x$. ICs: $C_1 = 3$, $C_2 = 9/2$. At $x = \pi/4$, $\cos(\pi/2) = 0$, $\sin(\pi/2) = 1$ so $y = C_2 = 9/2$.

RESULT

9/2.

WHY IT MATTERS

Classic resonance case (forcing matches a homogeneous mode $\rightarrow$ multiply by x). The clean evaluation point $\pi/4$ kills the $x \cos 2x$ term. Confirms the site ships resonance in the Hard Differential tier.

13. Separable ODE log squared plus Weierstrass arctan branch at $3\pi/2$ **DE3****EXAMPLE**

$(1 + \sin^2 y) \ln(1 + x) \sum_{n=1}^{\infty} x^n dx = (\sin(2y) \ln(1 + \sin^2 y) + 1) dy$, with $|x| < 1$ and $\lim_{y \rightarrow (3\pi/2)^-} x(y) = 0$; find the constant of integration C .

METHOD

Sum the geometric series ($\sum_{n \geq 1} x^n = x/(1 - x)$) and divide by $1 + \sin^2 y$ to separate. On the y -side, spot that $u = \ln(1 + \sin^2 y)$ has $du = \sin(2y) dy/(1 + \sin^2 y)$, so the first term integrates to $\frac{1}{2} \ln^2(1 + \sin^2 y)$, while the leftover $\int dy/(1 + \sin^2 y) = \frac{1}{\sqrt{2}} \arctan(\sqrt{2} \tan y)$ (Weierstrass/standard reduction). Apply the boundary condition at $y \rightarrow (3\pi/2)^-$: there $\sin^2 y \rightarrow 1$ so the log term $\rightarrow \frac{1}{2} \ln^2 2$, and $\tan y \rightarrow +\infty$ so the arctan term $\rightarrow \pi/(2\sqrt{2})$; the x -side antiderivative $\int_0^x t \ln(1 + t)/(1 - t) dt$ vanishes as $x \rightarrow 0$.

RESULT

$C = -\frac{1}{2} \ln^2 2 - \frac{\pi}{2\sqrt{2}} = -\frac{1}{2} \ln^2 2 - \frac{\pi\sqrt{2}}{4} \approx -1.35095$ (under the standard normalisation that the x -side antiderivative is taken to vanish at $x = 0$; note that x -side antiderivative is itself dilogarithmic and non-elementary, which is precisely why the question asks only for C)

WHY IT MATTERS

New DE3 mechanism - a separable ODE whose answer is the constant of integration, fixed by a one-sided boundary condition placed at a point where $\tan y$ blows up so that the arctan branch contributes exactly $\pi/2$. Two recognitions are stacked: a geometric series to be summed before separating, and $\sin(2y)/(1 + \sin^2 y)$ being the log-derivative of $1 + \sin^2 y$ (so the messy first term is really $\int u du$). Distinct from the logged Clairaut/Euler-Cauchy/branch-selected-by-asymptotics DE3 entries - here nothing is solved for $y(x)$ at all; only the additive constant is extracted, and the non-elementary x -side is a deliberate red herring.

14. Square independent variable reduces variable coefficient ODE to oscillator**DE3****EXAMPLE**

Solve $xy'' - y' + 4x^3y = 0$ with $y(\sqrt{\pi}) = -1$ and $y'(\sqrt{\pi}) = 2\sqrt{\pi}$; find $y(\sqrt{\pi}/2)$.

METHOD

Set $t = x^2$ and write $y(x) = Y(t)$. Then $y' = 2xY_t$ and $y'' = 2Y_t + 4x^2Y_{tt}$, so the first-derivative terms cancel and the ODE becomes $Y_{tt} + Y = 0$. Translate the initial data at $t = \pi$, solve the harmonic oscillator, and evaluate at $t = \pi/4$.

RESULT

$Y(t) = \cos t - \sin t$, hence $y(\sqrt{\pi}/2) = 0$.

WHY IT MATTERS

A variable-coefficient second-order equation can be a constant-coefficient oscillator in a disguised independent variable; the cancellation in $xy'' - y'$ is the cue for $t = x^2$.

15. Taylor series self recognition functional equation**DE3****EXAMPLE**

f entire, and for all $x > 0$: $\sum_{n=0}^{\infty} \frac{x f^{(n)}(x-1) - (e(x-1))^n}{n!} = 0$. Find $f(2)$.

METHOD

Split the sum; $\sum_n (e(x-1))^n/n! = e^{e(x-1)}$ is a plain exponential series, while $\sum_n f^{(n)}(x-1)/n! \cdot 1^n$ is exactly the Taylor series of f about $x-1$ evaluated one unit away, i.e. $f(x)$ itself (Taylor's theorem for an entire function). The equation collapses to the algebraic relation $xf(x) = e^{e(x-1)}$.

RESULT

$f(2) = \frac{e^e}{2}$.

CONTINUED

WHY IT MATTERS

A new DE3 mechanism distinct from the logged "sum of all derivatives at a single point equals a geometric series" D2/D3 entries - here the derivatives are evaluated at a variable, shifted point ($x - 1$, not a fixed point like 0), and recognizing the sum as literally $f(x)$ via Taylor's theorem converts an intimidating series/functional equation directly into a closed form for the unknown entire function.

16. Transcendental derivative equation with branch selected by asymptotic BC

DE3

EXAMPLE

$\sin(y') = \sin(y)$, $x > 0$, $y(0) = 8\pi$, $y(\infty) = 5\pi$; find $y(1)$.

METHOD

$\sin(A) = \sin(B)$ has branches $A = B + 2k\pi$ or $A = \pi - B + 2k\pi$. The boundary condition at infinity (a finite limiting value, forcing $y' \rightarrow 0$ there) selects the branch, collapsing to the linear ODE $y' = 5\pi - y$.

RESULT

$y(1) = \pi(5 + 3/e) \approx 19.175$.

WHY IT MATTERS

New DE3 mechanism - "the derivative satisfies a periodic/multivalued equation in itself and the dependent variable; a boundary condition at infinity, not algebra, selects the branch." Distinct from Euler-Cauchy, Bernoulli, Abel, and the hidden-classic-integral-identity shell already in the menu.

17. Variable coefficient ODE via power series

DE3

EXAMPLE

$y''(t) + 3t y'(t) - 6y(t) = 2$, $y(0) = 0$, $y'(0) = 0$. Find $y(\pi)$.

METHOD

Polynomial-coefficient (non-Euler, non-Bernoulli) second-order linear ODE; solved via Frobenius/Maclaurin series recurrence for the coefficients of $y(t) = \sum a_n t^n$, or by isolating a polynomial particular solution for the constant forcing combined with a series-built homogeneous complement.

RESULT

Not shown; requires deriving the recurrence and summing (or truncating) the series at $t = \pi$.

WHY IT MATTERS

A genuinely new "variable-coefficient ODE solved by power series, not by name-brand substitution" DE3 flavor - thin coverage in the existing menu beyond Airy/Hermite/Legendre-style Frobenius, and this one has a $t y'$ (not $t^2 y''$) coefficient structure.

Learned variants - Limit - Daily

30 distinct screenshot-learned variants are documented with their example, method, result when recorded, and distinguishing lesson.

1. 1^∞ exponential mean limit, solve for n

L1

EXAMPLE

$$\lim_{x \rightarrow 0} \left(\frac{e^x + e^{2x} + \dots + e^{nx}}{n} \right)^{1/x} = e^{10}; \text{ find } n.$$

METHOD

$\ln(\cdot) \rightarrow$ by L'Hopital the exponent $\rightarrow \frac{1}{n} \sum k = \frac{n+1}{2}$; set $\frac{n+1}{2} = 10$.

RESULT

$n = 19$

WHY IT MATTERS

the geometric/arithmetic-mean-of-exponentials raised to $1/x$ limits to $e^{\text{mean of slopes}} = e^{(n+1)/2}$; a clean " 1^∞ then solve a linear equation" item.

2. Alternating zeta loggamma Euler mascheroni limit

L1

EXAMPLE

$$\lim_{k \rightarrow \infty} \sum_{n=2}^k \frac{(-1)^n \zeta(n)}{n}$$

METHOD

Recognize the sum as the Taylor series of $\ln \Gamma(1+x) = -\gamma x + \sum_{n=2}^{\infty} \frac{(-1)^n \zeta(n)}{n} x^n$ evaluated at the boundary $x = 1$; since $\Gamma(2) = 1$, $\ln \Gamma(2) = 0$, isolating the sum as γ .

RESULT

γ (Euler-Mascheroni constant, ≈ 0.5772).

WHY IT MATTERS

A distinctive "named-constant-as-answer" L1 flavor via the $\ln \Gamma$ Taylor series at $x = 1$ - a different special-function route from the already-logged Euler sine-product L1 entry (which used the product formula for $\sin$, not the Gamma function's log-Taylor series) and from the Stirling/Wallis/Stolz-Cesàro/telescoping-product families.

3. Arc length limit via limiting shape

L1

EXAMPLE

$\lim_{n \rightarrow \infty}$ (arc length of $y = x^n$ on $[-1, 1]$).

METHOD

Do NOT push the limit inside $\int \sqrt{1 + (nx^{n-1})^2}$. The graph converges (in shape) to an L on $[0, 1]$ (unit horizontal + unit vertical) and its mirror, total polyline length 4.

RESULT

4.

WHY IT MATTERS

A limit whose trap is the non-uniform convergence of arc length - answer comes from the limiting polygonal shape, not from swapping limit and integral. Distinct from Riemann-sum/Stirling/Wallis limit entries.

4. Arithmetico geometric series times n limit

L1

EXAMPLE

$$\lim_{n \rightarrow \infty} n \left(\sum_{k=0}^{\infty} \frac{k}{n^k} \right).$$

METHOD

Closed form of the arithmetico-geometric series $\sum_{k \geq 0} k x^k = \frac{x}{(1-x)^2}$ with $x = 1/n$. So the inner sum is $\frac{1/n}{(1-1/n)^2}$ and

$$n \cdot (\text{sum}) = \frac{1}{(1-1/n)^2} \rightarrow 1.$$

RESULT

1

WHY IT MATTERS

A Limit whose payload is recognising the inner sum as an arithmetico-geometric series with the standard closed form $\sum kx^k = x/(1-x)^2$; the outer factor n exactly cancels the $1/n$ so the limit is a clean 1. Distinct from existing L1 entries (Riemann-sum-to-integral, Stirling/Wallis, Γ -Maclaurin, $\cosh + \cos$, Cauchy-repeated-integral) - here there is no integral or asymptotic, just an exact generating-function evaluation followed by $n \rightarrow \infty$.

5. Cauchy repeated integration of log limit

L1

EXAMPLE

For $x > 0$ and $n \geq 0$ let $F_0(x) = \ln x$ and $F_{n+1}(x) = \int_0^x F_n(t) dt$. Evaluate $\lim_{n \rightarrow \infty} \frac{n! F_n(1)}{\ln n}$.

METHOD

By Cauchy's formula for repeated integration, $F_n(x) = \frac{1}{(n-1)!} \int_0^x (x-t)^{n-1} \ln t dt$. At $x = 1$: $F_n(1) = \frac{1}{(n-1)!} \int_0^1 (1-t)^{n-1} \ln t dt = \frac{1}{(n-1)!} \cdot \left(-\frac{H_n}{n}\right)$ (the Beta-type integral $\int_0^1 (1-t)^{n-1} \ln t dt = -H_n/n$). So $n! F_n(1) = -H_n$, and since $H_n \sim \ln n$, the limit $\frac{-H_n}{\ln n} \rightarrow -1$.

RESULT

-1.

WHY IT MATTERS

A Limit whose payload is recognising iterated antidifferentiation as Cauchy's repeated-integral formula, then the Beta-derivative identity $\int_0^1 (1-t)^{n-1} \ln t dt = -H_n/n$, then the harmonic-number asymptotic $H_n \sim \ln n$. Three-stage: operator $\rightarrow$ special integral $\rightarrow$ asymptotic. Distinct from telescoping-sum and polylog-cancellation L1 entries; the $n!$ prefactor is engineered to leave exactly $-H_n$.

6. Conjugate rationalization square root difference limit

L1

EXAMPLE

$$\lim_{x \rightarrow 0} \frac{\sin^2 x}{\sqrt{2} - \sqrt{1 + \cos x}}$$

METHOD

Multiply num/denom by the conjugate $\sqrt{2} + \sqrt{1 + \cos x}$; the denominator becomes $2 - (1 + \cos x) = 1 - \cos x$. Then $\sin^2 x / (1 - \cos x) = 1 + \cos x$, leaving $(1 + \cos x)(\sqrt{2} + \sqrt{1 + \cos x})$.

RESULT

$$4\sqrt{2}$$

WHY IT MATTERS

Elementary conjugate-rationalization of a difference-of-square-roots denominator, paired with the $\sin^2 / (1 - \cos) = 1 + \cos$ identity. Distinct from the Taylor-leading-term and cube-root-minus-sqrt L1 entries - no series needed, pure algebra.

7. Cosh+cos averaging kills low orders > $e^{1/24}$

L1

EXAMPLE

$$\lim_{x \rightarrow 0} \left(\frac{\cosh x + \cos x}{2} \right)^{1/x^4}.$$

METHOD

$\cosh x + \cos x = 2 + \frac{2x^4}{4!} + O(x^8)$ - the x^2 terms cancel between cosh and cos, leaving a pure x^4 leading term. Base $= 1 + \frac{x^4}{24} + \dots$, so $\frac{1}{x^4} \ln(\text{base}) \rightarrow \frac{1}{24}$.

RESULT

$e^{1/24}$ (verified: at $x = 0.03$ the value is 1.0425469 vs $e^{1/24} = 1.0425469$).

WHY IT MATTERS

Flagship 1^∞ limit - the elegance is the x^2 -cancellation forcing a $1/x^4$ exponent. Strong new L1 anchor; pairs naturally with the half-angle/parity "cancellation" aesthetic.

8. Digamma reflection alternating quad sum

L1

EXAMPLE

$$\lim_{n \rightarrow \infty} \sum_{k=1}^n \left(\frac{1}{4k+1} + \frac{1}{4k+2} - \frac{1}{4k+3} - \frac{1}{4k+4} \right)$$

METHOD

Recognize each group of four consecutive-mod-4 terms as pieces of digamma-function asymptotic expansions; use the reflection formula $\psi(3/4) - \psi(1/4) = \pi$ together with $\psi(1/2) = -\gamma - 2 \ln 2$ and standard digamma special values to assemble the closed form.

RESULT

$\pi/4 + (\ln 2)/2 - 11/12 \approx 0.215$ (numerically verified)

WHY IT MATTERS

An advanced L1 family not in the current menu - digamma reflection/special-value bookkeeping applied to a sum with period-4 sign pattern, distinct from the catalogued Euler-Maclaurin, Stolz-Cesàro, and telescoping-product L1 flavors.

9. Dilog inversion limit

L1

EXAMPLE

$$\lim_{x \rightarrow \infty} \left(-\text{Li}_2(-x) - \frac{1}{2} \ln^2(x+1) \right)$$

METHOD

Use the dilogarithm inversion identity $\text{Li}_2(-x) + \text{Li}_2(-1/x) = -\frac{\pi^2}{6} - \frac{1}{2} \ln^2 x$; the divergent $\ln^2$ pieces cancel the subtracted counterterm, leaving a finite constant.

RESULT

$$\frac{\pi^2}{6}$$

WHY IT MATTERS

A limit whose entire payload is an asymptotic cancellation between a polylog and a hand-built $\ln^2$ counterterm - not L'Hôpital or series-leading-term. Rewards knowing special-function inversion formulas rather than standard limit machinery.

10. Dottie number limit of a parametric integral

L1

EXAMPLE

$$\lim_{n \rightarrow \infty} \int_0^1 \frac{1}{1+(x \sec x)^n} dx$$

METHOD

$(x \sec x)^n \rightarrow 0$ where $x/\cos x < 1$ and $\rightarrow \infty$ where > 1 ; the integrand $\rightarrow$ the indicator of $\{x \sec x < 1\} = \{x < x^*\}$ where $x^* = \cos x^*$. The limit equals x^* , the Dottie number.

RESULT

$x^* \approx 0.739085$ (root of $x = \cos x$).

WHY IT MATTERS

a "limit of an integral with an n th power" that converges to a measure/indicator; the threshold is the Dottie fixed point - a fresh way to source that constant.

11. Double limit Riemann sum telescoping reciprocal squares**L1****EXAMPLE**

$$\lim_{m \rightarrow \infty} \lim_{n \rightarrow \infty} \sum_{k=1}^m \sum_{r=kn+1}^{(k+1)n} \frac{n}{r^2}$$

METHOD

For fixed k , the inner sum is a Riemann sum: with $r = nx$, $\sum_{r=kn+1}^{(k+1)n} \frac{n}{r^2} \rightarrow \int_k^{k+1} \frac{dx}{x^2} = \frac{1}{k} - \frac{1}{k+1}$. Summing over $k = 1..m$ telescopes to $1 - \frac{1}{m+1} \rightarrow 1$.

RESULT

1

WHY IT MATTERS

"Daily Limit" flavor with a nested double limit: the inner $n \rightarrow \infty$ is a Riemann-sum-to-integral conversion, the outer $m \rightarrow \infty$ a telescoping series. Distinct from existing Riemann-sum limit entries because of the two-stage limit order and the telescoping reciprocal-squares structure.

12. Dual asymptotic sum ratio solve for parameter**L1****EXAMPLE**

For $a \in \mathbb{R}$, $|a| > 1$: $\lim_{n \rightarrow \infty} \frac{\sum_{r=1}^n r^{1/3}}{n^{7/3} \sum_{r=1}^n \frac{1}{(an+r)^2}} = 54$. Find the sum of squares of all possible a .

METHOD

Numerator is a Faulhaber/Euler-Maclaurin power-sum asymptotic ($\sum r^{1/3} \sim \frac{3}{4}n^{4/3}$); denominator's inner sum is a genuine Riemann sum in disguise ($\rightarrow \frac{1}{n} \int_0^1 \frac{dx}{(a+x)^2}$). The matching $n^{4/3}$ orders cancel, collapsing the limit to a pure algebraic equation in the parameter a , which is then solved as a quadratic.

RESULT

$a = 8$ or $a = -9$; sum of squares = 145.

WHY IT MATTERS

Distinct from the existing L1 "Riemann sum in disguise" and telescoping entries - here TWO different asymptotic mechanisms (a power-sum integral-approximation and a true Riemann sum) are pitted against each other in a ratio, and the payoff is solving for an unknown parameter rather than evaluating a fixed number.

13. Endpoint Laplace asymptotic weighted power integral limit**L1****EXAMPLE**

$$\lim_{n \rightarrow \infty} \left(n + \frac{1}{2}\right) \int_0^1 x^n e^x dx$$

METHOD

For large n , $\int_0^1 x^n f(x) dx$ is dominated by the behavior of f near $x = 1$ (Laplace/Watson's-lemma-style endpoint asymptotics); substituting $x = 1 - s/n$ gives $\int_0^1 x^n e^x dx \sim e/(n+1)$ to leading order, with an $O(1/n^2)$ correction. Multiplying by $(n + \frac{1}{2})$ leaves the leading term $e \cdot \frac{n+1/2}{n+1} \rightarrow e$ while the correction term vanishes in the limit.

RESULT

$e \approx 2.71828$ (verified numerically: the product converges slowly, from ≈ 2.394 at $n = 10$ up to ≈ 2.7175 at $n = 5000$).

WHY IT MATTERS

A new L1 family - the " $n \cdot \int_0^1 x^n f(x) dx \rightarrow f(1)$ " endpoint-asymptotic mechanism, here dressed with a $(n + \frac{1}{2})$ multiplier (rather than a bare n) that doesn't change the limit but is a recognizable convergence-acceleration flourish. Distinct from the catalogued Riemann-sum, Euler-Maclaurin, and dual-asymptotic-ratio L1 entries.

14. Euler sine product log sum limit

L1

EXAMPLE

$$\lim_{x \rightarrow 0} \frac{1}{x^2} \sum_{n=1}^{\infty} \ln \left(1 - \frac{x^2}{n^2 \pi^2} \right)$$

METHOD

Recognize $\prod_{n=1}^{\infty} \left(1 - \frac{x^2}{n^2 \pi^2} \right) = \frac{\sin x}{x}$ (Euler's infinite product for sine), so the sum of logs equals $\ln(\sin x/x)$; Taylor-expand $\ln(\sin x/x) \approx -x^2/6$ near 0 and divide by x^2 .

RESULT

$$-\frac{1}{6}.$$

WHY IT MATTERS

A genuinely distinctive L1 flavor - the sum-of-logs-as-log-of-infinite-product trick (via Euler's sine product) hasn't appeared in the existing L1 menu (which has Riemann sums, Stirling/Wallis asymptotics, Stolz-Cesàro, equidistribution, telescoping products, but no infinite-product-representation-of-sin route). Strong candidate for a future L1 slot.

15. Factorial series resolving to a sin/cos combination

L1

EXAMPLE

$$\sum_{n=0}^{\infty} (-1)^n \frac{n+1}{(2n+1)!}$$

METHOD

split $n+1 = \frac{1}{2}(2n+1) + \frac{1}{2}$; the first piece collapses the factorial to $\frac{1}{2} \sum \frac{(-1)^n}{(2n)!} = \frac{1}{2} \cos 1$, the second is $\frac{1}{2} \sum \frac{(-1)^n}{(2n+1)!} = \frac{1}{2} \sin 1$.

RESULT

$$\frac{1}{2}(\sin 1 + \cos 1) \approx \mathbf{0.69089} \text{ (verified).}$$

WHY IT MATTERS

Series-recognition Limit where the linear numerator is split so each half matches a known Maclaurin series at $x = 1$. Reward is the algebraic split, not convergence analysis.

16. Factorial square root product limit maclaurin

L1

EXAMPLE

$$\lim_{x \rightarrow 0} \frac{\sqrt{e^x - x!} \tan(x - \sin x)}{x}, \text{ where } x! = \Gamma(x+1).$$

METHOD

Near $x = 0$: $\Gamma(x+1) \approx 1 - \gamma x + O(x^2)$ (Euler-Mascheroni constant γ), so $e^x - x! \approx (1+x+\dots) - (1-\gamma x+\dots) = (1+\gamma)x + O(x^2)$. Thus $\sqrt{e^x - x!} \approx \sqrt{(1+\gamma)x}$. Also $x - \sin x \approx x^3/6$, so $\tan(x - \sin x) \approx x^3/6$. Product $\approx \sqrt{(1+\gamma)x} \cdot x^3/6 = (x^3/6)\sqrt{(1+\gamma)x}$ which goes to 0 as $x \rightarrow 0$ faster than x . So the limit is 0.

RESULT

0

WHY IT MATTERS

A limit requiring the Γ function expansion at $x = 0$ (i.e. $\Gamma(1+x) \approx 1 - \gamma x$) combined with the $x - \sin x \approx x^3/6$ standard series. Both factors vanish, making the overall expression $O(x^{7/2}/x) = O(x^{5/2}) \rightarrow 0$. New L1 sub-flavor: "factorial under a radical forces

Gamma-function Maclaurin at 0".

17. Fibonacci golden ratio power decomposition ratio limit

L1

EXAMPLE

Every positive power φ^n can be written as $a + b\varphi$ where φ is the golden ratio. Let φ_a^n, φ_b^n be the a and b parts of φ^n . Find

$$\lim_{n \rightarrow \infty} \left(\frac{\varphi_b^n}{\varphi_a^n} - \frac{\varphi_a^n}{\varphi_b^n} \right).$$

CONTINUED

METHOD

Recognise $\varphi^n = F_{n-1} + F_n\varphi$ (the classical Fibonacci decomposition of golden-ratio powers), so $a = F_{n-1}$, $b = F_n$. Since $F_n/F_{n-1} \rightarrow \varphi$ and $F_{n-1}/F_n \rightarrow 1/\varphi$ as $n \rightarrow \infty$, the expression tends to $\varphi - 1/\varphi$, which collapses via $1/\varphi = \varphi - 1$ to a clean integer.

RESULT

1.

WHY IT MATTERS

A new L1 family bridging the Fibonacci sequence and the golden-ratio power-decomposition identity - distinct from the catalogued dilog-inversion and telescoping-sum L1 flavors. The payoff is recognizing a, b as literally consecutive Fibonacci numbers in disguise, then using the standard Fibonacci-ratio-to- φ limit plus the defining quadratic $\varphi^2 = \varphi + 1$ to collapse to 1.

18. Harmonic number cesaro asymptotic

L1

EXAMPLE

$\lim_{n \rightarrow \infty} \frac{n^2}{H_n \sum_{k=1}^n \frac{k}{H_k}}$, H_n the harmonic number.

METHOD

$H_n \sim \ln n$, so $\frac{k}{H_k} \sim \frac{k}{\ln k}$ and (Stolz-Cesàro) $\sum_{k=1}^n \frac{k}{H_k} \sim \frac{n^2}{2 \ln n} = \frac{n^2}{2H_n}$. Denominator $\sim H_n \cdot \frac{n^2}{2H_n} = \frac{n^2}{2}$.

RESULT

2

WHY IT MATTERS

A Cesàro/Stolz asymptotic built specifically on harmonic numbers, with H_n appearing both inside the sum and as an outside factor that cancels - a cleaner-than-Stirling packaging.

19. Indeterminate power base near 1 limit large exponent

L1

EXAMPLE

The largest non-negative integer a for which $\lim_{x \rightarrow 1} \left\{ \frac{-ax + \sin(x-1) + a}{x + \sin(x-1) - 1} \right\}^{(1-x)/(1-\sqrt{x})} = \frac{1}{4}$.

METHOD

The exponent $(1-x)/(1-\sqrt{x}) = (1-\sqrt{x})(1+\sqrt{x})/(1-\sqrt{x}) = 1+\sqrt{x} \rightarrow 2$ as $x \rightarrow 1$. The base: at $x = 1$ numerator and denominator both vanish (use $\sin(x-1) \approx (x-1)$ for $x \approx 1$); numerator $\approx -a(x-1) + (x-1) = (1-a)(x-1)$ and denominator $\approx (x-1) + (x-1) = 2(x-1)$. So base $\rightarrow (1-a)/2$. The limit equals $((1-a)/2)^2 = 1/4$, so $(1-a)/2 = \pm 1/2$, giving $a = 0$ or $a = 2$. Largest non-negative: $a = 2$.

RESULT

 $a = 2$

WHY IT MATTERS

The exponent simplifies to 2 via the $(\sqrt{x})$ factorization, converting a complicated 1^∞ form into a^2 base. The near- $x = 1$ linearization of numerator and denominator is the key. New L1 flavor: "exponent simplification via radical factoring + linear-approximation on the base".

20. Lambert w maclaurin product with sin

L1

EXAMPLE

$\lim_{x \rightarrow 0} \frac{(W_0(x) - x + x^2)(\sin x - x)}{x^6}$

METHOD

Expand $W_0(x) = x - x^2 + \frac{3}{2}x^3 - \frac{8}{3}x^4 + \dots$ (Lambert W Maclaurin series). Then $W_0(x) - x + x^2 = \frac{3}{2}x^3 + O(x^4)$. And $\sin x - x = -x^3/6 + O(x^5)$. Product leading term: $\frac{3}{2}x^3 \cdot (-x^3/6) = -x^6/4$. Divide by x^6 .

RESULT

 $-1/4$

WHY IT MATTERS**CONTINUED**

A limit built from the product of two "Taylor-to-leading-order" series; each factor requires knowing the first non-trivial term of $W_0(x)$ and $\sin x$ respectively beyond their linear part. Distinct from the existing dilog-inversion and conjugate-rationalization L1 entries - the key skill is knowing $W_0(x) = x - x^2 + \frac{3}{2}x^3 - \dots$ and multiplying series to find the x^6 coefficient.

21. Ln over Lambert W ratio limit equals one**L1****EXAMPLE**

$\lim_{x \rightarrow \infty} \frac{\ln x}{W(x)}$, where W is the Lambert W function.

METHOD

Use the large-argument asymptotic $W(x) = \ln x - \ln \ln x + o(1)$. Then $\frac{\ln x}{W(x)} = \frac{\ln x}{\ln x - \ln \ln x + \dots} \rightarrow 1$ since $\ln \ln x = o(\ln x)$. (Convergence is slow: ≈ 1.02 even at $x = 10^{100}$.)

RESULT

1.

WHY IT MATTERS

A short "know the asymptotics of a special function" L1 - Lambert W appears as a denominator whose leading term is exactly $\ln x$, so the ratio tends to 1. Distinct from the logged Lambert- W entries (which are integrals or Maclaurin-product derivatives); this is purely the $W(x) \sim \ln x$ headline asymptotic. The slow convergence makes it resistant to naive numerical guessing.

22. Odd harmonic series via integral interchange**L1****EXAMPLE**

$\lim_{n \rightarrow \infty} \sum_{r=0}^n \frac{1}{2r+1} \int_0^1 \sin((2r+1)x) dx$

METHOD

evaluate the inner integral, recognise the resulting odd-indexed series as a Fourier/ ζ -type sum.

RESULT

$\pi^2/8$ (odd-square series $\sum 1/(2k+1)^2$)

WHY IT MATTERS

series-integral interchange landing on $\pi^2/8$; a clean way to source the $\pi^2/8$ constant in a limit slot.

23. Plain Taylor cancellation rational limit**L1****EXAMPLE**

$\lim_{x \rightarrow 0} \frac{1 - \cos x - \frac{1}{2}x^2}{x^4}$.

METHOD

Maclaurin to the first surviving order: $\cos x = 1 - \frac{x^2}{2} + \frac{x^4}{24} - \dots$, so the numerator $= -\frac{x^4}{24} + O(x^6)$ and the ratio $\rightarrow -\frac{1}{24}$.

RESULT

$-1/24$.

WHY IT MATTERS

A bare, un-disguised Taylor-cancellation limit (the engineered numerator subtracts off exactly the constant and quadratic terms of $\cos$). Distinct from the existing L1 entries, which are 1^∞ exponential-mean limits, Stirling/Wallis asymptotics, Lambert- W , Weyl equidistribution, or higher-order $\sin(\tan x) - \tan(\sin x)$ matches. This is the clean canonical "subtract the series prefix, read the next coefficient" anchor.

24. Poisson clt median limit sum is a probability**L1****EXAMPLE**

$$\lim_{n \rightarrow \infty} e^{-n} \sum_{k=0}^n \frac{n^k}{k!}$$

METHOD

Recognise the sum as $P(X \leq n)$ for $X \sim \text{Poisson}(n)$. Since a $\text{Poisson}(n)$ has mean n and standard deviation $\sqrt{n}$, the CLT puts the mass symmetrically about n and the limit is the median-at-the-mean value $\frac{1}{2}$. (Classically the Ramanujan Q -function / Szeg\H{o} result.)

RESULT

$$\frac{1}{2}$$

WHY IT MATTERS

FIRST probabilistic identity to resolve an L1 — the existing menu is Riemann sums, Stolz–Ces\aro, Stirling/Wallis, squeeze, endpoint-Laplace, telescoping products, equidistribution. The distinctive tell is the slow $\frac{1}{2} + O(n^{-1/2})$ approach ($n = 1000$ still gives only 0.5084), so a numerical probe alone will not reveal the answer. Naive dominant-term reasoning wrongly gives 0 or 1.

25. Ratio of $\sin / \sinh$ and $\arcsin / \operatorname{arcsinh}$ small argument limit**L1****EXAMPLE**

$$\lim_{x \rightarrow 0} \frac{\sin(\sqrt{x}) \arcsin(x^2)}{\sinh(\sqrt{x}) \operatorname{arcsinh}(x^2)}.$$

METHOD

Leading equivalents $\sin u \sim u \sim \sinh u$ and $\arcsin v \sim v \sim \operatorname{arcsinh} v$ cancel pairwise; limit is 1. (Next-order terms confirm the approach.)

RESULT

1 (verified numerically).

WHY IT MATTERS

A limit whose punchline is that circular and hyperbolic siblings share the same leading Taylor term, so the ratio is trivially 1 — a "spot the cancellation, don't grind" Limit anchor.

26. Reciprocal log limit with asinh expansion**L1****EXAMPLE**

$$\lim_{x \rightarrow 0} \left(\frac{1}{\ln(x + \sqrt{x^2 + 1})} - \frac{1}{\ln(1 + x)} \right).$$

METHOD

Recognise $\ln(x + \sqrt{1 + x^2}) = \operatorname{arsinh} x$, expand both logarithms through enough terms, then subtract reciprocal series so the $1/x$ singularities cancel.

RESULT

Finite constant from the next-order expansion.

WHY IT MATTERS

Limit problems can involve cancellation between two singular reciprocal logarithms; require expansions beyond first order.

27. Second order telescoping tan difference Riemann sum**L1****EXAMPLE**

$$\lim_{n \rightarrow \infty} \frac{1}{n^3} \sum_{k=1}^{n-1} \frac{\sin\left(\frac{(2k-1)\pi}{2n}\right)}{\cos^2\left(\frac{(k-1)\pi}{2n}\right) \cos^2\left(\frac{k\pi}{2n}\right)}$$

METHOD

Recognise $\sin(A-B)/(\cos^2 A \cos^2 B)$ -type terms as a second-order telescoping difference of $\tan$ (built from $\tan B - \tan A = \sin(B-A)/(\cos A \cos B)$, applied twice), so the sum collapses before the $1/n^3$ Riemann-sum scaling is applied.

RESULT

not fully derived in the source capture (flagged as advanced).

WHY IT MATTERS

New L1 family - a "telescoping product" cousin but for tan-difference sums at second order with cubic (not quadratic) normalization; distinct from the catalogued first-order telescoping-tan and Riemann-sum L1 families.

CONTINUED**28. Small parameter exponential to large variable growth****L1****EXAMPLE**

$g(x) = \lim_{r \rightarrow 0} ((x+1)^{r+1} - x^{r+1})^{1/r}$; find $\lim_{x \rightarrow \infty} g(x)/x$.

METHOD

Take logarithms in the inner $r \rightarrow 0$ limit, expand to first order, simplify the resulting closed form, then take the outer large- x limit.

RESULT

e .

WHY IT MATTERS

Nested limits should be completed in order; a 1^∞ form becomes an exponential of a logarithmic derivative.

29. Sophie germain telescoping sum**L1****EXAMPLE**

$$\lim_{k \rightarrow \infty} \sum_{r=1}^k \frac{4r^2 - 2}{4r^4 + 1}$$

METHOD

Sophie Germain identity $4r^4 + 1 = (2r^2 - 2r + 1)(2r^2 + 2r + 1) = (2(r-1)r + 1)(2r(r+1) + 1)$. The numerator $4r^2 - 2$ splits into a partial-fraction difference $1/(2(r-1)r + 1) - 1/(2r(r+1) + 1)$ which telescopes consecutively.

RESULT

1

WHY IT MATTERS

Existing L1 menu has telescoping infinite products via $k^3 \pm 1 = (k \pm 1)(k^2 \mp k + 1)$. The Sophie-Germain $4r^4 + 1$ factorisation produces a telescoping sum - same spirit, different identity, different slot flavor.

30. Variable term count sum squeeze to one**L1****EXAMPLE**

$$\lim_{x \rightarrow \infty} \left(\frac{1}{\sqrt{x^2 + 1}} + \frac{1}{\sqrt{x^2 + 2}} + \cdots + \frac{1}{\sqrt{x^2 + x}} \right)$$

METHOD

There are x terms, each between the smallest and largest: $\frac{x}{\sqrt{x^2 + x}} \leq S(x) \leq \frac{x}{\sqrt{x^2 + 1}}$. Both bounds $\rightarrow 1$ (e.g. $x/\sqrt{x^2 + x} = 1/\sqrt{1 + 1/x}$), so by the squeeze theorem $S(x) \rightarrow 1$. Equivalently, each term is $\frac{1}{x}(1 + O(1/x))$ and there are x of them.

RESULT

1 (verified numerically: $S(10^3) \approx 0.99975$, $S(10^4) \approx 0.999975$)

WHY IT MATTERS

The L1 menu lists "squeeze theorem" only as a bare bullet with no worked anchor - this is the canonical one, and its distinctive feature is that the number of summands is itself the limit variable (the sum's length grows with x), so the naive move of taking the limit termwise gives $0 + 0 + \cdots$ and loses the answer. Distinct from the catalogued Riemann-sum L1 entries because there is no $1/n$ weight to spot: the $1/x$ scale is hidden inside each radical, and no integral is needed - two crude bounds suffice.

Learned variants - Logic - Daily

40 distinct screenshot-learned variants are documented with their example, method, result when recorded, and distinguishing lesson.

1. AM-HM product invariant and geometric mean

P1

EXAMPLE

Tony writes the powers of 2 from 2^0 to 2^{15} on a board, then repeatedly erases two numbers and replaces them with their arithmetic mean and their harmonic mean. The numbers converge to a single common value; find it.

METHOD

PRODUCT INVARIANT - $AM(a, b) \cdot HM(a, b) = \frac{a+b}{2} \cdot \frac{2ab}{a+b} = ab$, so every move preserves the product of all 16 numbers.

Since $AM \geq GM \geq HM$ squeezes the spread to zero, the common limit L must satisfy $L^{16} = \prod_{k=0}^{15} 2^k = 2^{120}$, i.e. L is the geometric mean of the starting multiset.

RESULT

$$L = 2^{120/16} = 2^{15/2} = 128\sqrt{2} \approx 181.019$$

WHY IT MATTERS

Classic "find the monovariant/invariant" P1 where the invariant is multiplicative rather than additive, and it is exactly the pairing $AM \times HM = GM^2$ that makes it work; the triangular-number exponent sum $\sum_{k=0}^{15} k = 120$ then divides evenly enough to leave a half-integer power of 2. Distinct from the catalogued CRT / modular-order / digit-manipulation P1 families - this is a board-game process puzzle whose whole content is spotting that the product never changes, and it pairs naturally with the AM-GM-HM inequality chain to justify that a single limit value even exists.

2. $\arccos(\cos x)$ sawtooth linear intersection count

P1

EXAMPLE

Let $f : [0, 4\pi] \rightarrow [0, \pi]$ be defined by $f(x) = \cos^{-1}(\cos x)$. Find the number of points $x \in [0, 4\pi]$ satisfying $f(x) = \frac{10-x}{10}$.

METHOD

$f(x) = \cos^{-1}(\cos x)$ is a triangular sawtooth: it rises linearly from 0 to π on $[0, \pi]$, drops from π to 0 on $[\pi, 2\pi]$, and repeats with period 2π . The right side $g(x) = (10-x)/10$ is a decreasing line from 1 at $x=0$ to -3 at $x=40$. Count intersections graphically by checking each linear piece of the sawtooth against the line.

RESULT

Requires casework per period; 3 or 4 intersections depending on exact crossing pattern in $[0, 4\pi] \approx [0, 12.57]$ (the line g stays in $[0, 1]$ for $x \leq 10$, so intersections occur on the rising/falling sawtooth segments that lie in $[0, \pi] \approx [0, 3.14]$ range).

WHY IT MATTERS

The $\cos^{-1}(\cos x)$ "sawtooth wrapper" is a classic dailyintegral P flavor - it converts a trig identity into a piecewise-linear function whose intersections with a line must be counted carefully. Distinct from the Chebyshev-closure and hockey-stick P entries; the core skill is drawing the sawtooth and counting segment crossings.

3. Ceva altitude median angle bisector area

P1

EXAMPLE

In $\triangle ABC$, altitude AP , median BR , and angle bisector CQ are concurrent; $AC = 10$ and $BC = 26$. Express the area as $a\sqrt{b}/c$ in lowest square-free form and find $a + b + c$.

METHOD

Ceva gives $(BP/PC)(CR/RA)(AQ/QB) = 1$. The median gives $CR/RA = 1$ and the angle-bisector theorem gives $AQ/QB = AC/CB = 5/13$, hence $BP : PC = 13 : 5$. With $PC = 65/9$, use the right triangle APC to find the altitude, then the area.

RESULT**CONTINUED**

$[ABC] = 65\sqrt{155}/9$, so $a + b + c = 229$.

WHY IT MATTERS

Concurrency combines three different cevian theorems; Ceva converts the geometric roles into a side ratio, after which one Pythagorean calculation finishes the area.

4. Chebyshev closure root set count**P1****EXAMPLE**

Find the number of monic quadratics with real coefficients such that if X is a root then $X^2 - 2$ is also a root.

METHOD

The map $r \mapsto r^2 - 2$ must permute the (multiset) root set. Cases: fixed points $r = r^2 - 2 \Rightarrow r \in \{2, -1\}$ (double roots), a genuine 2-cycle $\{a, b\}$ with $a^2 - 2 = b$, $b^2 - 2 = a$ (Chebyshev/ $2 \cos \theta$ structure), and complex-conjugate pairs swapped by the map.

RESULT

6

WHY IT MATTERS

A dynamical-systems "root set closed under a quadratic map" count - distinct from existing P1 Vieta-coefficient and constraint-counting flavors. The Chebyshev $x = 2 \cos \theta$ conjugacy is the hidden structure.

5. Chinese remainder theorem puzzle**P1****EXAMPLE**

An integer N satisfies $N \equiv 6 \pmod{7}$ and $N \equiv 7 \pmod{8}$. Let M be the smallest positive such N . What is $M \bmod 9$?

METHOD

CRT trick: $N \equiv -1 \pmod{7}$ AND $N \equiv -1 \pmod{8} \Rightarrow N \equiv -1 \pmod{\text{lcm}(7, 8) = 56} \Rightarrow M = 55$. Then $55 \bmod 9 = 1$.

RESULT

1

WHY IT MATTERS

Existing P1 menu has Bayesian paradoxes, Vieta, constraint-counting, AM-GM - but classical CRT is genuinely missing. The " $\equiv -1$ in both moduli" cosmetic disguise is what makes it click in one line. Clean P1 quickie.

6. Circle tangent to two legs and circumcircle**P1****EXAMPLE**

Right triangle ABC , $\angle A = \pi/2$, $AB = 1$, $AC = 3$. A circle of radius r touches legs AB , AC and is internally tangent to the circumcircle. Find r .

METHOD

Place A at origin; the circle tangent to both legs has center (r, r) . Circumcircle has the hypotenuse as diameter: center $(\frac{1}{2}, \frac{3}{2})$, radius $R = \frac{\sqrt{10}}{2}$. Internal tangency: $\text{dist}((r, r), (\frac{1}{2}, \frac{3}{2})) = R - r$, a quadratic that factors.

RESULT

$r = 4 - \sqrt{10} \approx 0.838$

WHY IT MATTERS

Combines "tangent to both axes $\Rightarrow$ center (r, r) " with internal tangency to a circumcircle (hypotenuse-as-diameter). Distinct from the three-circle-coincident-tangent and inscribed-square P1 entries.

7. Circular seating conditional probability couple blocks

P1

EXAMPLE

One Indian couple and four American couples are seated uniformly at random around a circular table. Given that every American man sits next to his own wife, find the probability that the Indian man also sits next to his wife. (Answer as a decimal.)

METHOD

Condition by gluing each of the 4 American couples into a block (2^4 internal orders). The conditioned configuration is 6 units (4 blocks + Indian man + Indian wife) in a circle: $(6 - 1)! \cdot 2^4$. Favourable (Indian couple also adjacent): glue them too $\Rightarrow$ 5 units, $(5 - 1)! \cdot 2^4 \cdot 2$. Ratio = $\frac{4! \cdot 2}{5!} = \frac{2}{5}$.

RESULT

$$\frac{2}{5} = 0.4.$$

WHY IT MATTERS

A conditional-probability puzzle solved purely by couples-as-blocks circular counting - the 2^4 internal factors and the overall circular $(k - 1)!$ cancel in the ratio, leaving $2/5$. Distinct from existing P1 Bayesian / geometric-probability / constraint-counting entries: the conditioning event itself restructures the sample space into blocks, and the "decimal, not percentage" convention applies.

8. Clock letter walk with removals

P1

EXAMPLE

A clock numbered 1-12. A player chooses a random starting number n and moves n spaces clockwise — where n equals the number of letters in the English name of n (e.g. "three" = 5). Repeat the process from each new landing twice more. Then remove three random clock numbers (excluding the current). Move three spaces anticlockwise, skipping the gaps. Find the sum of all numbers the player could possibly have landed on.

METHOD

State-graph enumeration on a small finite map: build the deterministic "spell-and-walk" graph $n \mapsto (n + \ell(n)) \bmod 12$ where $\ell(n)$ is the letter-count of the spelling of n ; iterate three times; then for each reachable trajectory enumerate the $\binom{9}{3}$ removal choices and apply the gap-skipping anticlockwise step. Sum distinct landings.

RESULT

Requires casework; not derivable from the screenshot alone.

WHY IT MATTERS

Process / simulation puzzle with a letter-count edge function plus gap-skipping after removals. The P1 menu has constraint-counting and Bayesian paradoxes but no "spell-the-number, walk, remove, gap-step" iterated-process flavor. Highly characteristic dailyintegral aesthetic — absurd-sounding rules that yield a finite, enumerable answer.

9. Coefficient parity via p of plus minus one with vanishing factor

P1

EXAMPLE

$$P(x) = \prod_{k=1}^{2569} (x + (-1)^k k) = \sum_{i=0}^{2569} a_i x^i; \text{ find } \left| \frac{a_{2568} + a_{2566} + \dots + a_0}{a_{2569} + a_{2567} + \dots + a_1} \right|.$$

METHOD

Even/odd coefficient sums split as $\frac{P(1)+P(-1)}{2}$ and $\frac{P(1)-P(-1)}{2}$, so the ratio is $\left| \frac{P(1)+P(-1)}{P(1)-P(-1)} \right|$. The kicker: the factor $(x - 1)$ makes $P(1) = 0$, so the ratio is $\left| \frac{P(-1)}{-P(-1)} \right| = 1$.

RESULT

1

WHY IT MATTERS

New within the well-stocked Vieta family — no catalogued entry uses the $P(\pm 1)$ even/odd split, and none has the "one factor vanishes at $x = 1$ so the whole monstrosity is 1" punchline. Note 2569 is deliberately NOT year-derived, unlike the D3 constant in the same batch.

10. Common remainder gcd of differences

P1

EXAMPLE

1059, 1417, 2312 leave the same remainder r when divided by an integer $d > 1$. Find $d - r$.

METHOD

d divides all pairwise differences: $358 = 2 \cdot 179$, $895 = 5 \cdot 179$, $1253 = 7 \cdot 179$, so $d = \gcd = 179$; then $r = 1059 \bmod 179 = 164$.

RESULT

$$d - r = 179 - 164 = 15$$

WHY IT MATTERS

"Equal remainder under unknown modulus $\Rightarrow$ modulus divides the pairwise differences $\Rightarrow$ take the gcd." A classic number-theory P1 distinct from the CRT and modular-exponentiation-order entries; the differences sharing a large prime factor (179) is the engineered tell.

11. Concatenated integers number mod 45 (CRT)

P1

EXAMPLE

$N = 123456789101112 \dots 4344$ (integers 1..44 concatenated). Find $N \bmod 45$.

METHOD

CRT with $45 = 9 \cdot 5$; $\bmod 5 =$ last digit (4); $\bmod 9 =$ digit sum $\equiv \sum_{k=1}^{44} k = 990 \equiv 0$; solve $x \equiv 0(9), x \equiv 4(5)$.

RESULT

9

WHY IT MATTERS

huge concatenations are tractable by splitting the modulus into coprime parts (casting out nines + last-digit), never by direct division.

12. Conditional expectation of geometric race

P1

EXAMPLE

Airy and Barnes each flip a fair coin until they get heads for the first time. Given that Barnes takes strictly more flips than Airy, determine the expected number of flips taken by Airy.

METHOD

Let A, B be iid Geometric($\frac{1}{2}$) on $\{1, 2, \dots\}$, $P(A = k) = 2^{-k}$. Condition on $B > A$. By symmetry $P(B > A) = P(A > B)$ and $P(A = B) = \sum 4^{-k} = 1/3$, so $P(B > A) = 1/3$. Compute $E[A \cdot \mathbf{1}_{B>A}] = \sum_{k \geq 1} k 2^{-k} P(B > k) = \sum_{k \geq 1} k 2^{-k} 2^{-k} = \sum k 4^{-k} = \frac{4}{9}$. Then $E[A \mid B > A] = \frac{4/9}{1/3} = \frac{4}{3}$.

RESULT

$\frac{4}{3}$.

WHY IT MATTERS

A Logic/probability anchor: a "race of two geometric variables" with conditioning on a strict inequality. The clean hooks are $P(B > k) = 2^{-k}$ (memoryless tail) and the symmetry identity $P(B > A) = \frac{1}{2}(1 - P(A = B)) = \frac{1}{3}$. Distinct from existing P1 Bayesian-posterior and expected-value entries - here the conditioning event couples two independent runners.

13. Cube sum identity c equals minus one forces x plus y

P1

EXAMPLE

For distinct positive numbers x, y with $x^3 + y^3 + 3xy = 1$, determine $\frac{x^3 + 1}{x^2 - x + 1} + \frac{y^3 - 1}{y^2 + y + 1}$.

METHOD

Rewrite the constraint as $x^3 + y^3 + (-1)^3 - 3 \cdot x \cdot y \cdot (-1) = 0$, the $c = -1$ case of $a^3 + b^3 + c^3 - 3abc = (a + b + c)(a^2 + b^2 + c^2 - ab - bc - ca)$. Since $x, y > 0$ are distinct the second factor is positive, forcing $x + y - 1 = 0$, i.e. $x + y = 1$. Then $\frac{x^3 + 1}{x^2 - x + 1} = \frac{(x + 1)(x^2 - x + 1)}{x^2 - x + 1} = x + 1$ and $\frac{y^3 - 1}{y^2 + y + 1} = \frac{(y - 1)(y^2 + y + 1)}{y^2 + y + 1} = y - 1$, so the sum is $(x + 1) + (y - 1) = x + y = 1$.

RESULT

1

WHY IT MATTERS

A Logic/algebra anchor combining the $a^3 + b^3 + c^3 - 3abc$ factorisation (with the hidden $c = -1$) to force $x + y = 1$, then the sum/difference-of-cubes collapse $x^3 \pm 1 = (x \pm 1)(x^2 \mp x + 1)$ so each fraction telescopes to a linear term. Distinct from existing P1 number-theory (CRT, multiplicative-function) and probability entries - this is a pure symmetric-polynomial-identity puzzle whose two cubic factorisations chain into the constraint value $x + y$.

14. Digit shift multiplication mistake division algorithm

P1

EXAMPLE

Two naturals differ by 94; a multiplication mistake makes the product's tens digit 4 less than correct. Dividing the incorrect product by the larger factor gives quotient 52, remainder 107. Find the sum of the two numbers.

METHOD

Let the smaller number be x , larger $L = x + 94$. The digit slip means the incorrect product is $P' = xL - 40$. The division fact gives $P' = 52L + 107$. Equating: $xL - 40 = 52L + 107 \Rightarrow L(x - 52) = 147$; substituting $L = x + 94$ gives the quadratic $(x + 94)(x - 52) = 147 \Rightarrow x^2 + 42x - 5035 = 0 \Rightarrow x = 53$. Sum = $x + (x + 94) = 200$.

RESULT

200 (verified: $x = 53$, $L = 147$, correct product 7791, incorrect product $7751 = 52 \cdot 147 + 107$).

WHY IT MATTERS

A classic "digit-error in a product + division algorithm (quotient/remainder)" logic puzzle, distinct from the catalogued CRT, modular-order, and common-remainder-gcd P1 families - the mistake is modeled as a constant subtraction (-40) from the true product rather than a modular congruence, and the division fact pins down a second linear relation that turns the whole setup into a single solvable quadratic in one unknown.

15. Escalating die escalating stake recursion

P1

EXAMPLE

A die starts with n^2 faces ($n = 2$ initially), numbered $1, \dots, n^2$. Players alternate rolling; if a 1 comes up, the roller pays the other player the current value of n and the game ends; otherwise $n \rightarrow n + 1$ (die grows to $(n + 1)^2$ faces) and the other player rolls next. Alice rolls first. Find Alice's expected net profit.

METHOD

Recursive expectation where BOTH the win probability ($1/n^2$ at stage n) and the payoff magnitude (n) grow together each round; requires summing an alternating series over turn number with a shrinking-probability/growing-payoff trade-off.

RESULT

Not fully resolved on screen; requires the infinite series $\sum_{n \geq 2} (\pm n) \prod (\text{survival probs}) \cdot \frac{1}{n^2}$.

WHY IT MATTERS

Distinct from existing P1 menu (Vieta, CRT, Bayesian, combinatorics, clock-walk) - the "stake and odds both driven by the same escalating counter" mechanic is a fresh recursive-expectation flavor worth mining further.

16. Expected value of a difference of dice sums (logic)

P1

EXAMPLE

David rolls a 10-sided die 11 times (sum B); Dakota rolls an 11-sided die 10 times (sum A). Find $E[B - A]$.

METHOD

linearity of expectation; $E[B] = 11 \cdot \frac{1+10}{2} = 60.5$, $E[A] = 10 \cdot \frac{1+11}{2} = 60$.

RESULT

$E[B - A] = 0.5$.

WHY IT MATTERS

a one-line linearity-of-expectation puzzle dressed as a dice contest; reward is "don't simulate, just average".

17. Generalized slope 2 ballot prefix constraint

P1

EXAMPLE

14 people (10 VIP, 4 General Admission) queue in uniformly random order. Probability that at every prefix, $\#VIP \geq 2 \#GA$.

METHOD

Generalized ballot / lattice-path count with a 2:1 prefix constraint - count arrangements of 10 V and 4 G whose every prefix satisfies $\#V \geq 2\#G$ (DP over (v, g) states), divide by $\binom{14}{4} = 1001$. Distinct from slope-1 Dyck/ballot and the lattice-walk-with-floor entries.

RESULT

$$\frac{273}{1001} = \frac{3}{11} \approx 0.2727.$$

WHY IT MATTERS

"Group A always at least $m \times$ group B" is a slope- m ballot boundary; count by reflection/DP, not by the Catalan formula.

18. Hermite floor identity $\lfloor kx \rfloor - k\lfloor x \rfloor = \lfloor k\{x\} \rfloor$ + Farey step counting

P1

EXAMPLE

$f(x) = \sum_{k=2}^{10} (\lfloor kx \rfloor - k\lfloor x \rfloor)$; how many distinct values for $x \geq 0$?

METHOD

Each term depends only on $t = \{x\}$ via $\lfloor kt \rfloor$, so f is a step function on $[0, 1)$ with breakpoints at fractions j/k ($2 \leq k \leq 10$). Count distinct levels across consecutive Farey₁₀ intervals.

RESULT

distinct-level count over the Farey-10 partition (enumeration).

WHY IT MATTERS

Reduces a periodic floor-sum to fractional-part analysis + Farey breakpoints; clean constraint-counting Logic anchor.

19. Hockey stick binomial with diophantine target

P1

EXAMPLE

Find the smallest positive integer m such that the coefficient of x^2 in $\sum_{k=2}^{49} (1+x)^k + (1+mx)^{50}$ equals $(3n+1)\binom{51}{3}$ for some positive integer n . Find n .

METHOD

Hockey-stick identity $\sum_{k=2}^{49} \binom{k}{2} = \binom{50}{3}$. The x^2 -coefficient becomes $\binom{50}{3} + \binom{50}{2}m^2$. Match against $(3n+1)\binom{51}{3}$ as a Diophantine equation in (m, n) and find the minimal solution.

RESULT

smallest (m, n) from the resulting linear-in- n , quadratic-in- m Diophantine.

WHY IT MATTERS

Existing P1 has Vieta-on-roots and constraint-counting combinatorics, but the hockey-stick identity $\sum \binom{k}{r} = \binom{n+1}{r+1}$ coupled with a Diophantine target is a distinct combinatorial flavor. Rewards spotting the binomial-sum collapse.

20. Integer centroid lattice triple counting via residues mod 3

P1

EXAMPLE

On the 6×6 integer grid $\{1, \dots, 6\}^2$, count unordered triples $\{A, B, C\}$ whose centroid $(\frac{x_A+x_B+x_C}{3}, \frac{y_A+y_B+y_C}{3})$ has integer coordinates.

METHOD

Need $x_A + x_B + x_C \equiv 0$ and $y_A + y_B + y_C \equiv 0 \pmod{3}$, independently. Each residue class $(x \bmod 3)$ holds 2 of the 6 columns (12 points); per axis a triple's coordinate-sum $\equiv 0$ iff its three residues are all-equal or all-distinct. Count by residue-class multiplicities, combine the two axes.

RESULT

804 (verified by brute force over all $\binom{36}{3} = 7140$ triples; no collinearity exclusion - the problem only requires an integer centroid).

CONTINUED

WHY IT MATTERS

Characteristic constraint-counting Logic - reduces a geometric-sounding centroid condition to independent mod-3 residue bookkeeping per axis. Clean new P1 anchor; remember the all-equal-or-all-distinct rule for "sum of 3 residues $\equiv 0$ ".

21. Iterated digit count of a power tower

P1

EXAMPLE

Let $D(n)$ be the number of decimal digits of n . Find $D(D(D(7^{6^{54321}}))))$.

METHOD

$D(n) = \lfloor \log_{10} n \rfloor + 1$, so each application is essentially "take $\log_{10}$ ". For 7^E , $D = \lfloor E \log_{10} 7 \rfloor + 1$; iterating peels one exponent-tower layer at a time via $\log_{10}$. Track only the dominant term: with the tower $E = 6^{549}$ (since $2^1 = 2, 3^2 = 9, 4^9 = 262144$), compute $\log_{10} \log_{10} E = 262144 \log_{10} 5 + \log_{10}(\log_{10} 6) \approx 183230.68$. Then $D_3 = \lfloor \log_{10} \log_{10} E \rfloor + 1 = 183231$ (a 6-digit number), and the fourth digit-count $D_4 = \lfloor \log_{10} 183231 \rfloor + 1$.

RESULT

6.

WHY IT MATTERS

Classic "power tower + iterated digit function" logic problem: each $D(\cdot)$ is a disguised $\log_{10}$, so four nestings drill four levels down the tower's magnitude. Distinct from the logged modular-exponentiation ("last two digits") and CRT P1 entries - here the payoff is magnitude bookkeeping ($\log_{10} \log_{10}$) rather than residues. The final small integer answer (6) is the tell that the nesting depth is tuned to collapse the astronomical tower.

22. Lattice path expected diagonal visits

P1

EXAMPLE

A uniformly random monotone path from $(0,0)$ to $(4,4)$ uses unit right/up steps. A point is stable if $x = y$. Find the expected number of stable points on the path, expressed as m/n , then compute $m + n$.

METHOD

Use linearity of expectation over diagonal points (k,k) . The probability the path passes through (k,k) is $\binom{2k}{k} \binom{8-2k}{4-k} / \binom{8}{4}$.

RESULT

Sum the five diagonal-point probabilities and reduce.

WHY IT MATTERS

Logic screenshots now include lattice-path expectation puzzles; use indicator variables rather than enumerating paths.

23. Latus rectum normals tangent circle

P1

EXAMPLE

The normals to $y^2 = 4x$ at its latus-rectum endpoints are tangent to $(x-3)^2 + (y+2)^2 = r^2$. Find r^2 .

METHOD

Normal to $y^2 = 4ax$ at $(at^2, 2at)$ is $y = -tx + 2at + at^3$; with $a = 1$, the latus-rectum endpoints $t = \pm 1$ give lines $x + y - 3 = 0$ and $x - y - 3 = 0$. Set the perpendicular distance from $(3, -2)$ equal to r .

RESULT
 $r^2 = 2$
WHY IT MATTERS

Combines conic normal-line parametrization with a point-to-line tangency (distance) condition. Both normals are equidistant ($\sqrt{2}$) from the center, so the circle is tangent to both - beyond a plain Vieta/coordinate exercise.

24. Max determinant sign matrix hadamard bound

P1

EXAMPLE

Let P be a 3×3 matrix with every entry in $\{-1, 0, 1\}$. Find $\max(\det P)$.

METHOD

Hadamard's inequality bounds $|\det P| \leq (\sqrt{3})^3 = 3\sqrt{3} \approx 5.196$, but over the discrete set $\{-1, 0, 1\}$ the maximal 3×3 determinant is the known value 4 (attained e.g. by $\begin{pmatrix} 1 & 1 & -1 \\ -1 & 1 & 1 \\ 1 & -1 & 1 \end{pmatrix}$ -type circulants / maximal-determinant matrices).

RESULT

4.

WHY IT MATTERS

A maximal-determinant / Hadamard-bound extremal puzzle - across the ~ 26 logged P1 entries (Vieta, CRT, Bayesian, inscribed-figure, AM-GM, game theory) there is no sign-matrix determinant-maximization item. New extremal-combinatorics anchor; the discrete max (4) sits strictly below the continuous Hadamard bound, which is the intended insight.

25. Modular exponentiation last two digits

P1

EXAMPLE

Find the last two digits of 3^{999} .

METHOD

Work mod 100. The multiplicative order of 3 mod 100 is 20 (since $3^{20} \equiv 1$), so $3^{999} = 3^{20 \cdot 49 + 19} \equiv 3^{19} \pmod{100}$; reduce 3^{19} stepwise (or via CRT mod 4 and mod 25).

RESULT

27

WHY IT MATTERS

Distinct from the existing CRT puzzle entry: here the payoff is recognising the cyclicity/order of the base mod 100 and collapsing a huge exponent to a small residue, rather than reconstructing a number from simultaneous congruences. Classic "intimidating exponent, tiny order" flavor.

26. Monic quadratic iterated composition common root

P1

EXAMPLE

Let $f(x)$ be a monic quadratic with nonzero constant term and $f(1) = \frac{1}{3}$. If $f(x)$ and $f(f(f(f(x))))$ have a common real root, find $f(-3)$.

METHOD

If r is a common root of $f(x)$ and $f(f(f(f(x))))$, then $f(r) = 0$ and the orbit $r \rightarrow f(r) = 0 \rightarrow f(0) \rightarrow \dots$ must return. Write $f(x) = x^2 + bx + c$ (monic); $f(1) = 1 + b + c = 1/3$. A common root forces either $f(r) = 0$ (so r is a root of f) or r is a periodic point of period dividing 4. The fixed-point equation $f(r) = r$ combined with Vieta gives the constraints.

RESULT

Requires specific computation; the constraint system typically yields a clean rational, e.g. $f(-3) = 9 - 3b + c$.

WHY IT MATTERS

"Iterated composition $f^{(4)}$ shares a root with f " - a functional-iteration P flavor. The orbit structure of the quadratic map is constrained by the given value $f(1) = 1/3$. Distinct from the Chebyshev-closure-root-set-count P1 entry (which asks for a count) - here a specific function value is sought.

27. Multiplicative function diophantine uniqueness

P1

EXAMPLE

Let $f : \mathbb{Z}^+ \rightarrow \mathbb{Z}^+$ be strictly increasing with $f(mn) = f(m)f(n)$ for all $m, n \in \mathbb{Z}^+$, and such that $x^2 - f(y) = 1$ has a unique solution for integers $x, y > 1$. Find $f(4)$.

CONTINUED

METHOD

The multiplicativity forces $f(p^k) = f(p)^k$ for primes p . $f(1) = 1$. The Diophantine $x^2 - f(y) = 1$, i.e. $f(y) = x^2 - 1 = (x-1)(x+1)$, having a unique solution forces f to be the identity on primes (or a specific power map). Trying $f(n) = n$: $x^2 - y = 1$ has infinitely many solutions. Trying $f(n) = n^2$: $x^2 - y^2 = 1$ has unique solution $(x, y) = (1, 0)$ which fails $x, y > 1$... so try $f(p) = p$ for $p = 2$ giving $f(4) = 4$, but then check uniqueness. The unique-solution constraint pins down f .

RESULT

$f(4) = 9$ (if $f(p) = p^{\log_2 p}$ -type map, or more likely $f(n) = n^2$ gives no solution for $x, y > 1$ so f must be chosen so exactly one Pell-type solution exists).

WHY IT MATTERS

"Multiplicative function + unique Pell/Diophantine solution = pin down f " - a number-theory P flavor. The interplay between strict monotonicity, multiplicativity, and uniqueness of the equation $x^2 = f(y) + 1$ forces f to take a specific form on prime powers. Distinct from existing modular-exponentiation and CRT puzzle entries.

28. Multiplicative recurrence sequence exponent solve

P1

EXAMPLE

Positive integers $a_1, a_2, a_3, \dots$ satisfy $a_n = a_{n-2}a_{n-1}$ for $n \geq 3$, and $a_5 = 3200$. Find a_1 .

METHOD

Expand $a_3 = a_1a_2$, $a_4 = a_1a_2^2$, $a_5 = a_3a_4 = a_1^2a_2^3$; factor the target ($3200 = 2^7 \cdot 5^2$) and solve the resulting linear system in the prime exponents of a_1, a_2 separately for 2 and 5.

RESULT

$a_1 = 20$ (with $a_2 = 2$).

WHY IT MATTERS

A genuinely new P1 mechanism - a multiplicative (product, not sum) recurrence collapses to a monomial $a_1^2a_2^3$ in the last given term, turning a sequence puzzle into a prime-exponent linear system rather than the additive/Fibonacci-style recurrences, counting, or probability puzzles already in the P1 menu.

29. Paired Jensen min and AM-GM max combined via log arithmetic

P1

EXAMPLE

$m = \min \log_3(3^{y_1} + 3^{y_2} + 3^{y_3})$ s.t. $\sum y_i = 9$; $M = \max(\log_3 x_1 + \log_3 x_2 + \log_3 x_3)$ s.t. $\sum x_i = 9$, $x_i > 0$. Find $\log_2(m^3) + \log_3(M^2)$.

METHOD

A log-sum-exp/Jensen minimization (3^y convex, minimized at equal points) paired with a classic AM-GM product maximization (also equal points), recombined through a mismatched pair of log bases/powers engineered to land on clean integers.

RESULT

8.

WHY IT MATTERS

A new "twin extremal problem" P1 framing - a Jensen minimum and an AM-GM maximum computed side by side and stitched together with a final log-arithmetic combination step.

30. Probability of a posterior probability

P1

EXAMPLE

Two coins, one fair and one biased with heads-probability k . Pick one at random, flip, observe heads. Let p be the probability the same coin shows heads on the next flip. With $k \sim \text{Uniform}(0, 1)$, find $P(p > 2/3)$. (Answer as a decimal.)

METHOD

Build the Bayesian posterior over which coin was chosen given the first heads, giving $p(k) = \frac{0.25 + k^2}{0.5 + k}$; then treat p itself as a random variable and measure the k -set where $p > 2/3$ (Lebesgue measure on $[0, 1]$).

RESULT**CONTINUED**

numeric - solve $\frac{0.25 + k^2}{0.5 + k} > \frac{2}{3} \Rightarrow 3k^2 - 2k - 0.5 > 0 \Rightarrow k > \frac{1 + \sqrt{2.5}}{3} \approx 0.8604$, so $P \approx 0.1396$.

WHY IT MATTERS

The existing Bayesian-paradox family (Monty Hall, Boy-Girl, Three Prisoners) and Laplace's rule each compute a single posterior. This adds a second layer - the distribution OF a posterior probability over a uniform hyper-parameter - then asks a tail probability. A two-level Bayesian + geometric-probability hybrid not previously on the menu.

31. Regular heptagon reciprocal sine identity**P1****EXAMPLE**

Find the positive integer $n > 3$ satisfying $\frac{1}{\sin(\pi/n)} = \frac{1}{\sin(2\pi/n)} + \frac{1}{\sin(3\pi/n)}$.

METHOD

A classical heptagon identity. The reciprocal-sine relation among $\pi/7, 2\pi/7, 3\pi/7$ is one of the cleanest "named" trigonometric identities on the regular heptagon; equivalent forms involve the minimal polynomial of $2\cos(2\pi/7)$. Verify numerically: $1/\sin(\pi/7) \approx 2.305$; $1/\sin(2\pi/7) + 1/\sin(3\pi/7) \approx 1.279 + 1.026 = 2.305$. Competing $n = 5, 6$ both fail.

RESULT

$n = 7$

WHY IT MATTERS

A named-geometry trig identity, distinct from existing P1 Vieta-on-polynomial-roots and AM-GM flavors. The trick is recognising that " $\pi/n, 2\pi/n, 3\pi/n$ partition" is a regular-heptagon signature.

32. Rotated square stack union area by sector symmetry**P1****EXAMPLE**

Three side-6 squares share a centre; the middle is rotated 30° , the top 60° . The union is a 24-gon with area $a - b\sqrt{c}$; find $a + b + c$.

METHOD

The union is invariant under 30° rotation, so compute one of 12 congruent sectors: outer vertices at $R = 3\sqrt{2}$ (square corners), inner vertices where consecutive squares' edges cross at $r = 2\sqrt{3}$; area $= \frac{1}{2} \cdot 24 Rr \sin 15^\circ$.

RESULT

$108 - 36\sqrt{3} \Rightarrow a + b + c = 147$.

WHY IT MATTERS

Union-of-rotated-congruent-shapes puzzles collapse under the common rotational symmetry - find the fundamental sector and one inner/outer vertex pair.

33. Simon's Favorite Factoring Trick to divisor count of N^2 **P1****EXAMPLE**

$m = \#\text{ordered } (a, b) \in \mathbb{Z}_{>0}^2 \text{ with } \frac{1}{a} + \frac{1}{b} = \frac{1}{11}$; n likewise for $\frac{1}{60}$. Find mn .

METHOD

$\frac{1}{a} + \frac{1}{b} = \frac{1}{N} \Leftrightarrow (a - N)(b - N) = N^2$, so the number of ordered solutions is $d(N^2)$. $d(11^2) = 3$, $d(60^2) = d(2^4 3^2 5^2) = 5 \cdot 3 \cdot 3 = 45$.

RESULT

$mn = 3 \times 45 = \boxed{135}$ (verified).

WHY IT MATTERS

Highly characteristic dailyintegral number-theory puzzle - SFFT turns a unit-fraction equation into divisor counting of a perfect square. A clean new P1 anchor distinct from Vieta/lattice-walk families.

34. Tangency uniqueness by discriminant puzzle**P1****EXAMPLE**

Positive real n with $x^2 + y^2 = n(x + y)$ and $y^2 = x^2 + 2$ having a UNIQUE real solution; $n = \sqrt{p/q}$ coprime, find $p + q$.

METHOD

Substitute the hyperbola into the circle, reduce to one equation in x , impose discriminant = 0 (tangency) to force uniqueness; solve for $n^2 = p/q$.

RESULT

$p + q$ for the reduced fraction.

WHY IT MATTERS

"Daily Logic" uniqueness-by-tangency (circle vs hyperbola); the geometric reading (curves tangent) beats brute algebra. Distinct from Vieta/Bayes/game-theory logic entries.

35. Terminating decimal digit sum digital root via power of five order**P1****EXAMPLE**

Let k be the sum of the factorials of all positive integers from 1 to 67. Find the digital root of the sum of all decimal digits of $(1/2)^k$.

METHOD

Three-stage collapse. (1) $k \bmod 6$: for $i \geq 3$, $i!$ is divisible by 6, so $k \equiv 1! + 2! = 3 \pmod{6}$. (2) $(1/2)^k = 5^k/10^k$ is a terminating decimal whose digits (after the decimal point, padded with leading zeros to length k) are exactly the digits of 5^k ; leading zeros don't change the digit sum, so the digit-sum's digital root equals the digital root of 5^k itself. (3) 5 has multiplicative order 6 mod 9 (5, 7, 8, 4, 2, 1, ...), so $5^k \bmod 9$ depends only on $k \bmod 6 = 3$, giving $5^3 \equiv 8 \pmod{9}$.

RESULT

8.

WHY IT MATTERS

A new P1 family combining three separate number-theory facts - factorial divisibility mod a small modulus, the digit-sum-invariance of terminating decimals under the decimal-point shift $1/2^k \leftrightarrow 5^k$, and the multiplicative order of 5 mod 9 for a digital-root read-off. Distinct from the catalogued CRT and modular-order-of-3-mod-100 P1 entries because the exponent itself must first be tamed via factorial-sum divisibility before any modular-order argument applies.

36. Three circle coincident tangent radius**P1****EXAMPLE**

Two unit circles have centres 6 apart; P is their midpoint. A circle C is externally tangent to both. A line through P tangent to the first unit circle is also tangent to the second and to C . Find the radius of C .

METHOD

From P (distance 3 from each centre) a tangent to a unit circle makes $\sin \phi = 1/3$. Place C on the perpendicular bisector; impose (i) external tangency to both unit circles and (ii) distance-to-tangent-line = radius. Two constraints solve the radius.

RESULT

closed-form geometric (single positive root).

WHY IT MATTERS

A three-object tangency-coincidence configuration - distinct from inscribed-square/rectangle ratio and Sidon-ruler puzzles; rewards setting up "distance to a fixed tangent line = radius" simultaneously with external-tangency distance conditions.

37. Three player monty hall symmetric switch difference**P1****EXAMPLE**

Alice, Bob, Charlie each pick one of three doors (one hides a car). The host opens Charlie's door, which is empty. What is the difference between the probability Alice gets the car by switching and the probability she gets it by not switching?

CONTINUED

METHOD

Conditioning on "Charlie's door empty" leaves the car behind Alice's or Bob's door. Because all three players' choices were symmetric and the host's reveal carries no door-specific information distinguishing Alice from Bob, $P(\text{stay}) = P(\text{switch}) = \frac{1}{2}$. The difference is 0.

RESULT

0

WHY IT MATTERS

"Daily Logic" Monty-Hall variant engineered so the usual $\frac{1}{3}$ vs $\frac{2}{3}$ asymmetry collapses: with three players covering all doors and the host forced onto a specific (Charlie's) door, switching confers no advantage. The pedagogical trap is reflexively answering $\frac{1}{3}$; the correct answer is 0. Distinct from existing Bayes/probability logic entries.

38. Two-stage noisy-channel Bayes update

P1

EXAMPLE

A signal is green with probability $4/5$ and red with $1/5$. It is relayed through two independent stations, each of which transmits the colour correctly with probability $3/4$ and flips it with probability $1/4$. Station B reports green. What is the posterior probability the original was green? (Decimal.)

METHOD

$P(\text{B sees green} \mid \text{green sent}) = (3/4)^2 + (1/4)^2 = 5/8$ (zero or two flips). $P(\text{B sees green} \mid \text{red sent}) = 2 \cdot (3/4)(1/4) = 3/8$ (one flip). Bayes: $\frac{(4/5)(5/8)}{(4/5)(5/8) + (1/5)(3/8)} = \frac{20}{23} \approx 0.8696$.

RESULT
 $20/23 \approx 0.8696$
WHY IT MATTERS

A relay-channel / Markov-chain Bayes puzzle. Distinct from Boy-Girl-Tuesday (multi-condition) or Laplace's rule of succession. The trick is enumerating flip-parity, not naively multiplying $(3/4)^2$.

39. Uniform sum hitting time expectation equals e^r

P1

EXAMPLE

$U_1, U_2, \dots$ i.i.d. $\text{Unif}(0, 1)$; let $N(r) = \min\{n : \sum_{i=1}^n U_i > r\}$ for $r \in (0, 1]$. Find $E[N(r)]$, and hence $E[N(1)]$.

METHOD

Irwin-Hall on $[0, 1]$: for $0 \leq r \leq 1$, $P(\sum_{i=1}^n U_i \leq r) = \frac{r^n}{n!}$ (no min/floor terms in that range). Tail-sum expectation $E[N] = \sum_{n \geq 0} P(N > n) = \sum_{n \geq 0} P(\sum_{i=1}^n U_i \leq r) = \sum_{n \geq 0} \frac{r^n}{n!} = e^r$.

RESULT
 $E[N(r)] = e^r; E[N(1)] = e$.
WHY IT MATTERS

The classic "expected number of uniforms whose running sum first exceeds 1 is e ." Two clean recognitions - the simplex-volume CDF $r^n/n!$ and the $E[N] = \sum P(N > n)$ tail identity - chain to e^r . A strong non- π probability P1 whose headline constant is e ; distinct from the logged coupon-collector/Markov-hitting-time entries (those give rationals; this gives e).

40. Universal divisor of shifted linear product

P1

EXAMPLE

Find the largest possible $m \in \mathbb{N}$ such that m divides $n(n+1)(n+12)(n+123)(n+1234)(n+12345)$ for all $n \in \mathbb{N}$.

METHOD

For each prime p , determine the minimum guaranteed power of p dividing the product across all residues of $n \bmod p$ (and higher prime powers), by checking how the six shift amounts $\{0, 1, 12, 123, 1234, 12345\}$ distribute across residue classes mod p^k - a pigeonhole/CRT argument per prime, then multiply the guaranteed prime-power factors together.

RESULT

A specific integer m (computed via per-prime pigeonhole analysis over the six shifts; not a small "obvious" number since the shifts are not consecutive).

WHY IT MATTERS

Distinct from the catalogued "common-remainder-gcd-of-differences" P1 entry (which finds an unknown modulus from equal remainders) - here the modulus is universal-for-all- n divisibility of a product of six shifted linear factors, requiring per-prime worst-case residue analysis rather than a single gcd computation.

PART IV

Rapid-reference sheets

Integral recognition matrix

Inner derivative

Substitute the composite quantity; change bounds at once.

Polynomial times inverse trig/log

Integration by parts; rewrite the rational remainder.

Rational trig

Identities first, then $t = \tan(x/2)$ if necessary.

Symmetric interval

Form $f(x) + f(-x)$ before seeking an antiderivative.

Linear weight on symmetric kernel

Reflect $x \mapsto a - x$ and average.

Positive half-line with x and $1/x$

Pair reciprocal forms or introduce a Mellin parameter.

Gamma, beta, polylog

Use recurrences, reflection, or parameter differentiation.

Infinite series inside integral

Establish convergence, expand, separate, and integrate termwise.

Derivative and differential recognition matrix

High derivative at zero	Maclaurin coefficient or complex exponential.
Inverse or implicit relation	Locate the branch, differentiate, solve for the derivative, check nonzero denominator.
Variable power	Take logarithms.
Sampled derivatives inside definition	Name them, differentiate until the finite algebra closes.
$y' + p(x)y = q(x)$	Integrating factor.
$y' = F(y/x)$	Set $y = vx$.
$x^2y'' + axy' + by = 0$	Cauchy-Euler trial $y = x^m$.
Transport PDE	Characteristics.
Heat/wave/Laplace with Fourier data	Eigenmodes or d'Alembert.

Limit and logic recognition matrix

$1^\infty, 0^0, \infty^0$

Factorials

$n^{-1} \sum f(k/n)$

Concentrated powers

Recursive fixed point

Product over roots

Restricted paths

Repeated operation

Rotational equivalence

Large modular exponent

Take logarithms, determine the exponent limit, then exponentiate.

Stirling after logarithms.

Riemann sum.

Wallis, Laplace's method, or approximate identity.

Expand one order beyond linear and transform the error.

Evaluate the polynomial strategically.

Reflection/ballot/Catalan.

Invariant or monovariant.

Burnside.

Multiplicative order and CRT.

Final checking protocol

1. Differentiate every claimed antiderivative and inspect convergence of every improper integral.
2. Substitute every ODE/PDE solution into both the equation and all initial/boundary data.
3. For derivatives, check branch/domain conditions and whether the evaluation point was used early enough.
4. For limits, state the scale and the theorem that permits every exchange or asymptotic replacement.
5. For counting/probability, identify the sample space, forbidden-set bijection, group action, or posterior explicitly.
6. Compare signs, units, endpoint behaviour, and a numerical approximation whenever possible.

Name the structure. Write the decisive first line. Check by a different route.